Bahir Dar University

Bahir Dar Institute of Technology

Faculty of Computing

Requirement Analysis & Design Document for Industrial project on

Medicine recommender & reminder App

Submitted to the faculty of computing in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Systems

Group member

Name	ID no
Memar Alemneh	1507069
Dawit Zeleke	1505682
Addisu Gebeyehu	1504910

Adivisor: Ms. Birtukan

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Declaration

The Project is our own and has not been presented for a degree in any other university and all the sources of material used for the project have been duly acknowledged.

Name	Signture
Memar Alemneh	
Dawit Zeleke	
Addisu Gebeyehu	

Faculty: Computing

Program: Information Systems

Project Title: Medicine Recommender & Reminder App

This is to certify that I have read this project and that in my supervision and the students' performance, it is fully adequate, in scope and quality, as a project for the degree of Bachelor of Science.

Name of Advisor **Signature**

Ms.Birtukan

Examining committee members	**signature**	**Date**
1, Examiner1 ____________________	______________	_____________
2, Examiner2 ____________________	______________	_____________

It is approved that this project has been written in compliance with the formatting rules laid down by the faculty

Roles and Responsibilities of the Group Members

The following table presents the major tasks involved in producing this Requirement Analysis Document and indicates the participation of each group member. A checkmark (√) shows that the member actively contributed to the corresponding task. Each member participated collaboratively and gained a full understanding of the entire project.

List of Tasks	Memar	Dawit	Addisu
Requirement Gathering (Interview, Observation, Document Analysis)	√	√	√
Problem Identification & Statement Preparation	√		√
Literature Review (Medication Systems, Reminder Apps, AI Recommendation Research)	√	√	
Functional & Non-Functional Requirement Identification	√	√	√
Use Case Identification & Documentation	√	√	
System Modeling (Use Case, Activity, Sequence, State Diagrams)		√	√
Class Diagram & Logic Modeling		√	√
User Interface Prototyping	√		√
Feasibility Study Preparation	√	√	
System Features Description (Existing & Proposed System)	√	√	√
Business Rule Documentation		√	√
Writing and Drafting Chapters	√	√	√
Reviewing, Editing & Finalizing	√	√	√

the Document			
Formatting & Structuring the Report	√		√
Presentation Preparation	√	√	√

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List of Acronyms

Acronym	Meaning
RAD	Requirement Analysis Document
UML	Unified Modeling Language
AI	Artificial Intelligence
ML	Machine Learning
DBMS	Database Management System
SQL	Structured Query Language
CRUD	Create, Read, Update, Delete
UI	User Interface
UX	User Experience
SMS	Short Message Service
CRC	Class–Responsibility–Collaborator
CPU	Central Processing Unit
RAM	Random Access Memory
API	Appcation programming interface

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Abstract

Medication adherence is one of the most critical challenges affecting patients worldwide, particularly individuals managing chronic illnesses, elderly patients, and those with complex daily routines. Forgetting medication doses, misunderstanding dosage instructions, taking incompatible medicines together, and failing to follow prescribed schedules are common issues that lead to reduced treatment effectiveness, increased health risks, and preventable medical complications. In response to these challenges, digital health solutions have emerged as reliable support systems that assist individuals in managing their medication routines more effectively.

This project, titled **"Medicine Recommend and Reminder App,"** aims to design and analyze a user-friendly, intelligent, and efficient mobile-based system that assists users in managing their medication schedules, tracking dosage adherence, and receiving personalized health-support recommendations. The system incorporates features such as medicine scheduling, automated notifications, symptom reporting, AI-powered medicine recommendation, drug-allergy filtering, drug-interaction checking, and access to basic medicine information. In addition, the system enables users to build and manage a comprehensive health profile, including age, allergies, chronic diseases, and medical history, which allows the recommendation engine to generate safer and more accurate suggestions.

The core of the system is an AI-driven recommendation module that analyzes reported symptoms and maps them to relevant medicines or drug categories. This module integrates safety constraints derived from user health profiles to prevent risky drug interactions or allergic reactions. The app also provides additional functionalities including dose tracking, missed-dose alerts, and optional integration of home remedy suggestions. Together, these features aim to minimize human error, enhance medication adherence, and support users in maintaining a healthier and more organized lifestyle.

This Requirement Analysis Document presents a detailed breakdown of the system's functional and non-functional requirements, modeling techniques, feasibility analysis, business rules, and user interface prototypes. It includes a systematic exploration of the existing challenges in medication management, the proposed technological solution, and the analytical procedures used to translate user needs into actionable system requirements. By examining the problem domain and defining the conceptual, logical, and behavioral models, this document lays a strong foundation for the design and future development of the Medicine Reminder App.

Ultimately, the system is intended to benefit a wide range of users—patients, caregivers, family members, and healthcare institutions—by providing a reliable digital platform that promotes timely medication intake, improves treatment outcomes, and contributes to safer and more informed health practices. The findings and specifications outlined in this document will serve as a guiding blueprint for the subsequent phases of system design, implementation, and evaluation.

Chapter One: Introduction

1.1 Background

Medication adherence is a critical component of effective healthcare, particularly for patients with chronic conditions, elderly individuals, and those required to follow strict dosing schedules. Yet, despite medical advancements and increased awareness, a large number of individuals fail to take their medications correctly. Globally, non-adherence remains a silent obstacle in healthcare systems, contributing to worsened conditions, increased treatment failures, and preventable medical complications. Studies indicate that more than half of prescribed medications are not taken as directed, highlighting a major gap between medical instruction and real-world patient compliance.

Several factors contribute to poor medication adherence:

Forgetfulness and Routine Disruption: Many patients struggle to remember medication schedules, especially when taking multiple pills daily.

Misinterpretation of Dosage Instructions: Incorrect timing or dosing significantly reduces treatment effectiveness and can lead to medical emergencies.

Lack of Awareness About Drug Interactions: Patients often combine medications unknowingly, exposing themselves to severe allergic reactions or interaction-based risks.

Limited Medical Knowledge: Many individuals lack health literacy, making them unaware of how crucial medication timing and consistency are to recovery.

So, medication adherence represents one of the most persistent challenges in modern healthcare, particularly for patients who rely on long-term medical treatment. A simple missed dose may seem harmless, yet repeated lapses can escalate into life-threatening complications. The burden of improper medication use affects not only individuals but entire healthcare systems, increasing hospitalization rates, treatment costs, and long-term morbidity.

However, just as maternal health challenges drove innovation through digital solutions, medication management too stands at a turning point. The rapid global expansion of smartphones and network access presents an opportunity for technological intervention. Mobile-based healthcare tools have shown measurable success in self-care support, daily monitoring, and medical guidance. Today, millions of people use mobile health (mHealth) platforms—yet most existing systems remain limited to simple reminder alerts without intelligence, personalization, or clinical-safety measures.

In response to this gap, the **Medicine Recommend and Reminder Application** is designed as a comprehensive digital solution that supports safe, structured medication adherence. The system allows users to create personal health profiles containing allergies, age, pregnancy status, chronic conditions, and medical history, enabling personalized medication recommendations tailored to individual needs. Beyond reminder functions, the app integrates **AI-powered symptom reporting**, allowing users to record symptoms and receive preliminary guidance based on recognized clinical patterns. In addition, drug-interaction and allergy-filtering mechanisms reinforce safety by preventing harmful medication combinations.

Evidence from global Mobile Health implementations shows the impact of such digital intervention:

In chronic-care patients, reminder-based mobile systems improved adherence by up to **40%**, reducing relapse and complication rates.

In elderly populations, daily mobile monitoring lowered missed-dose frequency by **30–50%**.

Intelligent recommendation systems have been shown to improve treatment response time and prevent medication conflicts significantly.

These outcomes demonstrate how Mobile Health technologies can transform medication behavior, reduce preventable medical risk, and support long-term wellness.

1.2 Statement of the Problem

Medication management remains a pressing challenge across various patient groups, especially among individuals with chronic illnesses, elderly users, and those who are required to take multiple prescriptions daily. Despite its importance in treatment success, medication adherence is often disrupted due to a lack of structured systems and reliable support tools. As a result, many patients experience delayed recovery, reduced treatment effectiveness, and in some cases, serious health complications.

1. Lack of Reliable Medication Adherence Support

Many patients rely solely on memory or informal reminders to take their medication, making them prone to forgetting doses, taking the wrong amount, or missing medication schedules entirely. Without a dependable mechanism to organize as-needed intake, patients face increased health risks and prolonged illness.

2. Absence of Centralized Medication Management Tools

There is no unified digital solution that allows patients to store medication details, track dosage history, or manage schedules in one place. Most patients do not record their medicine intake, nor do they receive notifications when a dose is due or overdue, resulting in inconsistent adherence and ineffective treatment progress.

3. Limited Clinical Awareness and Unsafe Drug Use

A lack of medical literacy prevents many individuals from understanding proper dosage, side effects, or potential risks associated with mixing medications. Patients taking multiple prescriptions without guidance are at high risk of harmful drug interactions, allergic reactions, or overdose. In the absence of cross-checking systems, users may unknowingly consume medication that contradicts their conditions or medical history.

4. Insufficient Access to Medicine Information and Alternatives

Users often struggle to find reliable information regarding prescribed medications, substitutes, or usage guidelines. Existing information sources are scattered, unverified, and rarely personalized, leaving patients to make uninformed decisions. This challenge intensifies when managing multiple drugs or when the user lacks health-related knowledge.

5. Lack of Personalized and Automated Medication Recommendation and Assistance

Current medication management practices rarely offer personalized recommendations based on a user's condition, medication history, or availability of drugs. Elderly users, busy individuals, and those managing complex treatments receive little automated assistance beyond basic reminders. Additionally, users may face challenges identifying suitable substitute medications when a prescribed drug is unavailable. The absence of an intelligent recommendation feature limits the system's ability to support safe, efficient, and personalized medication management.

1.3 Objectives

1.3.1 General Objective

To design and develop a Medicine Recommend and Reminder App that enables users to effectively manage medication schedules while providing AI-assisted symptom analysis and personalized medicine recommendations for improved treatment adherence and safety.

1.3.2 Specific Objectives

- ✓ To implement automated medicine reminders.
- ✓ To allow creation of health profiles including allergies and chronic conditions.
- ✓ To provide AI-supported symptom analysis and medicine suggestions.
- ✓ To prevent drug conflicts and recommend safe alternatives.
- ✓ To track missed doses and notify the user.
- ✓ To provide access to medicine information and home remedies.

1.4 Methodology

The development of the Medicine Recommend and Reminder App follows a structured and systematic approach, beginning from requirement collection to implementation. The methodology ensures that the system meets user needs, functions effectively, and remains scalable for future enhancements.

1.4.1 Requirement Gathering Methods

To understand the needs of users and determine the system features, several requirement collection techniques were used:

Interviews with potential users
Direct conversations were held with patients, caregivers, and pharmacy users to identify daily challenges in medication management. Their feedback helped define core features such as reminders, medicine lists, and symptom-based recommendations.

Literature review on medical reminder systems
Previous studies, applications, and academic articles related to medication adherence tools were reviewed. This provided insight into existing solutions, their strengths, and their limitations.

Observation of common medication routines
Real-world medication practices were observed to understand how individuals manage dosage timing, track prescriptions, and respond to symptoms. These observations helped shape workflow design.

Document review of symptom-medicine mapping
Medical reference documents and symptom databases were analyzed to build logical links between symptoms and suggested medicines, forming the basis of the AI recommendation logic.

After comprehensive requirements gathering via **i**nterviews, literature review, observation, and document analysis, the diverse needs were consolidated and refined. The identified core features were structured into clear categories based on functionality and priority. This synthesis facilitated the mapping of software models and necessary application features. The collected data was then formally documented as user stories and functional specifications to guide the design and development process

1.4.2 Analysis and Design Methodology

After gathering the requirements, the system was analyzed and designed using structured modeling techniques:

Object-Oriented Analysis and Design (OOAD)
The system was broken down into interacting objects such as User, Medicine, Profile, Schedule, and AI Engine.

The **Object-Oriented Analysis and Design (OOAD)** methodology was chosen for the project due to its structured approach, which is ideal for a user-centric system like the **Medicine Recommend and Reminder App.**

OOAD was selected to effectively model the system's dynamic nature and extensive data management needs by breaking it down into interacting components such as User, Medicine, Profile, Schedule, and AI Engine.

The specific benefits for choosing the Object-Oriented approach are:

Modularity: It allows features, such as the Reminder or AI Engine components, to be developed as independent, reusable modules, improving system organization.

Data Handling: The app needs to handle complex user data, including health profiles, medical conditions, schedules. The OO approach provides efficient management of these interconnected data relationships and complex user interactions.

Maintainability: By compartmentalizing features into objects, changes or updates to one module (e.g., updating the dosage tracking logic) do not affect the others, making long-term maintenance easier.

Scalability: The modular structure ensures that new features (such as those identified in the future change cases), can be added without disrupting the existing system architecture.

UML Diagrams

Different diagrams were used to visualize system structure and behavior:

Use Case Diagrams – to represent system interactions with users and external components.

Sequence Diagrams – to describe communication flow between components.

Activity Diagrams – to illustrate process workflow (e.g., symptom analysis and reminder handling).

Class Diagrams – to define attributes, operations, and relationships between classes.

Prototyping for UI validation

Low-fidelity user interface prototypes were designed to test usability and layout. Feedback was gathered and used to improve navigation, screen structure, and feature accessibility.

1.4.3 Implementation Methodology

The implementation phase focuses on building the application using modern tools and development practices:

Frontend Development:Flutter

Chosen for cross-platform capability, enabling deployment on both Android and iOS devices using a single codebase.

Backend Development: Node.js / Express

Used to handle authentication, scheduling logic, user data processing, and communication with the AI engine.

Database: MongoDB

Selected for secure and flexible storage of medical profiles, reminders, logs, and user information.

AI Component: Rule-based combined with ML enhancement

The system analyzes user-entered symptoms using rule-based logic, while machine learning enhances accuracy and adaptability over time.

Project Management: Agile

Development is carried out in iterative sprints, enabling continuous testing, evaluation, and improvement until final delivery.

1.5 Feasibility

A feasibility assessment was conducted to determine whether the development of the Medicine Recommend and Reminder App is practical, achievable, and sustainable under the available resources, skill level, technology, and project timeframe. The feasibility study evaluates the project from three perspectives: economic, technical, and time feasibility. The goal is to ensure that the application can be successfully developed and deployed without exceeding financial or operational limits.

1.5.1 Economic Feasibility

The economic feasibility examines the financial requirements of the project in relation to the resources available. This system is designed to be developed using low-cost and open-source technologies, meaning no large investment is required for licensing, subscription, or specialized software purchases. Development tools such as Flutter for the frontend, Node.js for backend services, and MongoDB for database storage are freely available, which reduces overall cost to near zero for core development.

The only potential costs may be basic expenses such as internet connectivity during development, cloud hosting if the system is later deployed online, and optional access to pharmacy APIs if a commercial database is involved. Even these expenses are relatively minimal and manageable within an academic or small-scale project budget. Compared to the cost of existing commercial healthcare systems, this project offers a cost-effective solution that can be developed, tested, and used without financial strain.

Therefore, the project is economically feasible because the cost of tools, deployment, and maintenance remains low, and the development team does not require paid software or costly infrastructure.

1.5.2 Technical Feasibility

Technical feasibility determines whether the team has the required skills, tools, and infrastructure to build the application successfully. The technologies needed for developing the application are modern, widely taught, and well-documented.Flutter enables development for both Android and iOS, eliminating the need to build separate applications for different platforms. Node.js and Express provide a strong backend environment for handling authentication, reminders, scheduling, and communication with the AI module.

The database (MongoDB) supports large-scale storage of user details, medication schedules, symptom logs, and recommendation data. These databases are reliable, secure, and capable of handling real-time queries, which is essential for maintaining correct reminder triggers and profile filtering. The AI component can be implemented using a rule-based system, which is simple and technically achievable at this stage, and later expanded into machine learning when required.

Furthermore, the development team possesses the necessary programming knowledge and can access online learning resources if technical challenges arise. No specialized or high-end hardware is required — standard laptops, mobile devices for testing, and a stable development environment are sufficient. Since all required technologies and skill sets are available, the project is technically feasible.

1.5.3 Time Feasibility

Time feasibility evaluates whether the project can be completed within the allocated academic duration. The development plan is divided into phases, following the Agile development approach. This structure allows developers to build the application in repeated iterations, delivering modules step-by-step rather than all at once. Features such as login, reminder scheduling, symptom input, AI recommendation, and pharmacy lookup can be developed and tested in separate sprints.

Because each module is implemented and evaluated independently, errors can be fixed early, preventing delays later in the project. Prototyping for UI and system testing can be conducted in parallel to reduce development time. The features listed for the project are well-defined and achievable within a typical semester or development cycle if tasks are distributed and monitored effectively.

Given the realistic scope of work and structured approach, the project can be completed within the designated timeframe. Therefore, the project is time-feasible and development can proceed without exceeding the academic schedule.

1.6 Beneficiaries

The Medicine Reminder and Recommender App is designed to benefit a wide range of individuals who experience difficulties with medication adherence or require intelligent support in managing their health routines. In addition to reminder services, the system provides **symptom-based and profile-based medicine recommendations**, safe dosage guidance, and health tracking features. The following groups benefit from the implementation of this project:

1.6.1 Patients with Chronic Diseases

Patients living with chronic conditions such as diabetes, hypertension, asthma, and heart disease often depend on long-term medication schedules. Missing doses or taking incorrect medicines can result in severe health complications. The application assists these patients by providing **timely medication reminders**, maintaining personalized medication schedules, and **recommending safe medicines based on their specific conditions, allergies, and chronic illnesses**. By filtering unsafe drugs and suggesting appropriate alternatives, the system improves treatment adherence and reduces medication-related risks.

1.6.2 Elderly People

Elderly individuals commonly face memory challenges and frequently manage multiple medications simultaneously. The app supports them by sending **automated reminders**, recording medication history, and preventing accidental double-dosing. In addition, the **recommendation feature helps elderly users identify safe medicines by considering age-related risks, chronic conditions, and known allergies**, thereby reducing harmful drug interactions and supporting safer self-medication.

1.6.3 Students and Individuals with Busy Schedules

Students and working individuals often forget medication due to academic pressure, work demands, or irregular daily routines. The application helps them stay consistent by organizing reminders and tracking missed doses. Furthermore, the **medicine recommender system allows users to quickly receive appropriate medicine**

suggestions based on entered symptoms, helping them make informed choices without spending excessive time searching for health information.

1.6.4 Caregivers

Caregivers responsible for children, elderly family members, or chronically ill patients benefit from the application as a reliable monitoring and decision-support tool. The app not only automates reminders and logs medication intake but also **assists caregivers by recommending suitable medicines and warning against unsafe options based on the patient's health profile**. This reduces confusion, improves care quality, and lowers caregiver workload.

1.6.5 Health Institutions

Health institutions such as hospitals, clinics, and community health centers can **indirectly benefit** from the Medicine Reminder App through improved patient medication adherence and better self-care practices after treatment. Although health professionals and institutions are not direct users of the system, the application supports post-treatment care by helping patients follow prescribed medication schedules accurately.

1.6.6 Pharmacies

Pharmacies benefit **indirectly** from the Medicine Reminder App through improved medication awareness and responsible usage by patients. Although pharmacies do not directly interact with the system as actors, the application assists users in understanding prescribed medicines, dosage schedules, and general usage guidelines, which can reduce medication misuse and confusion at the point of purchase.

1.7 Limitations

Although the Medicine Recommend and Reminder App aims to improve medication management and provide intelligent support to users, there are certain limitations that affect the overall capability and scope of the system in its current form:

1.7.1 Not a Replacement for Professional Medical Diagnosis

The system is not intended to function as a certified medical diagnostic tool. The AI-based symptom recommendations are only supportive suggestions and may not fully interpret complex health conditions. Users are still required to consult healthcare professionals for accurate medical advice, prescriptions, or treatment decisions.

1.7.2 Dependence on Dataset Accuracy and Availability

The quality and safety of the recommendations rely on the accuracy of the symptom–medicine mapping and medical information stored in the system. If the dataset is incomplete, outdated, or contains inaccuracies, the suggestion engine may provide limited or less reliable outcomes.

1.7.3 Lack of Emergency Response Features

The application does not include emergency assistance such as direct calls to medical services or alerting family members in life-threatening conditions. In urgent health situations, users must seek immediate help from emergency medical providers rather than relying on the app.

These limitations highlight the areas where the system can be improved or extended in future development stages. Despite these constraints, the application remains a useful tool for increasing medication adherence and supporting everyday health management.

1.8 Scope of the Study

This project focuses on the design and development of a Medicine Reminder and Recommendation Application that assists users in managing prescribed medication schedules and provides AI-generated suggestions based on reported symptoms. The scope outlines the boundaries of what the system will accomplish, the features it will offer, and the areas it will cover during implementation.

1.8.1 Core Functional Coverage

The primary scope of this system includes:

Medication Scheduling and Alerts – Users can set reminders for daily, weekly, or specific-time medications. The system will send timely notifications to prevent missed doses.

Symptom-Based Medicine Recommendation – The application analyzes symptoms provided by the user and suggests over-the-counter medicine options supported by its dataset.

Pill Tracking and Dosage Monitoring – The system records taken and missed doses to help users track medication adherence over time.

Medicine Information Database – A searchable database is included where users can view dosage instructions, side effects, uses, and precautions for common medications.

User Profile and Medical History Storage – The app is able to store personal health profiles, previous illnesses, and past medication logs for more personalized suggestions.

1.8.2 Target Users and Usage Environment

The application is designed for:

- ✓ Patients who need daily medication reminders
- ✓ Individuals managing multiple prescriptions at once
- ✓ Users experiencing mild symptoms and needing guidance on basic medicine options
- ✓ Caregivers who monitor medication intake of family members

The app is intended to run on **mobile devices**, accessible anytime provided the user has internet access for AI-related features.

1.8.3 Technological Scope

The system will be developed using:

- ✓ A mobile platform (Android/iOS)
- ✓ AI-based recommendation algorithms
- ✓ A backend database for storing medicine and user records
- ✓ Push notification services for reminders

The scope includes system design, interface creation, algorithm development, and usability testing.

1.8.4 Geographical & Functional Boundaries

This study focuses on providing symptom-based recommendations only for common and mild health conditions such as headache, fever, flu, body pain, allergies, and minor digestive problems. The system is designed to support users through medicine reminders and general guidance, not to diagnose diseases or prescribe restricted medications. It does not cover complex, chronic, or life-threatening illnesses that require advanced medical examination. When the system detects symptoms that are severe, long-lasting, or potentially serious, it clearly advises the user to visit a qualified doctor or healthcare facility. This recommendation acts as a safety measure to protect users from relying solely on the application. The system serves as a supportive self-care tool and does not replace professional medical consultation. Geographic coverage is general, though some features may depend on local data availability.

1.8.5 End-User Interaction Scope

Users will be able to:

- ✓ Input symptoms and receive suggested medicines
- ✓ Add, edit, or delete medication schedules
- ✓ View reminders in calendar format
- ✓ Receive alert notifications for intake
- ✓ Track dosage completion and progress history

1.9 Organization of the Project

The project is organized into several chapters, each addressing specific aspects of the development
process. Below is a summary of the structure:

Chapter One: Introduction

This chapter introduces the project, outlining its background, the problem it seeks to solve, and the
objectives of the system. It includes the methodology used for requirements gathering, system analysis,
design, and implementation. Additionally, the chapter discusses the feasibility of the project, its
significance to the stakeholders, and the limitations and scope of the project.

Chapter Two: System Features

This chapter compares the existing system with the proposed system and analyzes the functional and
non-functional requirements of the app. It includes use case documentation, business rules, user interface
prototypes, system diagrams (e.g., activity, sequence, class, state chart), and other related models. It also
describes the system requirements in terms of hardware and software.

Chapter Three: System Design and Architecture

This chapter provides a detailed description of the system architecture, design patterns, and database
schema. It discusses the system's structure, the flow of data, and the relationships between various
components, ensuring that the design meets both functional and non-functional requirements.

Chapter Four: Implementation

This chapter explains the development process, including the tools and technologies used to implement
the system. It details the code structure, logic, and integration of different components such as the
backend, user interface, and database.

Chapter Five: Testing and Evaluation

This chapter outlines the testing methodology, including test cases, results, and how the system was
evaluated. It includes both functional testing to ensure the app meets requirements and performance
testing to assess the system's reliability and scalability.

Chapter Six: Conclusion and Recommendations

This chapter summarizes the project's achievements, discusses the challenges faced during development,
and suggests areas for future improvement. It also highlights the significance of the project and its
potential impact on the target users.

References

A comprehensive list of all sources cited throughout the project.

Appendice

additional information for our project (questionnaire)

Chapter Two: System features

2.1 Existing System

The current approach to medication management in most households and healthcare environments is still largely manual. Patients are often responsible for remembering their own medication schedules, relying on memory, written notes, or verbal instructions from healthcare providers. Although some individuals use basic reminder tools such as phone alarms or calendars, these methods often lack accuracy, personalization, and monitoring features. As a result, medication adherence remains a common challenge, especially for individuals taking multiple prescriptions or long-term treatment.

In many cases, patients forget to take their medicine at the required time or miss doses entirely, which can reduce the effectiveness of treatment. Traditional reminder methods also do not provide medical guidance or recommend medicines based on symptoms. They simply notify users of time, without considering dosage intervals, drug type, side effects, or patient history. This manual system heavily depends on human memory and discipline, making it prone to errors and inconsistency.

Furthermore, individuals experiencing mild symptoms such as headaches, flu, or stomach discomfort must often rely on self-diagnosis or general internet searches. While information is widely available, it is not always reliable or medically filtered, increasing the risk of taking inappropriate medication. There is no centralized digital solution that combines reminders, dosage tracking, medicine database access, and AI-based recommendations in one platform.

In summary, the existing system is:

- ✓ **Unguided**, as users must decide for themselves which medicine to take.
- ✓ **Unreliable**, because manual reminders and memory-based tracking often fail.
- ✓ **Unstructured**, lacking a dedicated database for medicine information.
- ✓ **Non-intelligent**, since no decision-support or symptom-based suggestions are available.

- ✓ **Difficult for elderly or busy individuals**, who are more likely to forget scheduled doses.

This gap highlights the need for a **Medicine Reminder and Recommendation Application** that not only alerts users about medication schedules but also assists in selecting suitable medicines using AI and well-structured medical data.

2.2 Proposed System

The proposed system is a **digital AI-powered Medicine Reminder and Recommendation Application** designed to improve medication adherence, reduce missed doses, and guide users in selecting safe medicines based on symptoms, allergies, and medical history. Unlike the traditional manual system, this application automates the entire medication management process and ensures that users follow their prescriptions properly and on time.

The system allows users to create a personalized medical profile where they can record chronic illnesses, allergies, and medications currently in use. Based on this information, the application analyzes symptoms entered by the user and generates medicine recommendations using AI logic and rule-based matching. To ensure safety, the system filters suggested medicines against allergies, drug interactions, and chronic conditions before presenting them to the user.

Furthermore, the application provides **automated reminders** for each scheduled dose, alerting the user at the right time with dose details such as medicine name, amount, and frequency. It also tracks missed or delayed doses, allowing users and caregivers to monitor medication adherence more effectively. In addition to recommendation services, the system offers access to medicine information, alternative treatment options, and nearby pharmacy availability for improved convenience.

Key functionalities of the proposed system include:

- ✓ Automated medication reminders with dosage schedule.
- ✓ AI-based symptom analysis to suggest possible medicines.
- ✓ Safety filtering using allergy, age, and medical profile data.

- ✓ Missed dose tracking and adherence monitoring.
- ✓ User-friendly interface for recording and managing medication history.
- ✓ Access to drug alternatives, side effects, and usage guidelines.

This system reduces dependency on memory or manual reminders and introduces a reliable, intelligent, and easily accessible platform for users. It enhances patient safety, supports self-care, and creates a structured environment where medication management becomes effortless and risk-free.

2.3 Requirement Analysis

Requirement analysis helps define what the system must accomplish and how it should behave. It clarifies the expectations of users, the functions the application needs to perform, and the constraints that guide its development. For this project, the Medicine Recommend and Reminder App requires both **functional** and **non-functional** requirements that ensure successful implementation, usability, and performance.

The requirements were identified through interviews, observation, literature review, and analysis of existing challenges in medication management. The following subsections present the detailed requirements the proposed system must satisfy.

2.3.1 Functional Requirements

Functional requirements describe the core operations the system should perform. These represent actions users can complete within the application and the responses expected from the system.

ID	Requirement Description	Priority
FREQ-1	User registration and secure login.	High
FREQ-2	Create and edit personal health profile.	High
FREQ-3	Add, modify, and delete medicine schedules.	High
FREQ-4	Automatic notification and reminder for each dose.	High
FREQ-5	AI-based symptom input and analysis.	High
FREQ-6	Provide recommended medicines based on symptoms.	High
FREQ-7	Filter medicines against allergies and drug interactions.	High
FREQ-8	Track missed doses and notify user accordingly.	Medium
FREQ-9	Display drug information, alternatives, and usage guidelines.	Medium

Table 1: Functional Requirements

2.3.2 System Use Case

2.3.2.1 Use Case Diagram

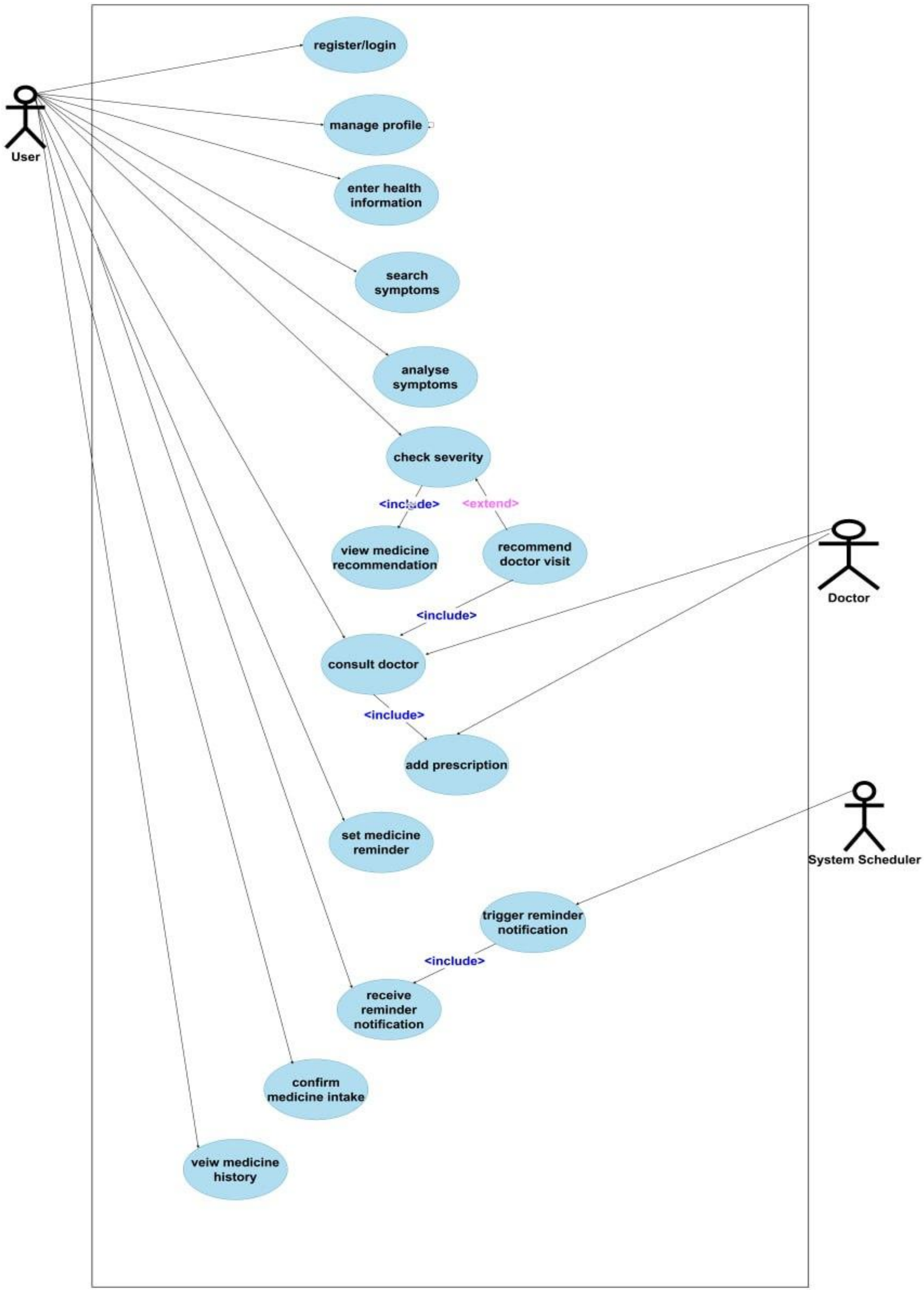

Figure 1;use case diagram

2.3.2.2 Use Case Documentation

Use Case 1	
Field	**Description**
Name	Register / Login
Actor	User
Description	Allows a user to register a new account or log in to the Medicine Recommend & Reminder App.
Pre-condition	User is not logged in to the system.
Post-condition	User is successfully authenticated and gains access to the application.
Priority	High
Basic Course of Action	1. User opens the application. 2. User selects Register or Login. 3. User enters required credentials (email/phone and password). 4. System validates the credentials. 5. System grants access to the user dashboard.
Alternate Course of Action	1. Invalid credentials entered → System displays an error message. 2. Network/system error → System prompts user to retry later.

Table 2:use case register/login

Use Case 2	
Field	**Description**
Name	Manage Profile
Actor	User
Description	Allows the user to manage and update personal profile information.
Pre-condition	User is logged in.
Post-condition	User profile information is updated successfully.
Priority	Medium
Basic Course of Action	1. User navigates to Profile section. 2. User edits personal details. 3. User saves the changes. 4. System updates the profile.
Alternate Course of Action	1. Invalid input → System requests correction. 2. System error → User is prompted to try again later.

Table 3:use case description for manage profile

Use Case 3	
Field	**Description**
Name	Enter Health Information
Actor	User
Description	Allows the user to enter health-related data such as age, allergies, chronic conditions, or pregnancy status.
Pre-condition	User is logged in and profile exists.
Post-condition	Health information is stored in the system.
Priority	High
Basic Course of Action	1. User selects Health Information option. 2. User enters health details. 3. System validates the information. 4. System saves the data.
Alternate Course of Action	1. Missing or invalid data → System requests correction.

Table 4: use case description for Enter Health Information

Use Case 4	
Field	**Description**
Name	Search Symptoms / Medicine
Actor	User
Description	Allows the user to search symptoms or medicines for recommendations.
Pre-condition	User is logged in.
Post-condition	Symptoms or medicine query is submitted for analysis.
Priority	High
Basic Course of Action	1. User enters symptoms or medicine name. 2. System receives the query. 3. System forwards the query for analysis.
Alternate Course of Action	1. Invalid input → System displays an error message

Table 5: use case description for search Symptoms/Medicine

Use Case 5	
Field	**Description**
Name	Analyze Symptoms
Actor	System
Description	System analyzes the symptoms using stored health information.
Pre-condition	Symptoms are submitted by the user.
Post-condition	Analysis results are generated.
Priority	High
Basic Course of Action	1. System processes symptoms. 2. System checks medical rules and constraints. 3. System generates recommendations.
Alternate Course of Action	1. Insufficient data → System requests more input.

Table 6: use case description for Analyze symptom

Use case 6	
Field	**Description**
Name	Check Severity
Actor	System
Description	The system evaluates the severity of user-entered symptoms using predefined medical rules and health profile data to determine whether the condition is mild or severe.
Pre-condition	User has entered symptoms and symptom analysis has been performed.
Post-condition	Symptom severity level (mild or severe) is determined and stored temporarily for decision-making.
Priority	High
Basic Course of Action	1. System receives analyzed symptom data. 2. System checks symptom severity rules and thresholds. 3. System classifies symptoms as mild or severe. 4. System forwards the result to the next decision process.
Alternate Course of Action	1. Insufficient symptom data → System requests additional input from the user. 2. Rule conflict detected → System defaults to safety recommendation (doctor visit).

Table 7: use case description for severity check

Use Case 7	
Field	**Description**
Name	View Medicine Recommendation
Actor	User
Description	Allows the user to view recommended medicines based on symptom analysis.
Pre-condition	Symptoms are analyzed successfully.
Post-condition	Medicine recommendations are displayed.
Priority	High
Basic Course of Action	1. System displays recommended medicines.2. User reviews dosage and warnings.
Alternate Course of Action	1. No suitable medicine found → System advises consulting a doctor.

Table 8: use case description for View Medicine Recommendation

Use case 8	
Field	**Description**
Name	Recommend Doctor Visit
Actor	System
Description	The system advises the user to consult a medical professional when symptoms are detected as severe, unclear, or potentially risky.
Pre-condition	Symptom severity has been checked and classified as severe or uncertain.
Post-condition	Doctor visit recommendation is displayed to the user.
Priority	High
Basic Course of Action	1. System receives severity result. 2. System determines that symptoms require professional medical attention. 3. System displays a message recommending doctor consultation. 4. System provides an option to proceed with doctor consultation.
Alternate Course of Action	1. User ignores recommendation → System records the action and continues normal operation. 2. Network/system error → System retries displaying recommendation.

Table 9:use case description for recommend doctor visit

Use case 9	
Field	**Description**
Name	Consult Doctor
Actor	User, Doctor
Description	Allows the user to consult a doctor for professional medical advice and receive a prescription when required.
Pre-condition	System has recommended a doctor visit or user chooses to consult a doctor manually.
Post-condition	Consultation is completed and prescription may be added to the system.
Priority	Medium
Basic Course of Action	1. User selects the option to consult a doctor. 2. System connects the user with a doctor. 3. Doctor reviews user symptoms and health information. 4. Doctor provides medical advice. 5. Doctor adds a prescription to the system.
Alternate Course of Action	1. Doctor unavailable → System notifies the user and suggests retrying later. 2. User cancels consultation → No changes are made.

Table 10: use case description for consult doctor

Use case 10	
Field	**Description**
Name	Add Prescription
Actor	User
Description	Allows the user to add a selected medicine to their prescription list.
Pre-condition	Medicine recommendation is available.
Post-condition	Prescription is added to the user account.
Priority	Medium
Basic Course of Action	1. User selects a medicine.2. User confirms adding it to prescription.3. System saves the prescription.
Alternate Course of Action	1. User cancels action → No changes are made.

Table 11: use case description for add prescription

Use case 11	
Field	**Description**
Name	Set Medicine Reminder
Actor	User
Description	Allows the user to schedule reminders for medicine intake.
Pre-condition	Prescription exists.
Post-condition	Reminder schedule is created.
Priority	High
Basic Course of Action	1. User sets reminder time and frequency.2. System saves the reminder.
Alternate Course of Action	1. Invalid time → System prompts correction.

Table 12: use case description for Set Medicine Reminder

Use case 12	
Field	**Description**
Name	Trigger Reminder Notification
Actor	System Scheduler
Description	Automatically triggers reminder notifications at scheduled times.
Pre-condition	Reminder schedule exists.
Post-condition	Notification is sent to the user.
Priority	High
Basic Course of Action	1. Scheduler checks reminder time.2. System sends notification to user.
Alternate Course of Action	1. Network failure → System retries delivery.

Table 13: use case description for Trigger Reminder Notification

Use case 13	
Field	**Description**
Name	Receive Reminder Notification
Actor	User
Description	User receives medicine reminder notification.
Pre-condition	Reminder is triggered by the system.
Post-condition	User is notified.
Priority	High
Basic Course of Action	1. User receives notification alert.
Alternate Course of Action	1. Notifications disabled → User misses alert.

Table 14: use case description for Receive Reminder Notification

Use case 14	
Field	**Description**
Name	Confirm Medicine Intake
Actor	User
Description	Allows the user to confirm that medicine has been taken.
Pre-condition	Reminder notification received.
Post-condition	Intake status is recorded.
Priority	Medium
Basic Course of Action	1. User confirms medicine intake.2. System updates intake record.
Alternate Course of Action	1. User ignores reminder → Intake marked as missed.

Table 15: use case description for Confirm Medicine Intake

Use case 15	
Field	**Description**
Name	View Medicine History
Actor	User
Description	Allows the user to view past medicine intake and reminders.
Pre-condition	User is logged in.
Post-condition	Medicine history is displayed.
Priority	Medium
Basic Course of Action	1. User opens medicine history.2. System displays intake records.
Alternate Course of Action	1. No history available → System displays empty state message.

Table 16: use case description for View Medicine History

2.3.3 Business Rule Documentation

Business Rule 1	
Field	**Description**
Identifier	BR-PR-001
Name	User Profile Creation
Description	Users must create a personal profile to access the features of the Medicine Recommender and Reminder App. The profile collects essential information such as name, age, contact details, medical conditions, and allergies. This information is used to provide personalized medicine recommendations and reminders. **Mandatory Fields:** • Name • Contact information (email or phone number) • Age and gender • Medical conditions and allergies **Features:** • Profile can be edited after creation. • Secure storage of personal and medical data.
Reference	Supports FREQ-1: User Profile Management, as defined in Section 2.3.1 of the project documentation.

Table 17: business rule for profile creation

Business Rule 2	
Field	**Description**
Identifier	**BR-HI-001**
Name	**Health Information Management**
Description	**Users must enter and maintain accurate health information, including chronic illnesses and allergies. This information is used to prevent unsafe medicine recommendations and improve personalization.**
Reference	**Based on FREQ-2: Personalized Health Information Management, described in Section 2.3.1 of the project documentation.**

Table 18: business rule for health information management

Business Rule 3	
Field	**Description**
Identifier	BR-MR-001
Name	Medicine Recommendation
Description	The system must generate medicine recommendations based on user-entered symptoms, health information, and stored medical rules. Recommendations must consider dosage safety, allergies, and known conditions.
Reference	Supports FREQ-3: Intelligent Medicine Recommendation, outlined in Section 2.3.1 of the project documentation.

Table 19: business rule for medicine recommendation

Business Rule 4	
Field	**Description**
Identifier	BR-SM-001
Name	Symptom Analysis
Description	The system must analyze symptoms entered by the user before generating medicine recommendations. If symptoms are severe or unclear, the system must advise the user to consult a healthcare professional.
Reference	Aligned with FREQ-3: Symptom-Based Recommendation, as described in Section 2.3.1.

Table 20: business rule for symptom analysis

Business Rule 5	
Field	**Description**
Identifier	BR-PRSC-001
Name	Prescription Management
Description	Users must be able to add, view, and manage prescriptions manually or based on recommendations. Prescription details include medicine name, dosage, frequency, and duration.
Reference	Supports FREQ-4: Prescription Management, as defined in Section 2.3.1.

Table 21: business rule for prescription management

Business Rule 6	
Field	**Description**
Identifier	BR-RM-001
Name	Medicine Reminder Scheduling
Description	Users must set reminders for medicine intake based on prescription details. Reminders must support multiple schedules per day and recurring notifications.
Reference	Supports FREQ-5: Medicine Reminder Scheduling, as outlined in Section 2.3.1.

Table 22: business rule for medicine reminder scheduling

Business Rule 7	
Field	**Description**
Identifier	BR-SCH-001
Name	Automated Reminder Triggering
Description	The system scheduler must automatically trigger reminder notifications at the scheduled times without user intervention.
Reference	Aligned with FREQ-6: Automated Notification System, described in Section 2.3.1.

Table 23: business rule for Automated Reminder Triggering

Business Rule 8	
Field	**Description**
Identifier	BR-NOT-001
Name	Reminder Notification Delivery
Description	Reminder notifications must be delivered through supported channels (in-app alerts, SMS, or push notifications). Failed notifications must be retried automatically.
Reference	Supports FREQ-6: Notification Delivery, as outlined in Section 2.3.1.

Table 24: business rule for Reminder Notification Delivery

Business Rule 9	
Field	**Description**
Identifier	BR-CMI-001
Name	Confirmation of Medicine Intake
Description	Users must confirm medicine intake after receiving a reminder. The system records intake status as taken, skipped, or missed.
Reference	Supports FREQ-7: Medicine Intake Tracking, described in Section 2.3.1.

Table 25: business rule for Confirmation of Medicine Intake

Business Rule 10	
Field	**Description**
Identifier	BR-HST-001
Name	Medicine History Tracking
Description	The system must store and display a history of medicine intake, reminders, and missed doses for user review and adherence tracking.
Reference	Aligned with FREQ-8: Medicine History Management, outlined in Section 2.3.1.

Table 26: business rule for Medicine History Tracking

Business Rule 11	
Field	**Description**
Identifier	BR-LG-001
Name	Secure Login
Description	Users must log in using a unique email or phone number with a password or OTP. The system verifies credentials securely before granting access. Accounts are temporarily locked after multiple failed attempts.
Reference	Supports FREQ-9: Secure Authentication, as described in Section 2.3.1.

Table 27: business rule for Secure Login

2.3.4 User Interface Prototype

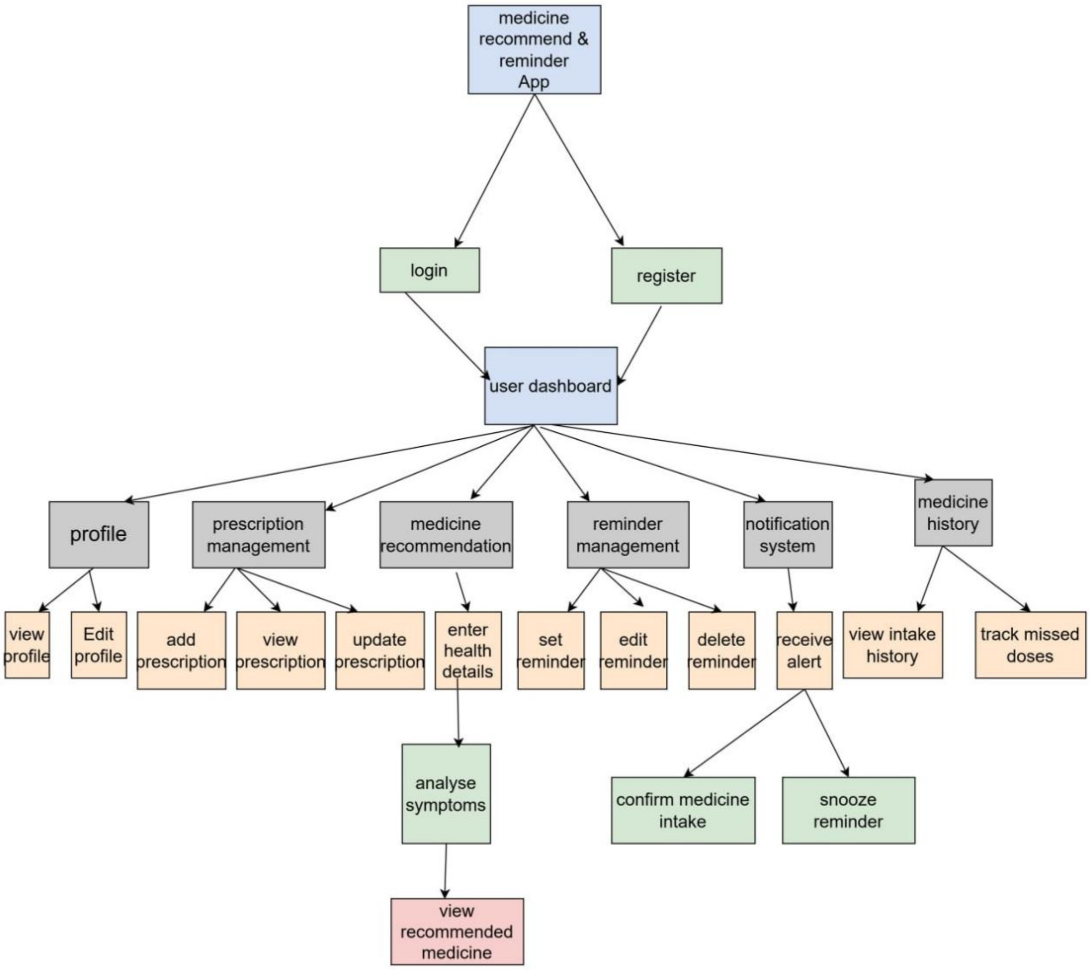

Figure 2: UI prototype

2.3.5 State chart diagram

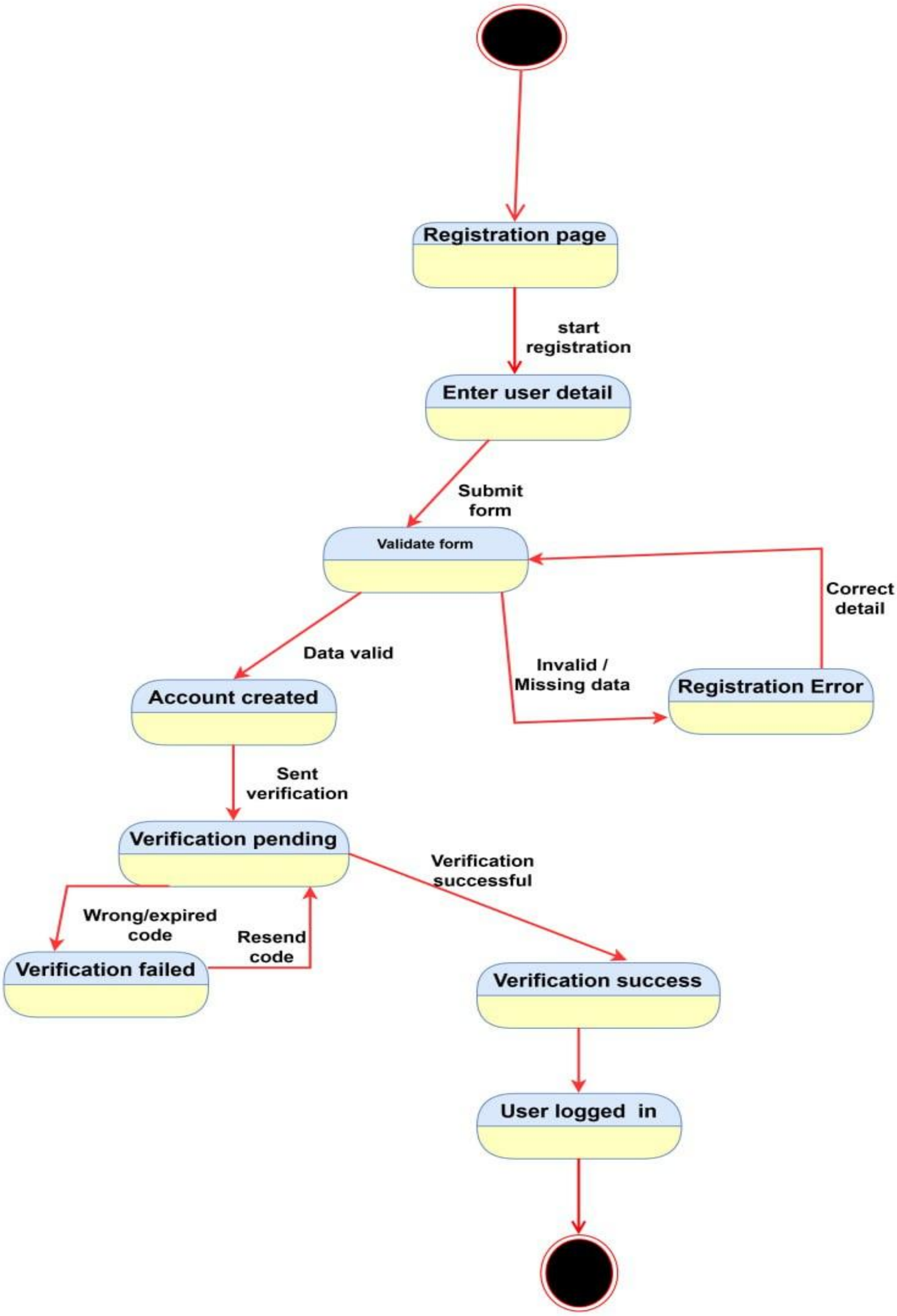

Figure 3: state chart diagram for user registration

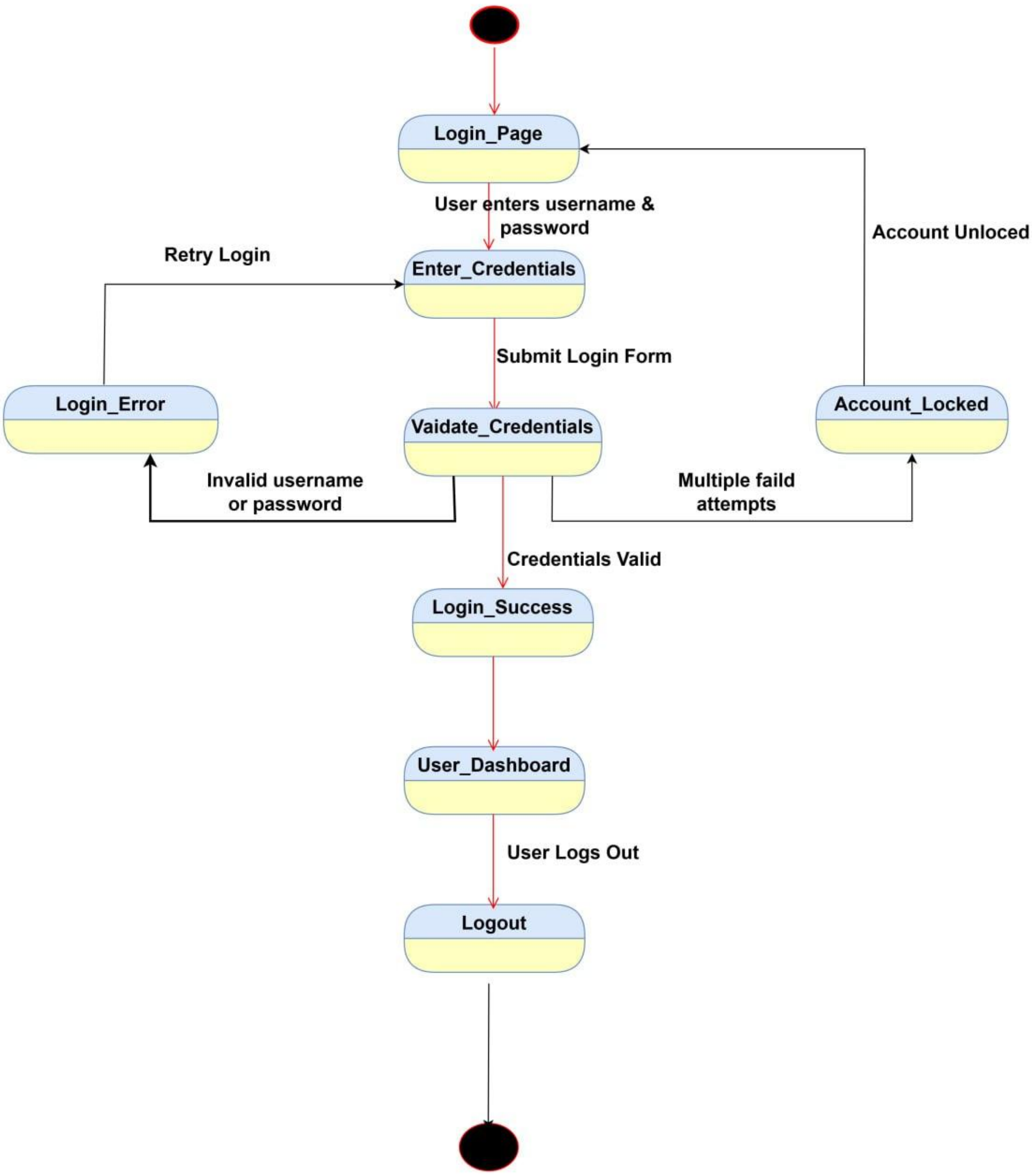

Figure 4: state chart diagram for user login

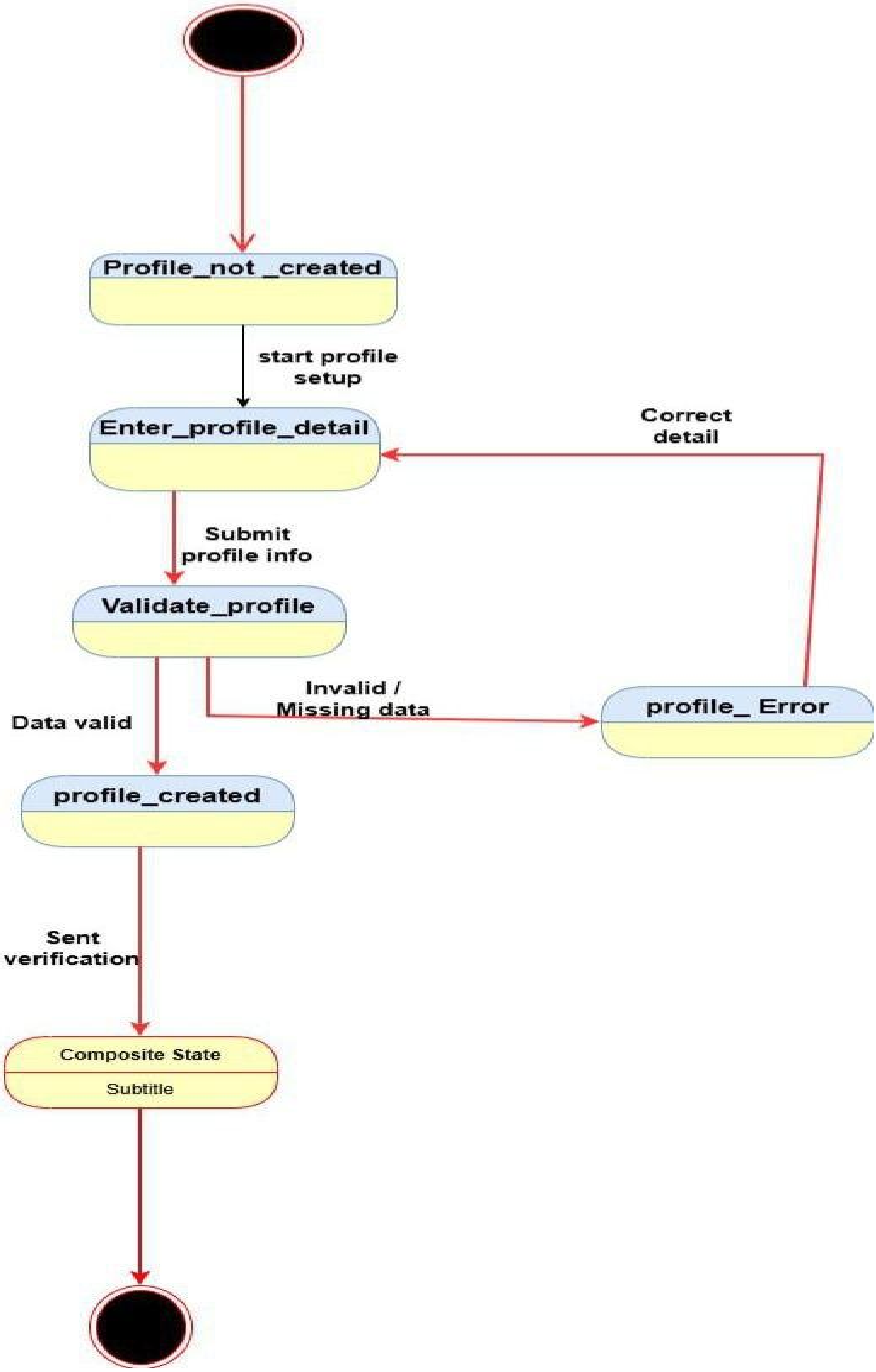

Figure 5: state chart diagram for profile creation

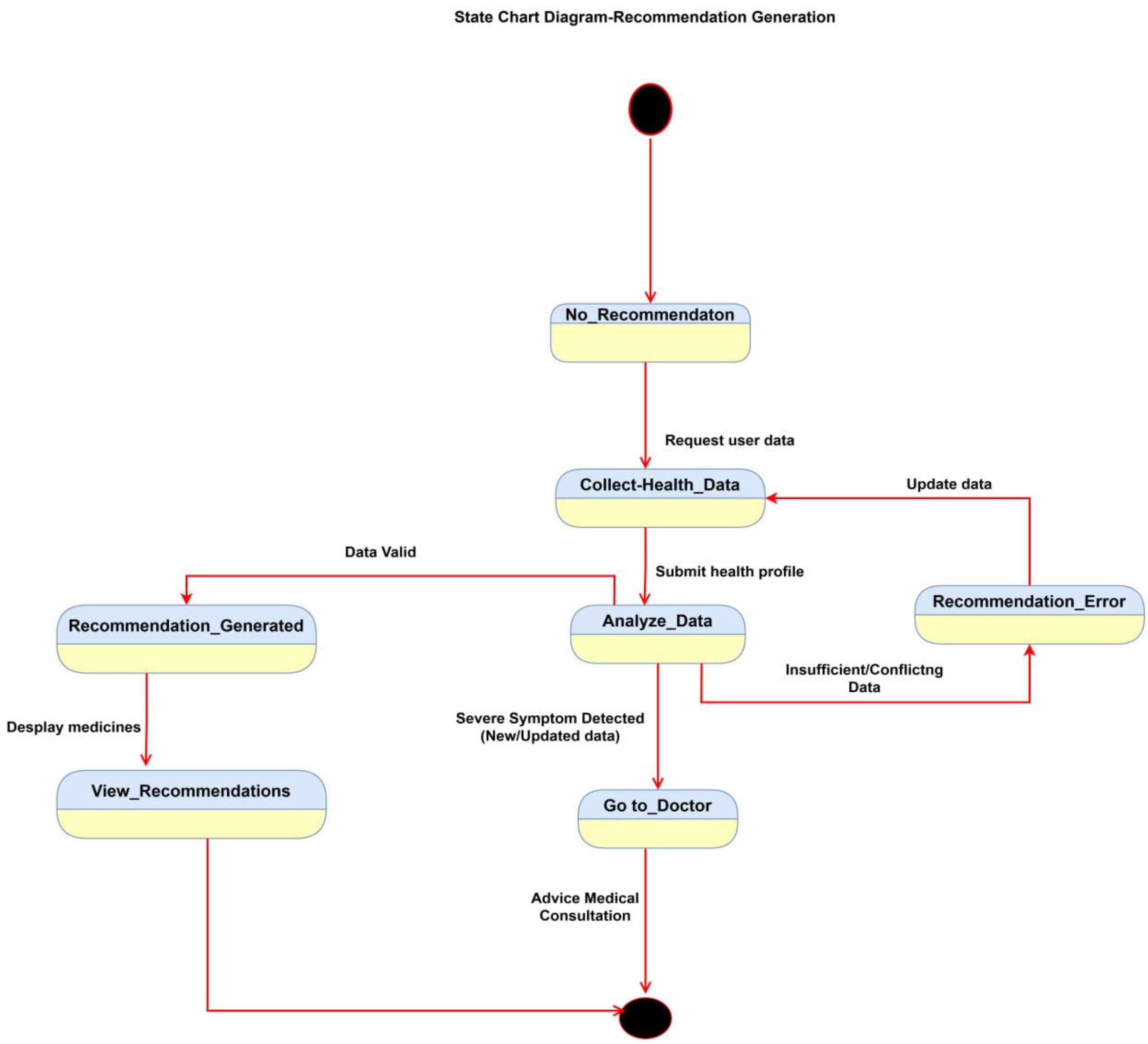

Figure 6: state chart diagram for Recommendation

state chart diagram :-Reminder setting

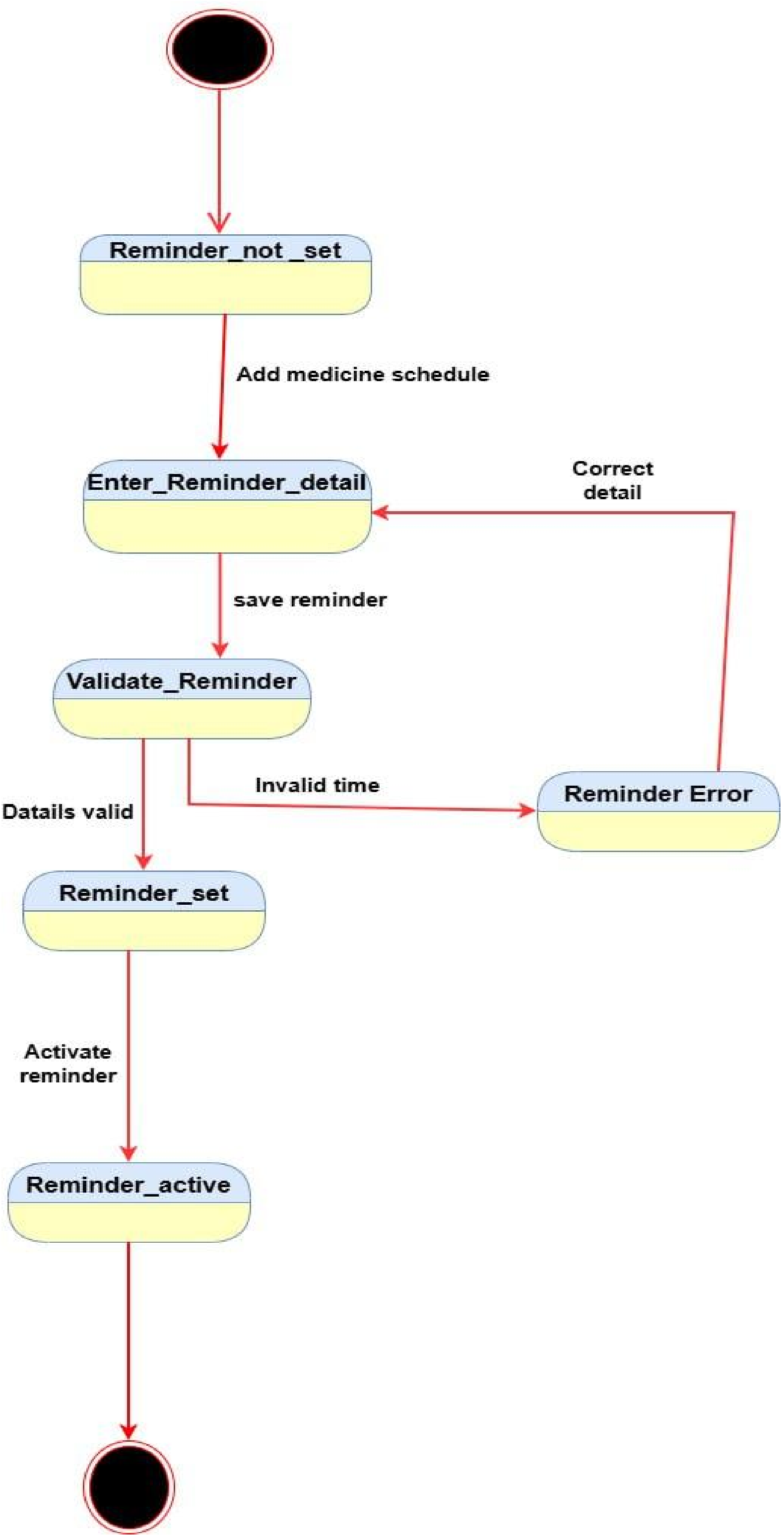

Figure 7: state chart diagram for reminder setting

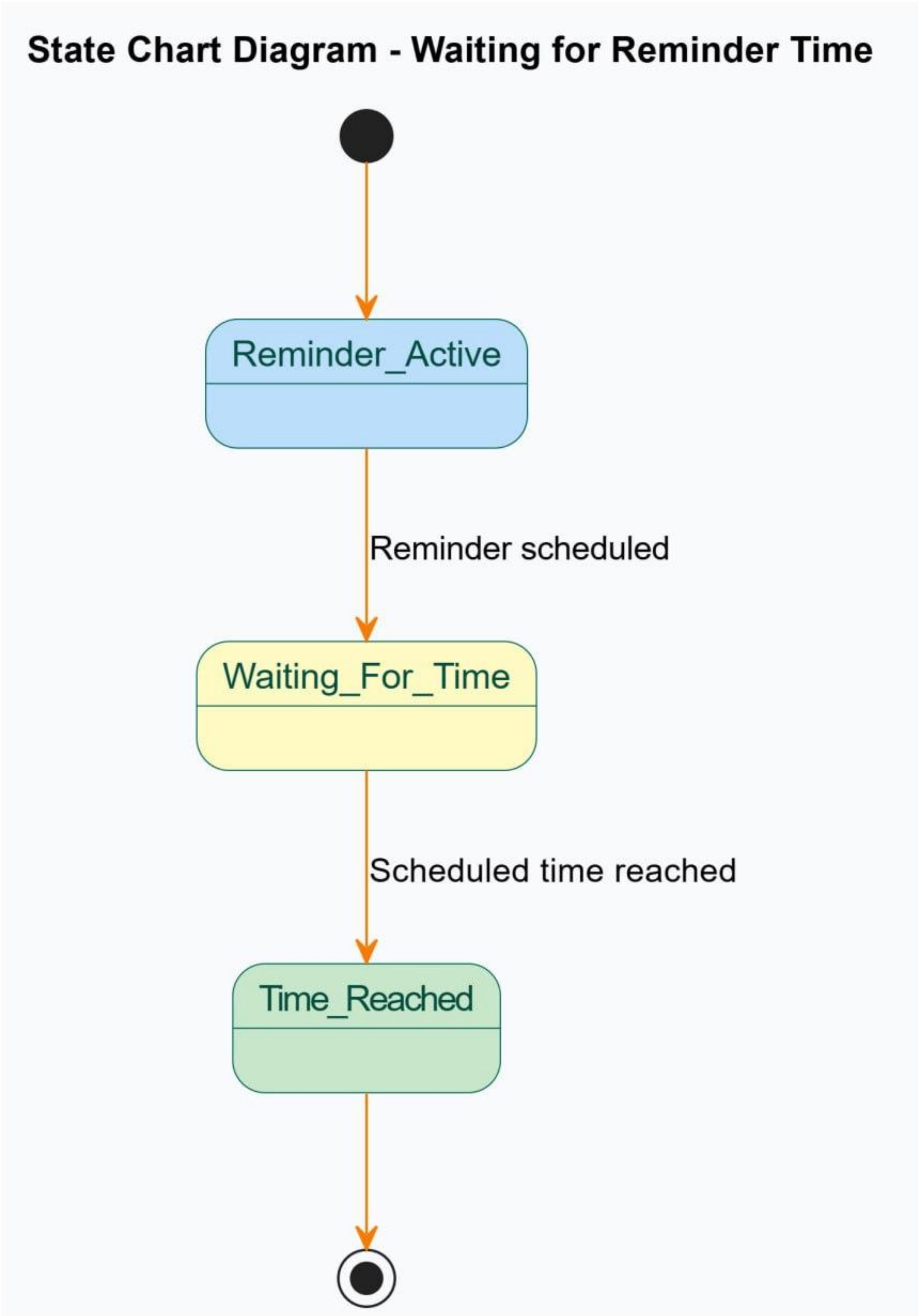

Figure 8: state chart diagram for waiting for reminder time

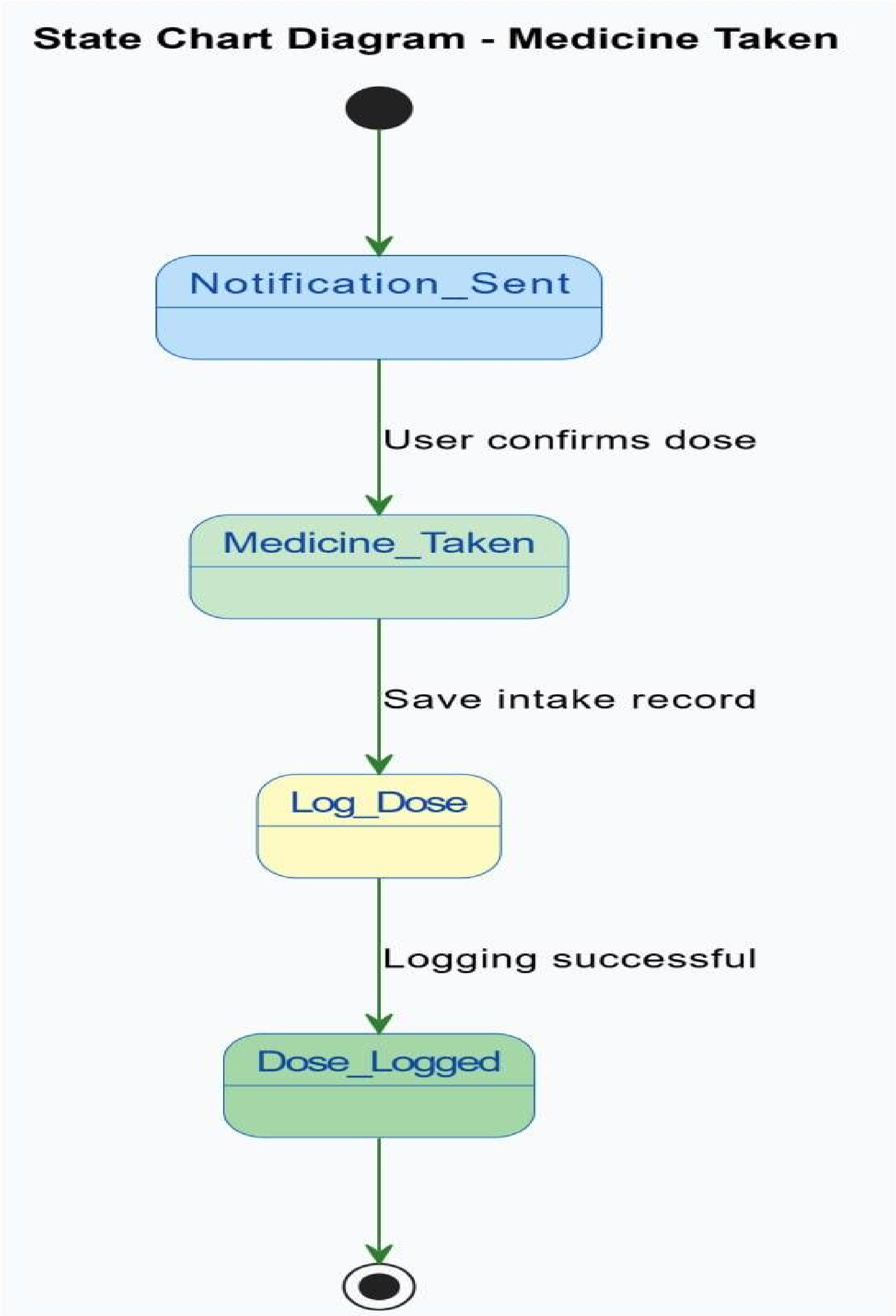

Figure 9: state chart diagram for medicine taken

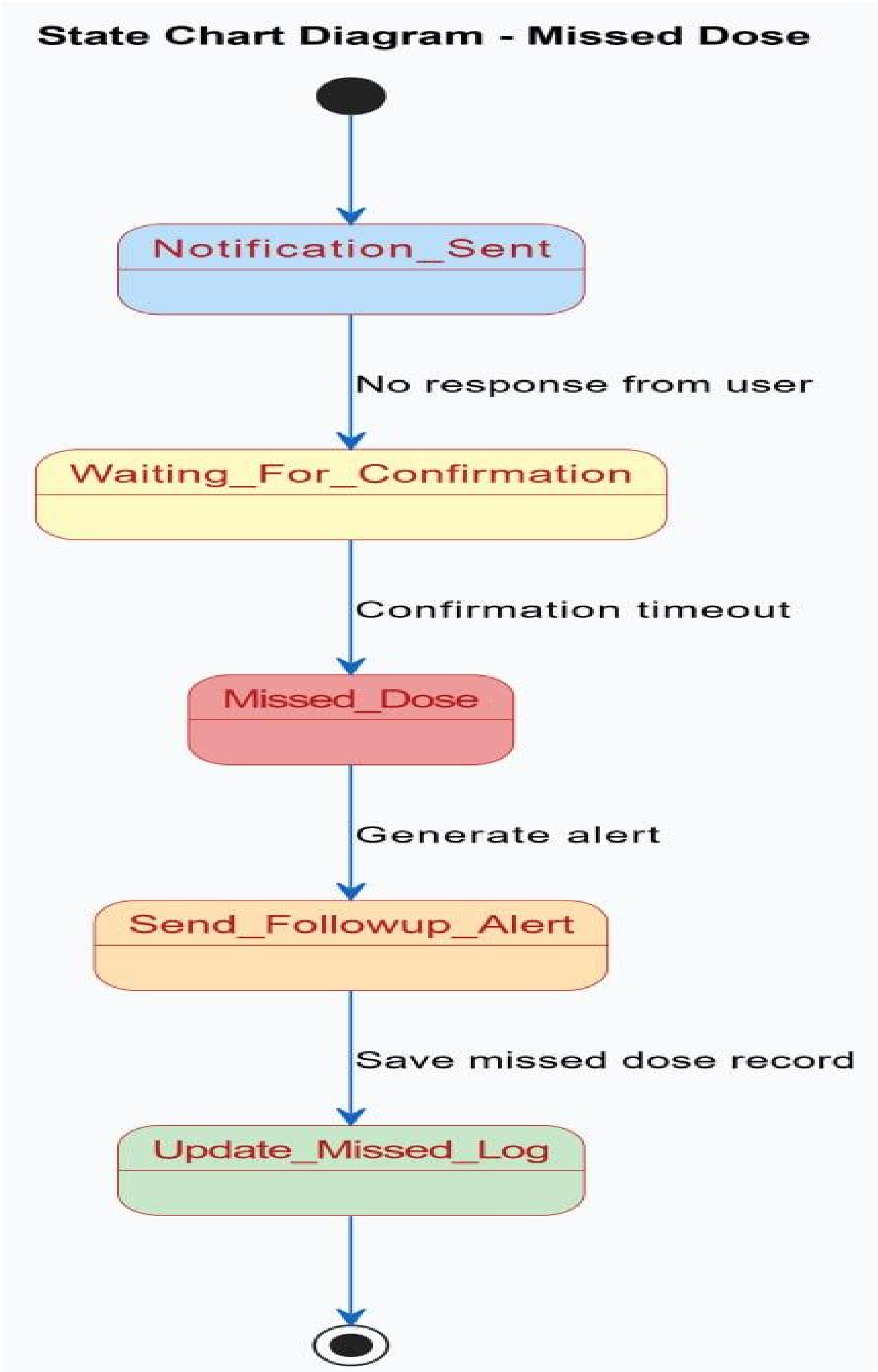

Figure 10:state chart diagram for medicine dose

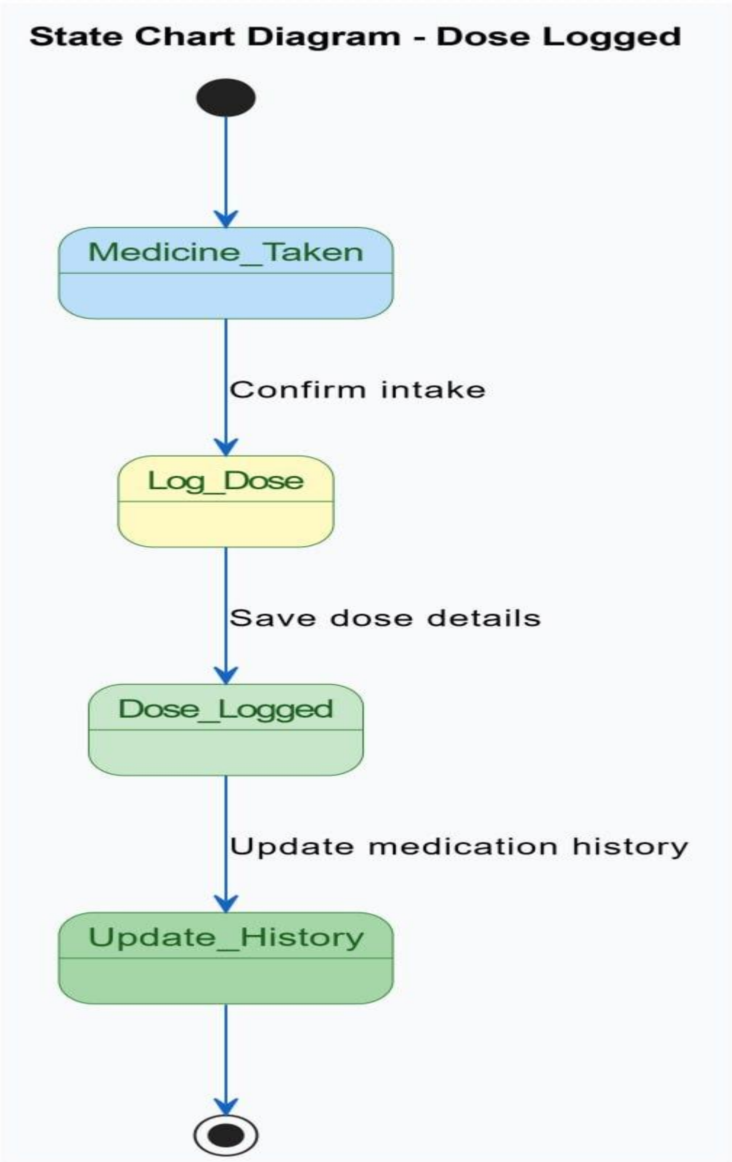

Figure 11: state chart diagram for dose logged

2.3.6 Activity Diagram

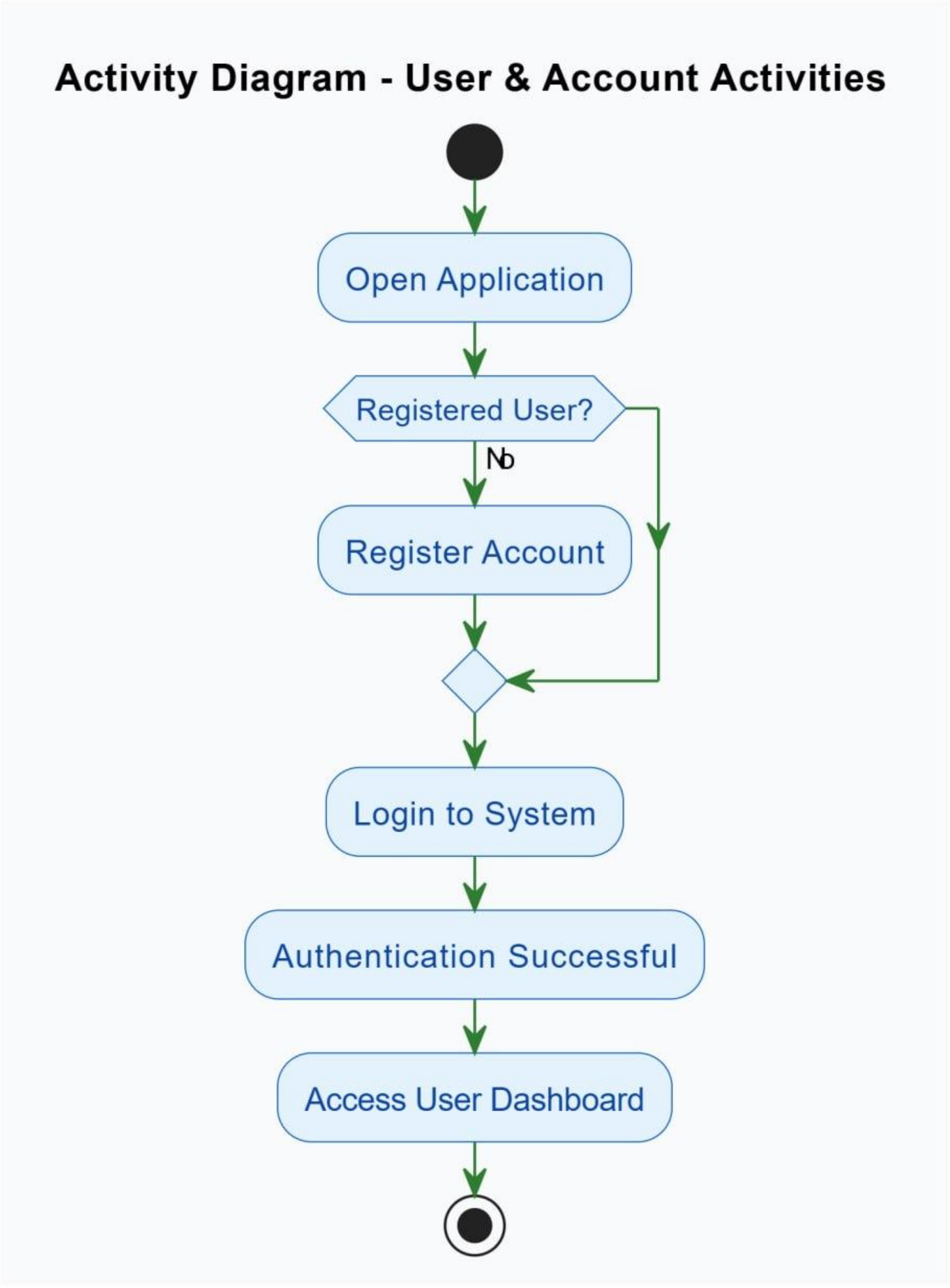

Figure 12: activity diagram for user & account activities

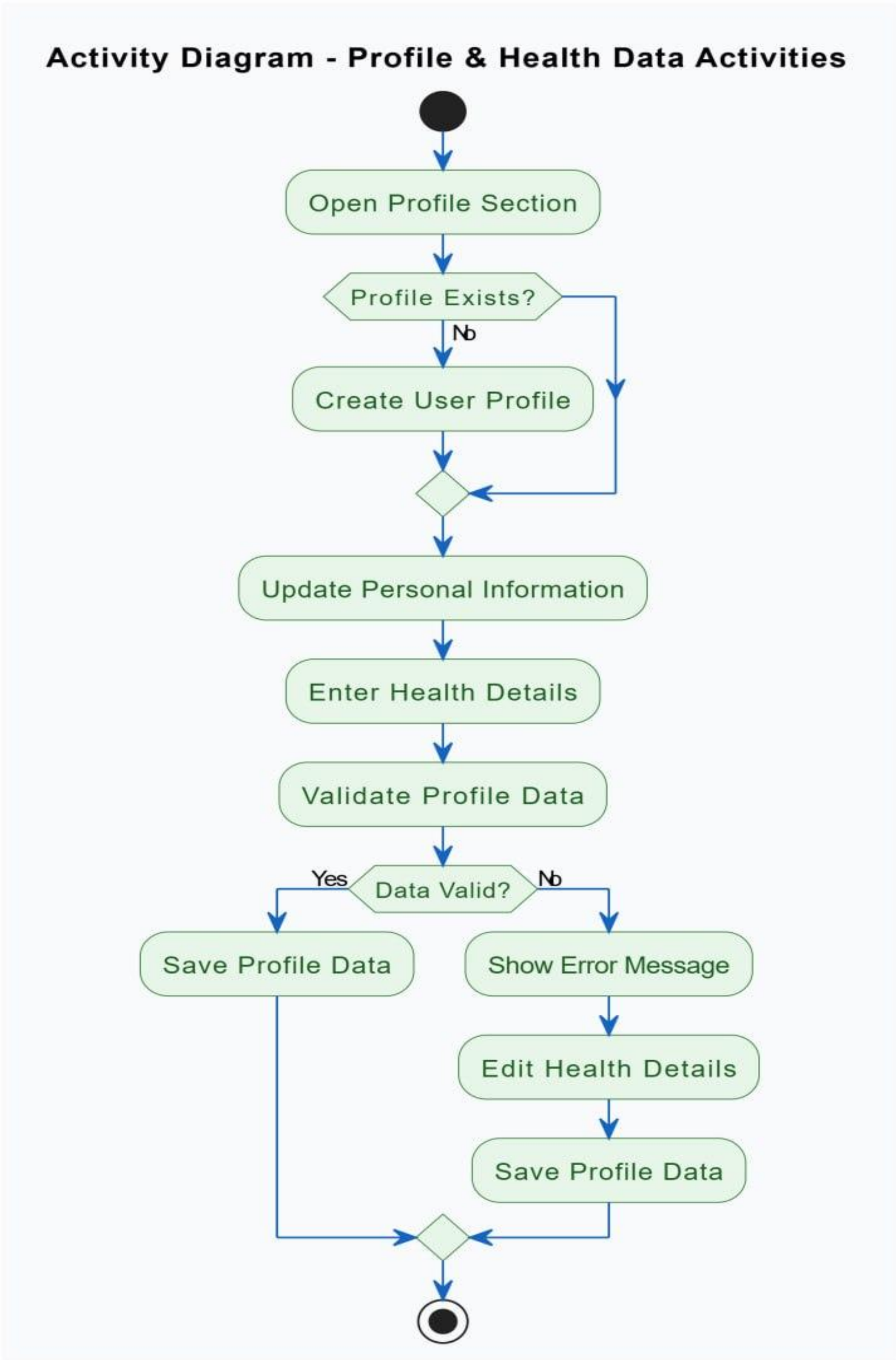

Figure 13: Activity diagram for profile and health data activity

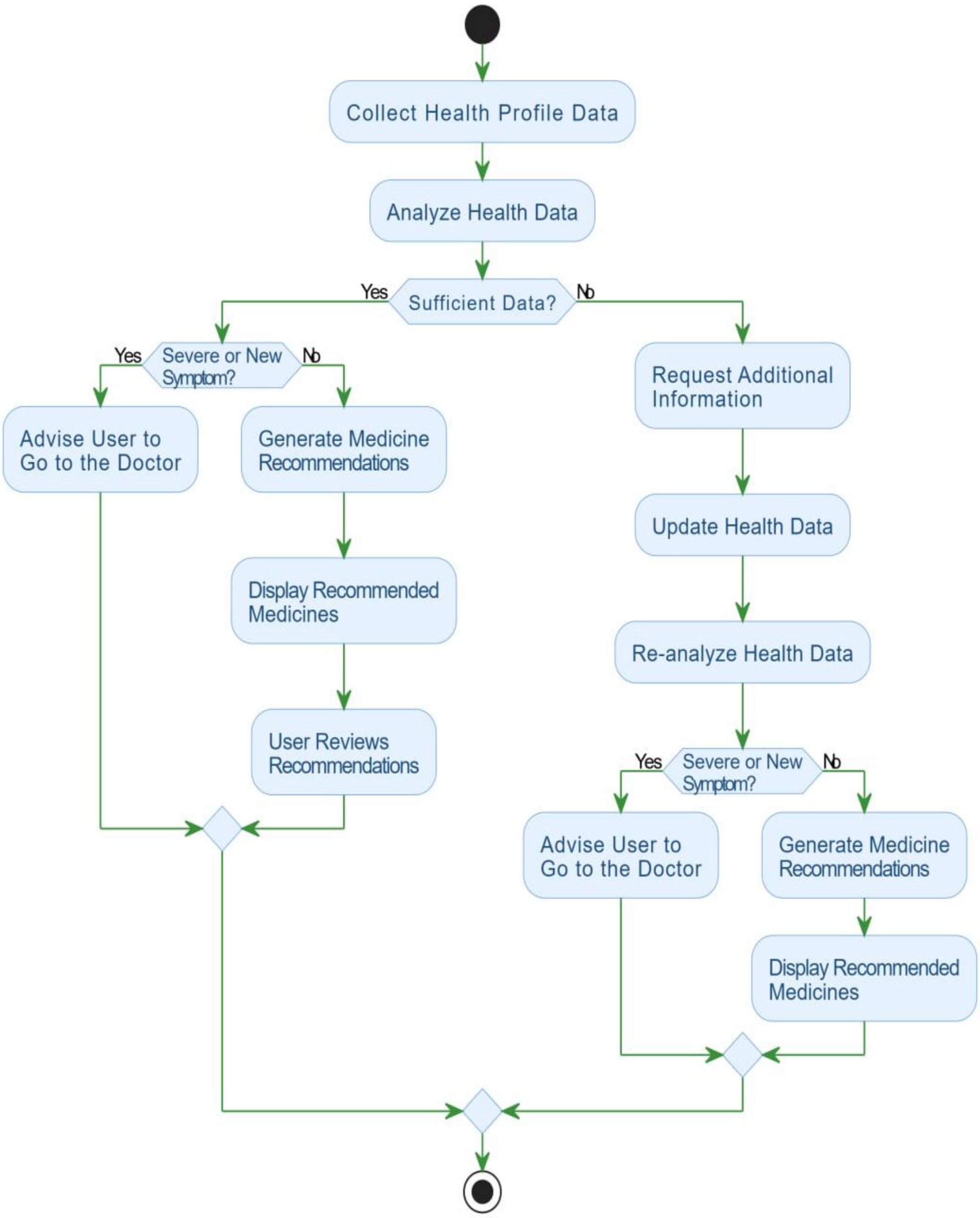

Figure 14: activity diagram for recommendation activities

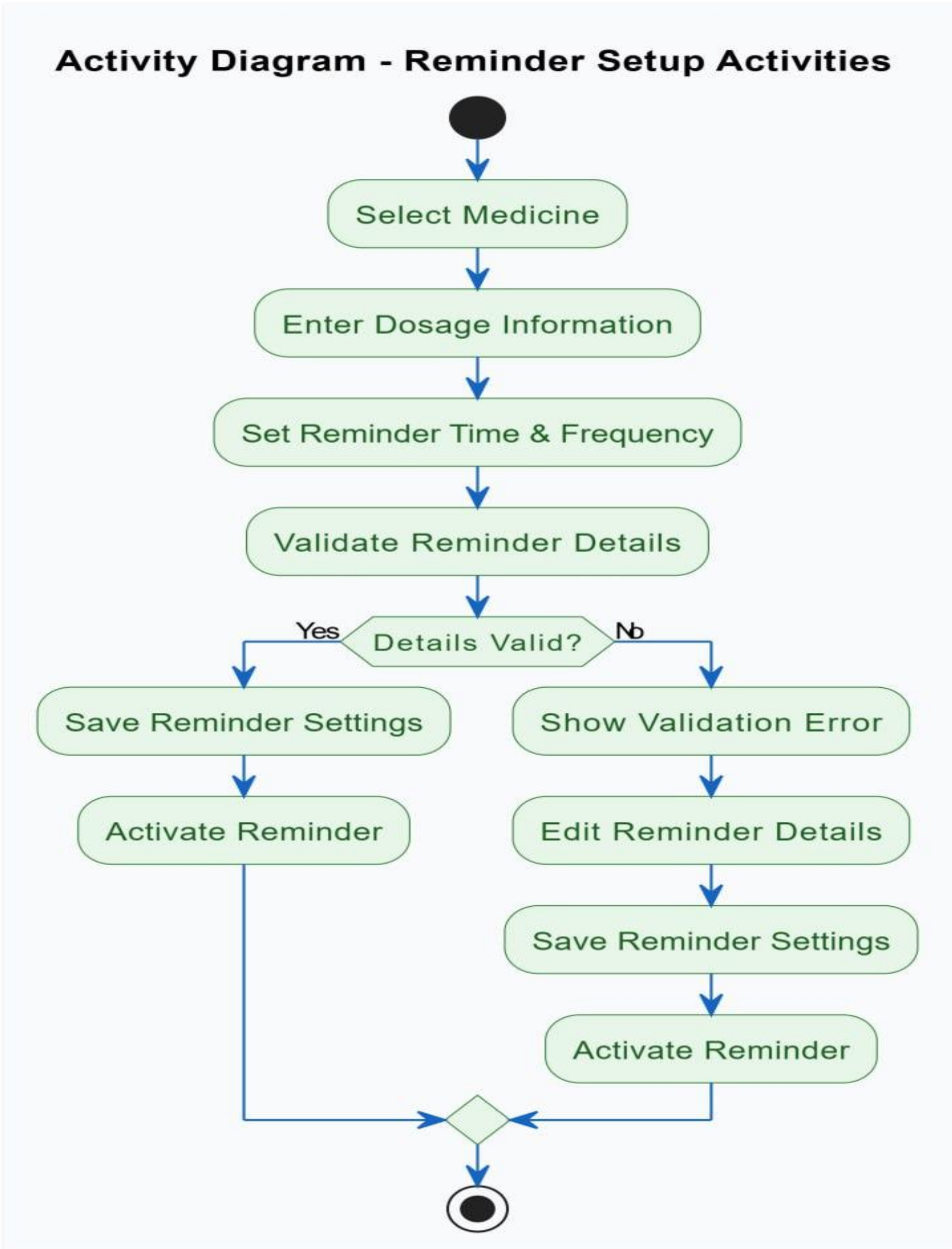

Figure 15: activity diagram for reminder setup activities

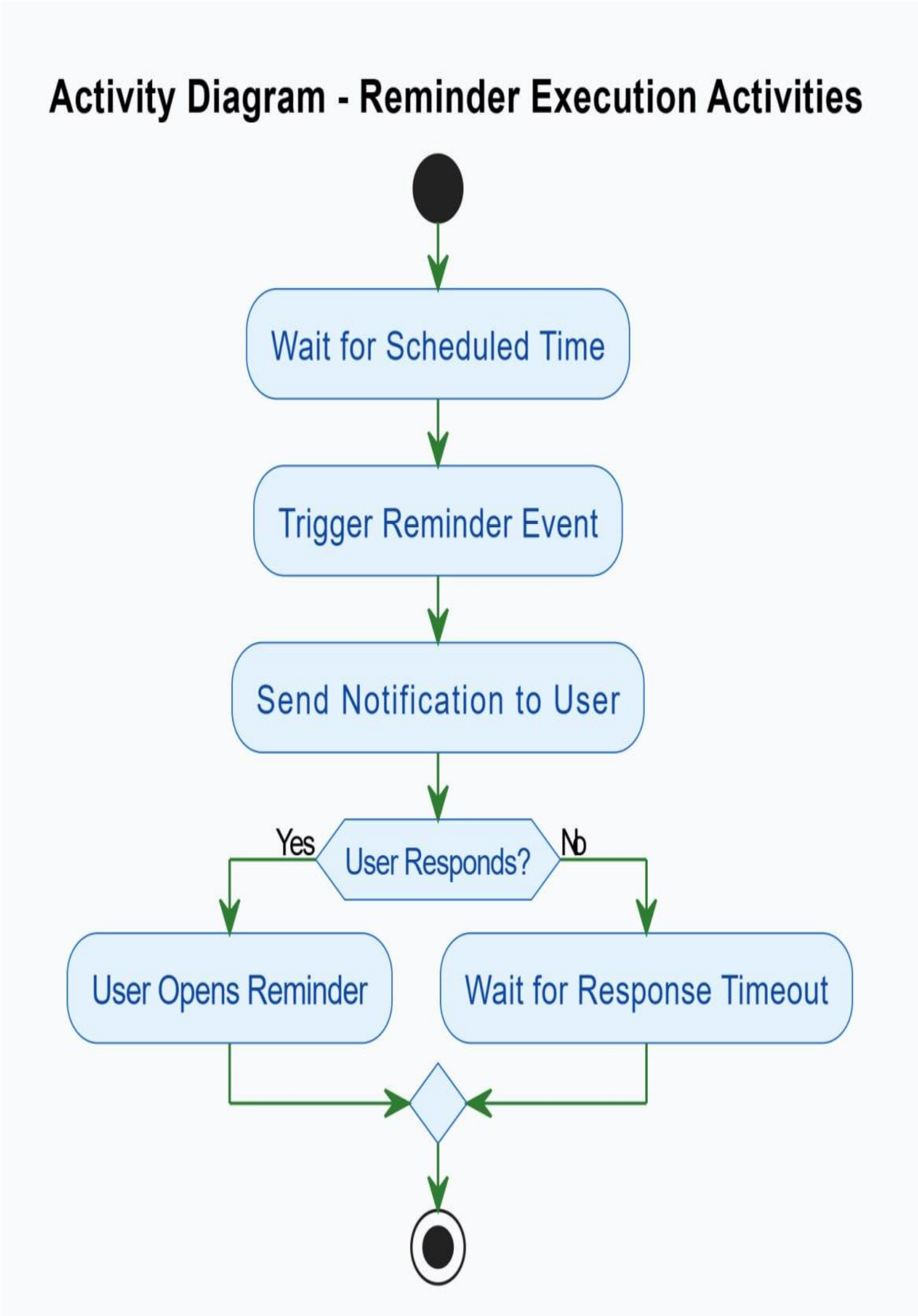

Figure 16: activity diagram for reminder execution activities

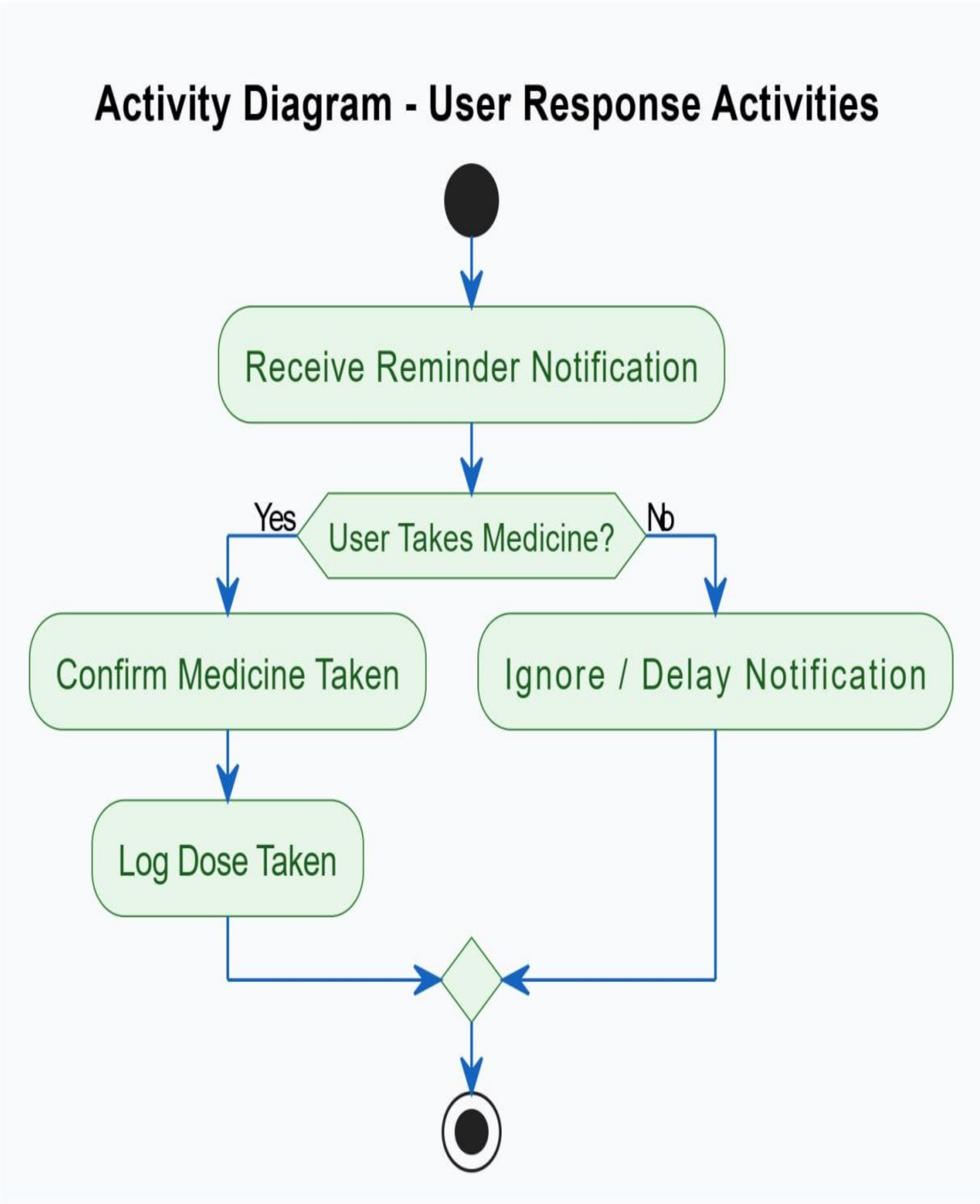

Figure 17:activity diagram for user response activities

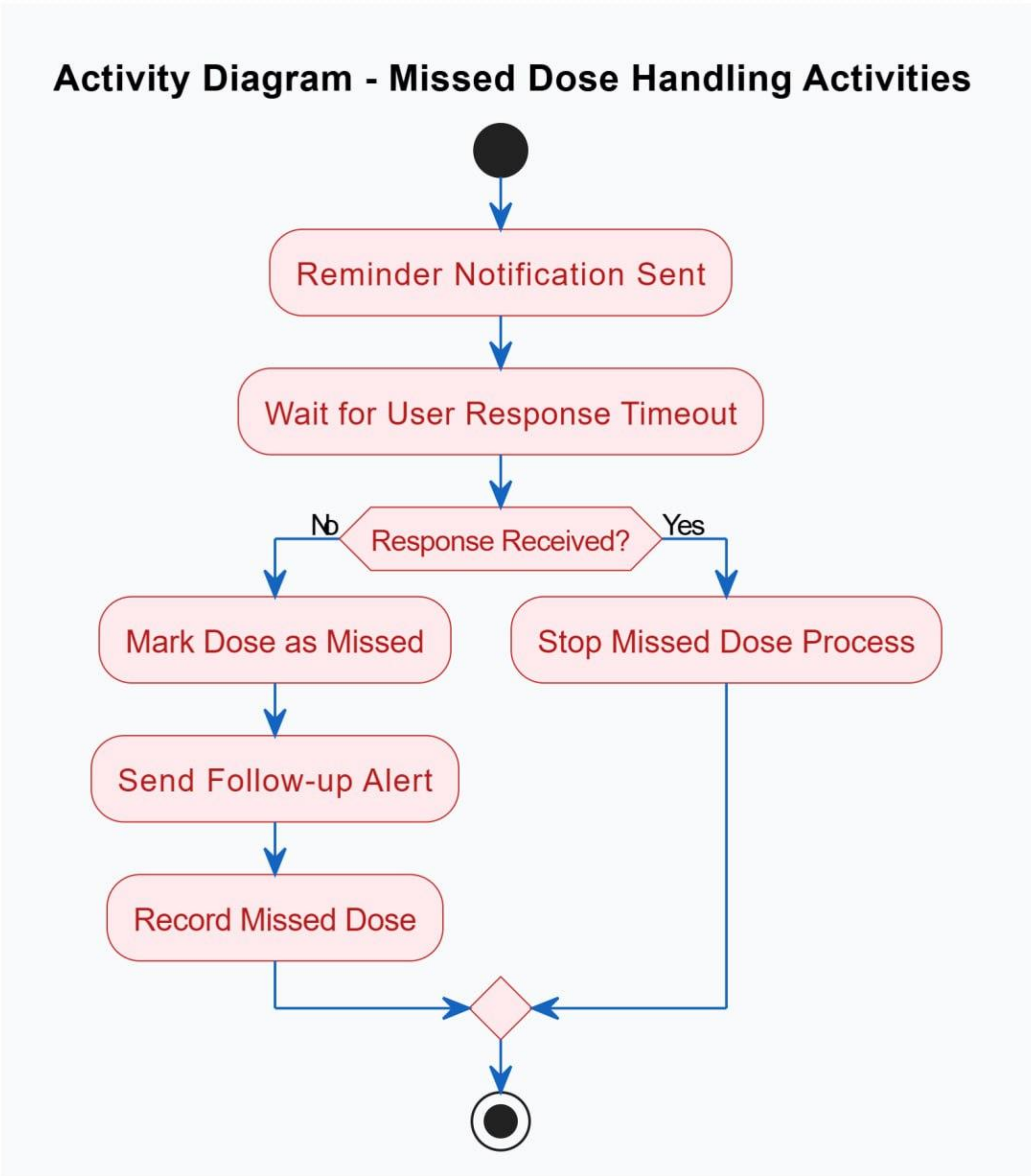

Figure 18: activity diagram for missed dose handling activities

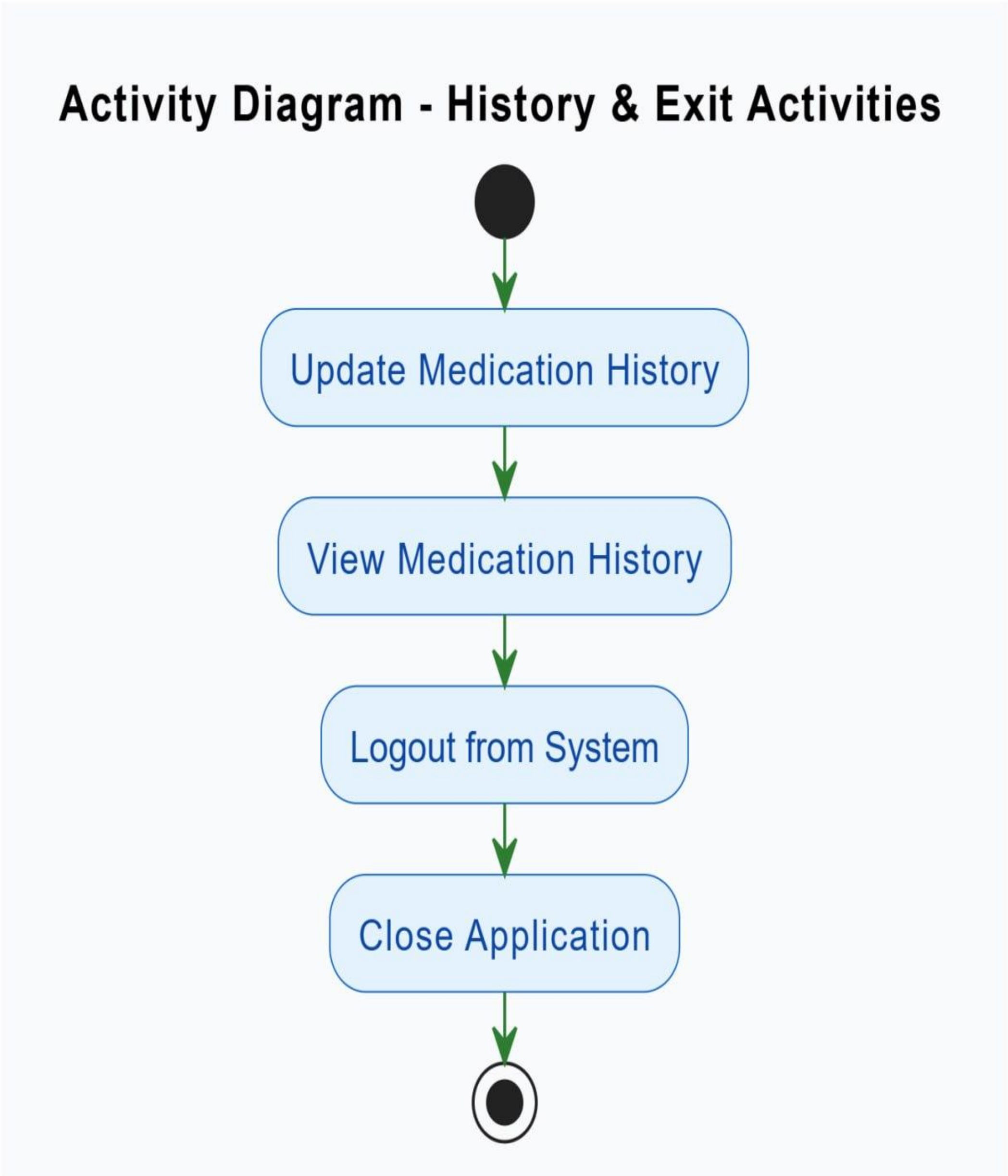

Figure 19: activity diagram for history & exit activities

2.3.7 Sequence diagram

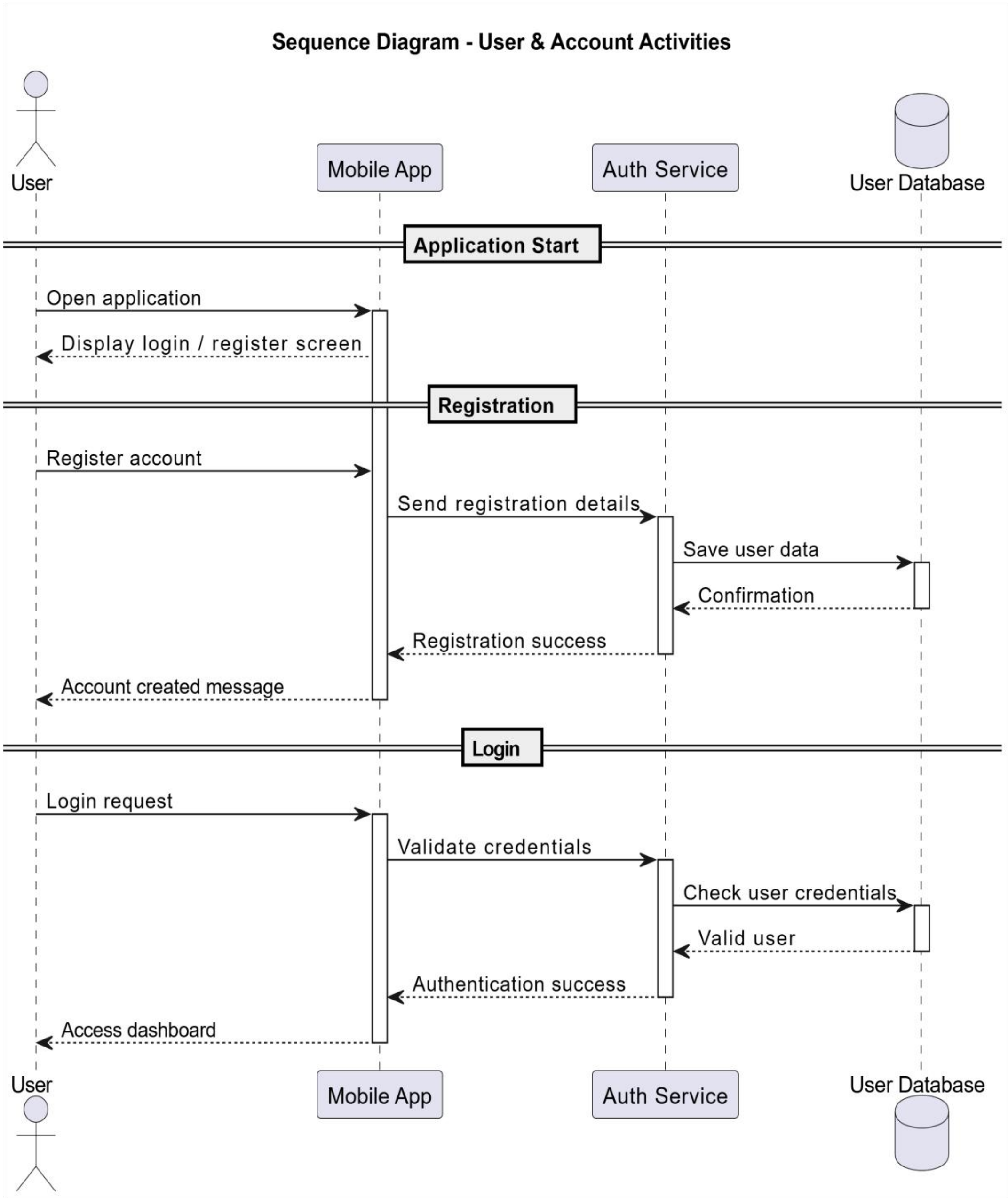

Figure 20: sequence diagram for user & account activities

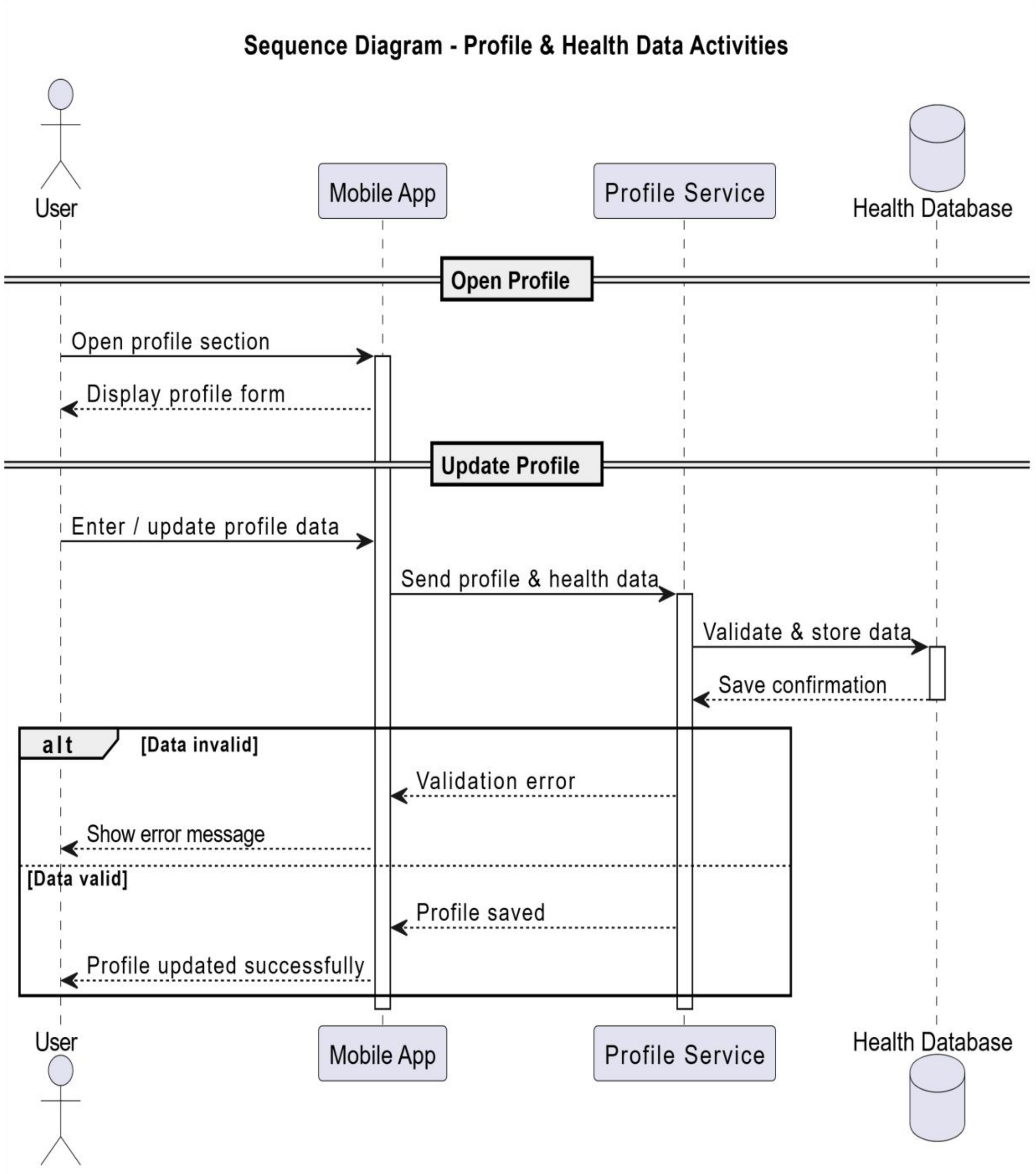

Figure 21: sequence diagram for profile & health data acivities

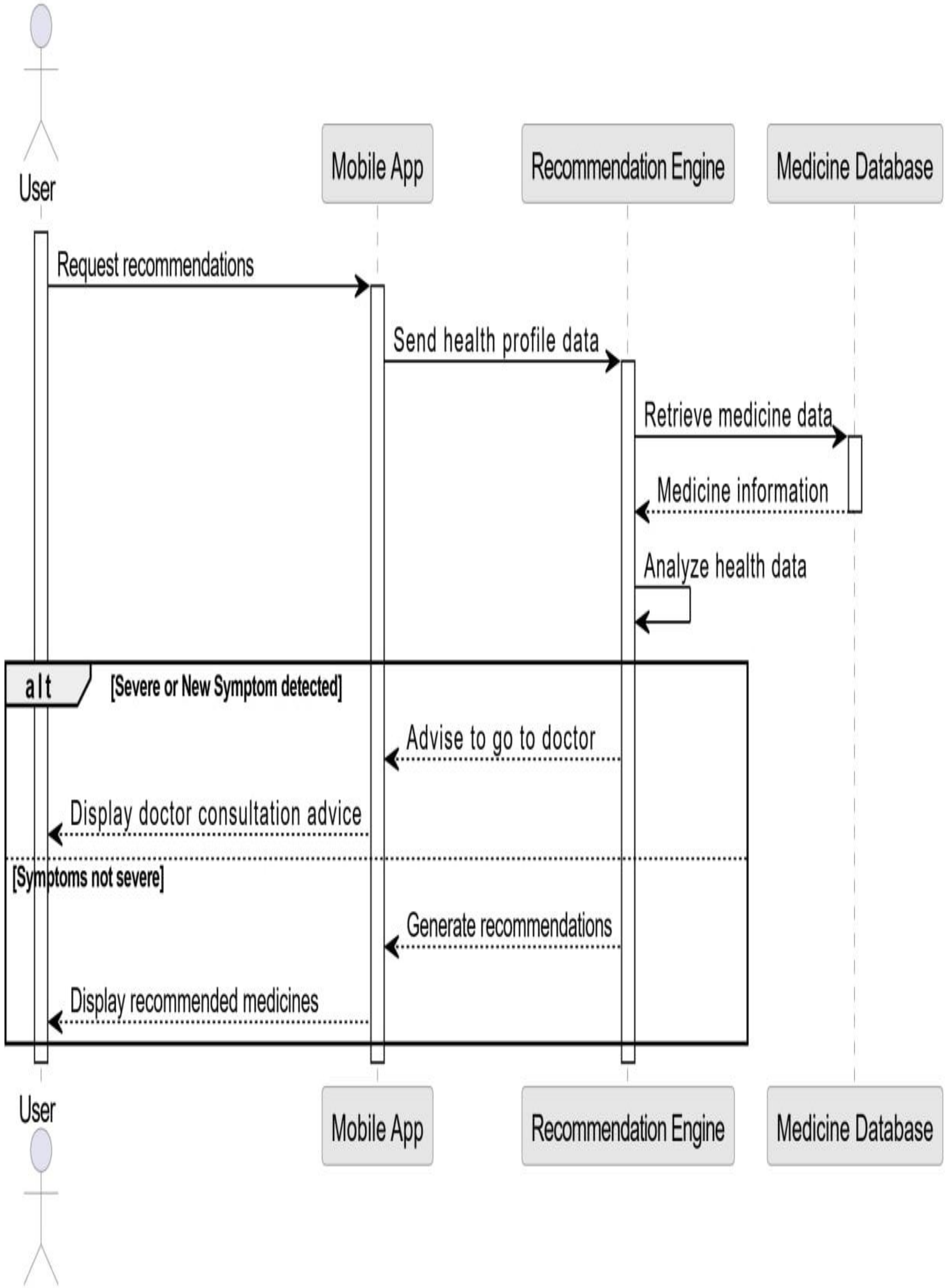

Figure 22: sequence diagram for recommendation activities

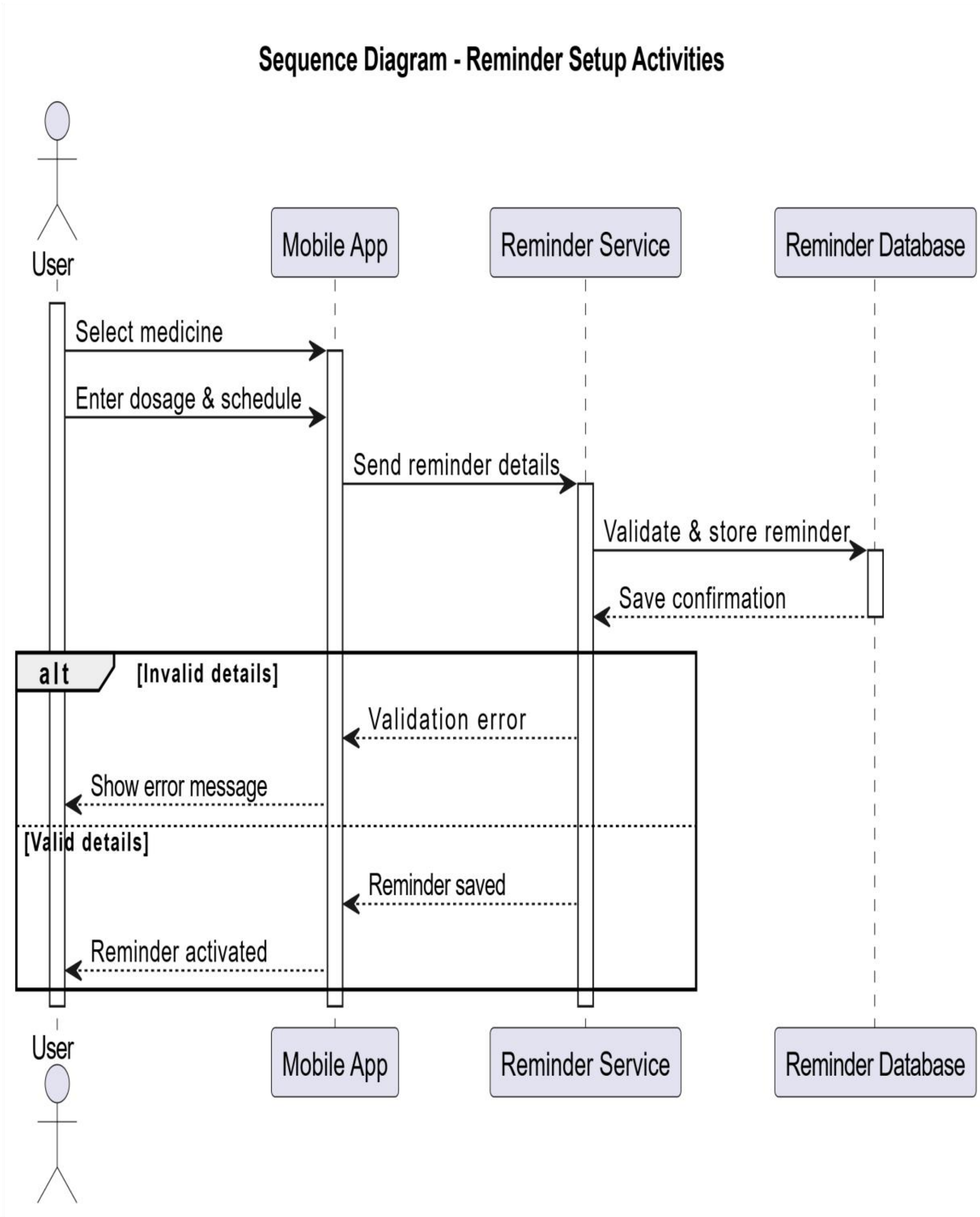

Figure 23: sequence diagram for reminder setup activities

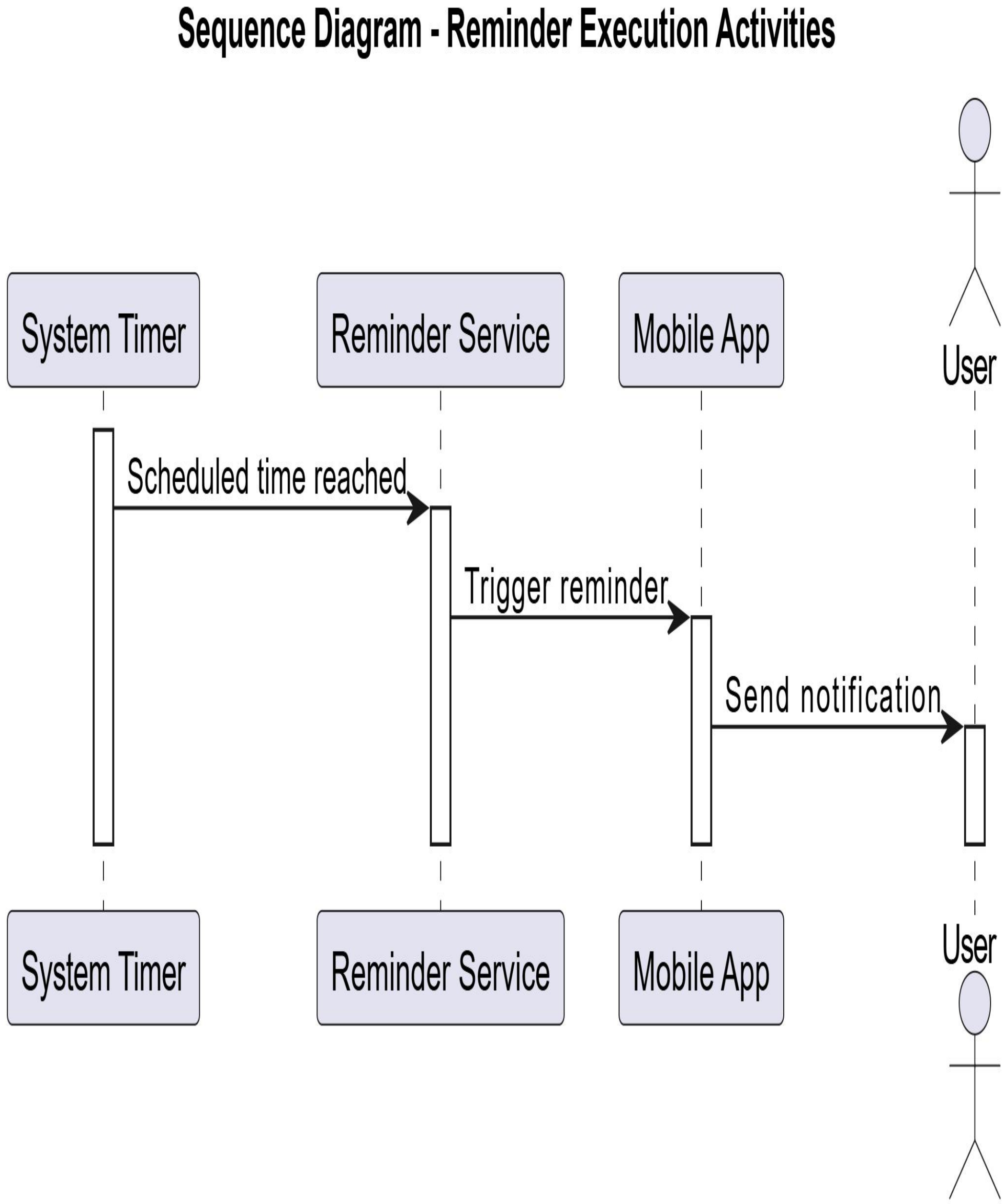

Figure 24: sequence diagram for reminder execution activities

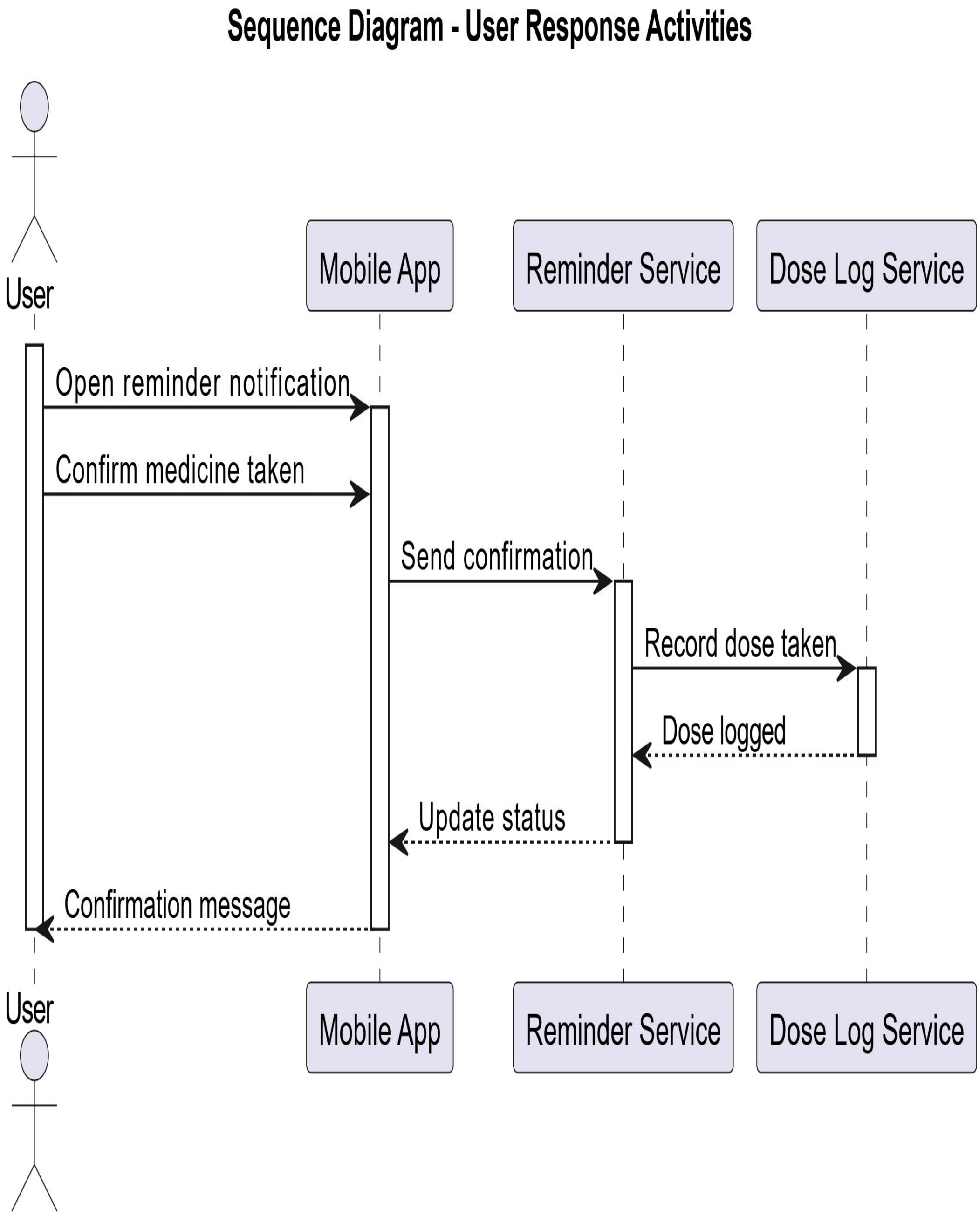

Figure 25 : sequence diagram for user response activities

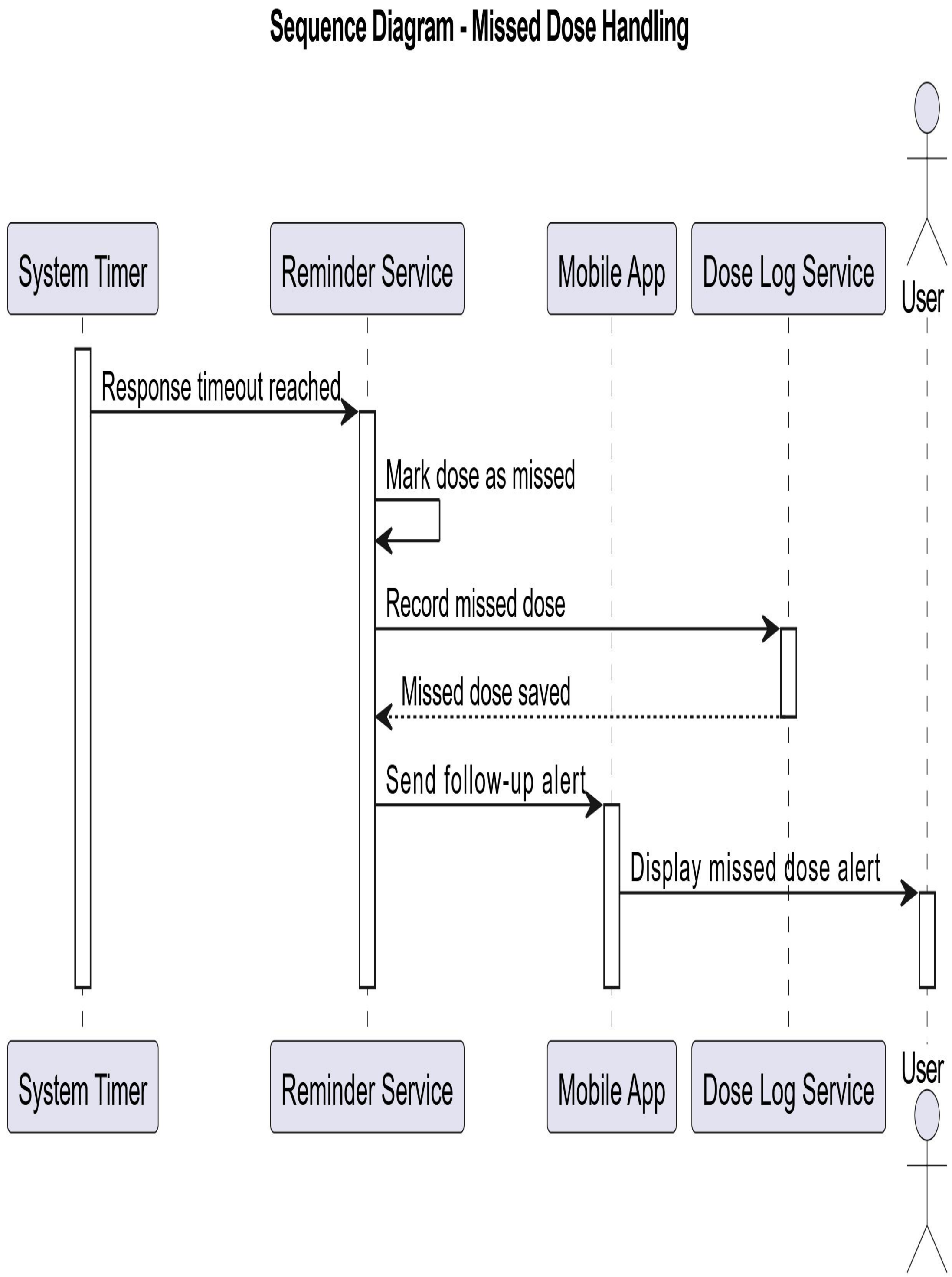

Figure 26: sequence diagram for missed dose handling

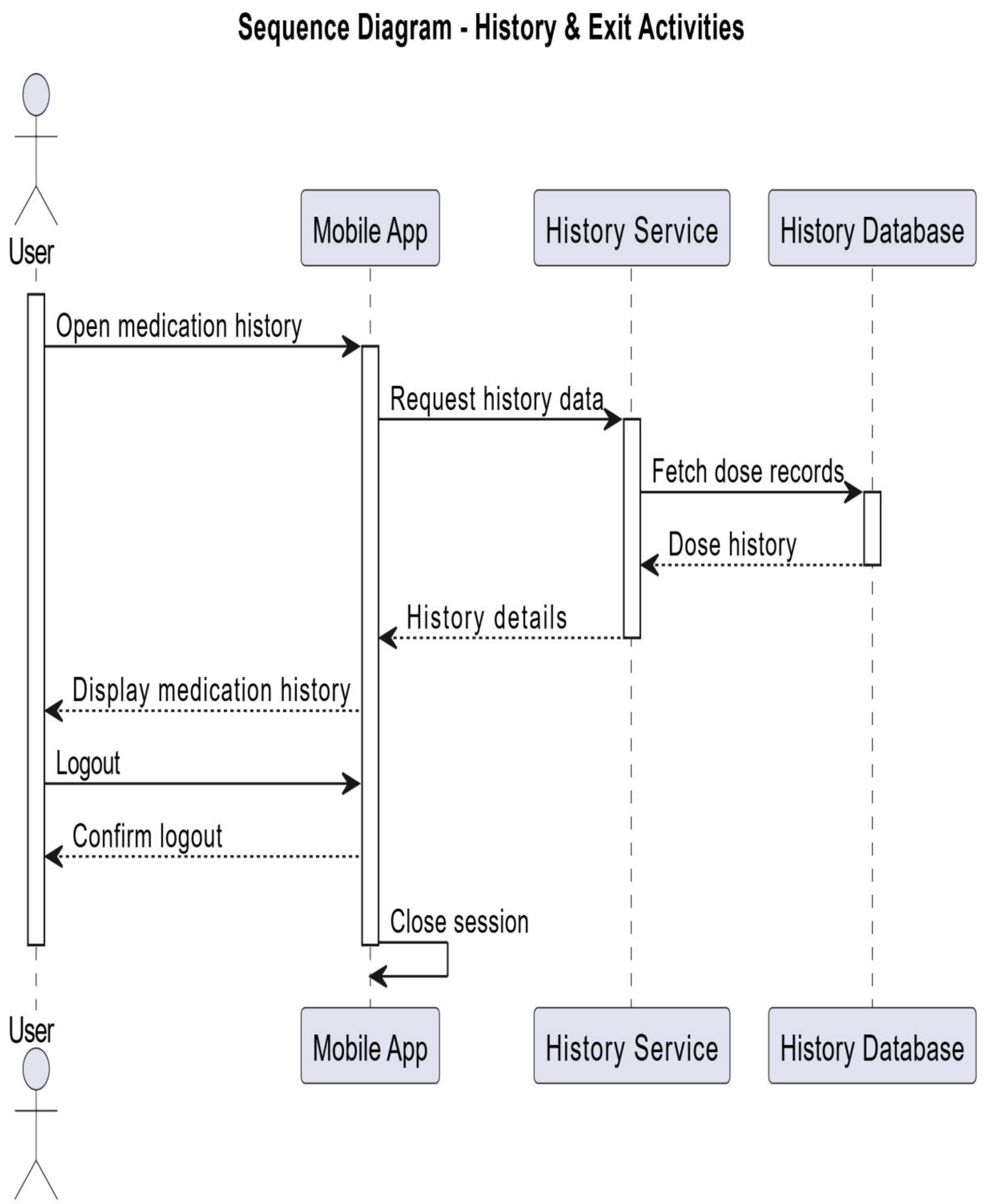

Figure 27: sequence diagram for history & exit activities

2.3.8 Analysis Class Model

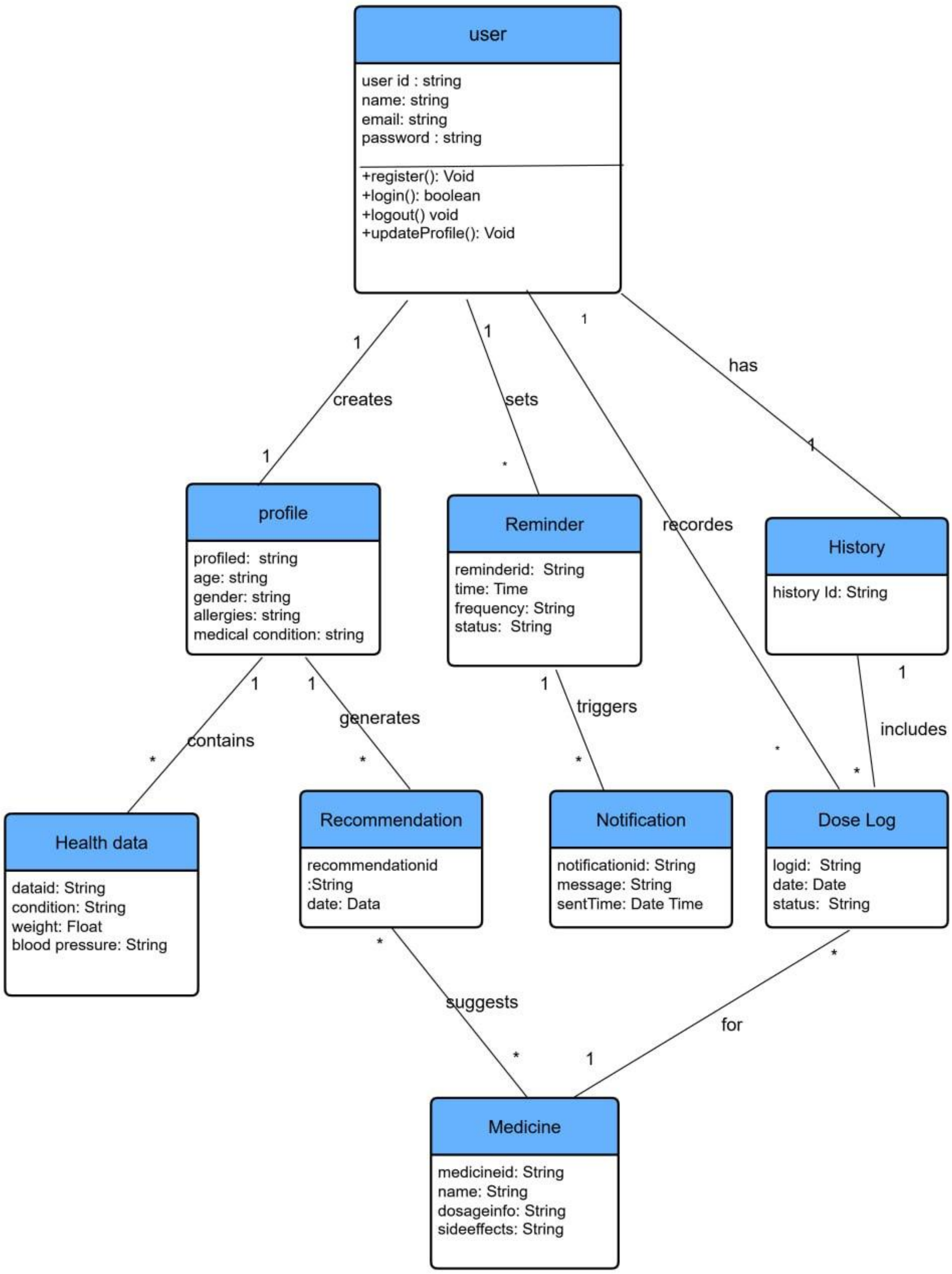

Figure 28 : analysis class model diagram

2.3.9 Logic Model

Common Logic Model

For Login

```
FUNCTION Login(email, password)
  GET email AND password
  VALIDATE email AND password AGAINST UserDatabase
  IF VALIDATION SUCCESSFUL
    CREATE userSession
    DISPLAY "Login Successful"
    REDIRECT TO Dashboard
  ELSE
    DISPLAY "Invalid Credentials"
    INCREMENT loginAttempts
    IF loginaAttempts > MAX_ALLOWED_ATTEMPTS
      LOCK account TEMPORARILY
    END IF
  END IF
END FUNCTION
```

For Sign-Up (Registration)

```
FUNCTION SignUp(userDetails)
  DISPLAY RegistrationForm
  GET userDetails (name, email, password)
  IF email ALREADY EXISTS IN DATABASE
    DISPLAY "Email already exists"
  ELSE
    VALIDATE userDetails
    IF VALIDATION SUCCESSFUL
      SAVE userDetails TO UserDatabase
      DISPLAY "Registration Successful"
      REDIRECT TO LoginPage
    ELSE
      DISPLAY "Invalid Registration Details"
    END IF
  END IFEND FUNCTION
```

For Logout

```
FUNCTION Logout(userID)
  VALIDATE activeSession FOR userID
  TERMINATE session
  DISPLAY "Logged Out Successfully"
  REDIRECT TO LoginPageEND FUNCTION
```

User Logic Model

For Profile Management

```
FUNCTION ManageProfile(userID, profileDetails)
  FETCH existingProfile FOR userID
  DISPLAY existingProfile
  GET updated profileDetails (age, gender, allergies, conditions)
  VALIDATE profileDetails
  IF VALIDATION SUCCESSFUL
    UPDATE profileDetails IN Database
    DISPLAY "Profile Updated Successfully"
  ELSE
    DISPLAY "Invalid Profile Details"
  END IFEND FUNCTION
```

For Health Data Management

```
FUNCTION UpdateHealthData(userID, healthData)
  GET healthData (weight, bloodPressure, condition)
  VALIDATE healthData
  IF VALIDATION SUCCESSFUL
    SAVE healthData TO Database
    DISPLAY "Health Data Saved"
  ELSE
    DISPLAY "Invalid Health Data"
  END IFEND FUNCTIO
```

Recommendation Logic Model

For Medicine Recommendation

```
FUNCTION GenerateRecommendation(userID)
FETCH healthProfile FOR userID
IF healthProfile IS NOT COMPLETE THEN
DISPLAY "Please complete your health profile"
RETURN
END IF
ANALYZE healthProfile USING RecommendationRules
IF healthProfile.hasSevereSymptom = TRUE OR healthProfile.hasNewSymptom
= TRUE THEN
DISPLAY "Your symptoms may require medical attention. Please consult a
doctor."
RETURN
END IF
FETCH suitableMedicines FROM MedicineDatabase
GENERATE recommendationList
DISPLAY recommendationList TO user
END FUNCTION
```

Reminder Logic Model

For Setting Reminder

```
FUNCTION SetReminder(userID, medicineID, dosage, time, frequency)
  VALIDATE reminderDetails
  IF VALIDATION SUCCESSFUL
    SAVE reminderDetails TO ReminderDatabase
    ACTIVATE reminder
    DISPLAY "Reminder Set Successfully"
  ELSE
    DISPLAY "Invalid Reminder Details"
  END IFEND FUNCTION
```

For Reminder Execution

```
FUNCTION ExecuteReminder(reminderID)
  WAIT UNTIL scheduledTime
  SEND notification TO user
  START responseTimer
END FUNCTION
```

User Response Logic Model

For Medicine Taken

```
FUNCTION ConfirmMedicineTaken(userID, reminderID)
  RECORD dose AS "Taken"
  SAVE record TO DoseLog
  UPDATE reminderStatus
  DISPLAY "Dose Logged Successfully"
END FUNCTIO
```

For Missed Dose Handling

```
FUNCTION HandleMissedDose(reminderID)
  IF userResponse NOT RECEIVED WITHIN timeLimit
    MARK dose AS "Missed"
    SAVE missedDose TO DoseLog
    SEND followUpNotification
  END IFEND FUNCTION
```

History & Exit Logic Model

For Viewing Medication History

```
FUNCTION ViewMedicationHistory(userID)
  FETCH doseLogs FOR userID
  DISPLAY medicationHistoryEND FUNCTION
```

For Exit Application

```
FUNCTION ExitApplication(userID)
  SAVE currentState
  TERMINATE session
  CLOSE applicationEND FUNCTION
```

2.4 Non-Functional Requirements

The non-functional requirements of the **Medicine Reminder and Recommender App** define the quality attributes that ensure the system operates efficiently, securely, and reliably while providing a smooth and user-friendly experience. These requirements focus on performance, reliability, security, scalability, usability, maintainability, and availability of the system.

Performance Requirements:

- The system must respond to user interactions such as setting reminders, viewing recommendations, and logging doses within two seconds for at least 95% of operations.
- Medication reminders and notifications must be delivered in real time without noticeable delay to ensure timely medicine intake.
- In addition, the recommendation engine must generate appropriate medicine suggestions within three seconds after processing user health data.

Reliability Requirements:

- The system must maintain an uptime of at least 90% to ensure continuous access to medication reminders, recommendations, and history tracking.
- Users must also be able to view previously saved reminders and medication records reliably without data loss or interruption.

Security Requirements:

- All user data, including personal information, medication schedules, and health-related data, must be securely encrypted during both transmission and storage.
- The system must implement secure authentication mechanisms such as one-time passwords (OTP) or email verification to prevent unauthorized access.
- Role-based access control must be enforced to ensure that only authorized users can access sensitive data and system functionalities.

Furthermore, logging and monitoring mechanisms must be implemented to detect suspicious activities or potential security breaches within five minutes of occurrence.

Scalability Requirements:

- The system architecture must support horizontal scalability to handle a growing number of users and increasing volumes of medication and reminder records.
- A cloud-based infrastructure should allow automatic allocation of system resources during peak usage periods without degrading overall performance.
- The database must be capable of efficiently managing large volumes of reminder logs and recommendation data.

Usability Requirements:

- The application must provide a simple, intuitive, and user-friendly interface that is suitable for elderly users and individuals with limited technical experience.
- The system must support clear visual notifications and alerts to ensure users do not miss medication reminders.
- Additionally, the application should support multiple languages, including English and Amharic, to improve accessibility for local users.

Maintainability Requirements:

- The system must be developed using modular, well-structured, and well-documented coding standards to simplify future maintenance and feature enhancement.
- Continuous Integration and Continuous Deployment (CI/CD) pipelines should be implemented to enable smooth and reliable system updates.

- The application must be easily updatable to accommodate new medicines, health rules, or changes in recommendation logic.

Availability Requirements:

- The system must be available 24 hours a day, seven days a week, allowing users to access medication information and receive reminders at any time.
- Scheduled system maintenance must be conducted during off-peak hours, and users must be notified at least 24 hours in advance to minimize inconvenience.

2.5 System Requirements

2.5.1 Hardware Requirements

- **CPU:**
  Development: Intel Core i5 (or higher) for running IDEs, emulators, and testing tools.
  Deployment: Cloud servers with at least 4-core CPUs to host backend services.
- **RAM:**
  Development: Minimum 8 GB, with 16 GB recommended.
  Mobile Devices: Minimum 4 GB RAM for smooth application performance.
- **Storage:**
  Development Environment: Minimum 256 GB SSD.
  Mobile Devices: At least 32 GB internal storage for application installation and local caching.
- **Mobile Devices:**
  Android devices running Android 8.0 (Oreo) or newer.

2.5.2 Software Requirements

- **Operating System:**
  Development: Windows 10/11 or Linux distributions.
  User Devices: Android 8.0 or newer, iOS 13.0 or newer.
- **Development Tools:**
  IDEs: Android Studio, Visual Studio Code.
  Programming Languages: Dart (Flutter), JavaScript / Node.js (backend services).
- **Database:**
  Primary Database: Firebase Realtime Database or Firestore.
  Backup Database: MongoDB Atlas or PostgreSQL.
- **API and Integration Tools:**
  Notification Services: Firebase Cloud Messaging (FCM).
  Health Data Integration: APIs compatible with health metrics where applicable.

- **Version Control:**

  Git and GitHub for source code management and collaboration.

- **Browser Requirements:**

  Latest versions of Google Chrome or Microsoft Edge for admin dashboards.

- **Security Tools:**

  JWT for secure API communication.

  Encryption tools for protecting sensitive data.

- **Testing Tools:**

  Flutter testing framework for unit and integration testing.

  Postman for API testing.

2.6 Key Abstraction with CRC Analysis

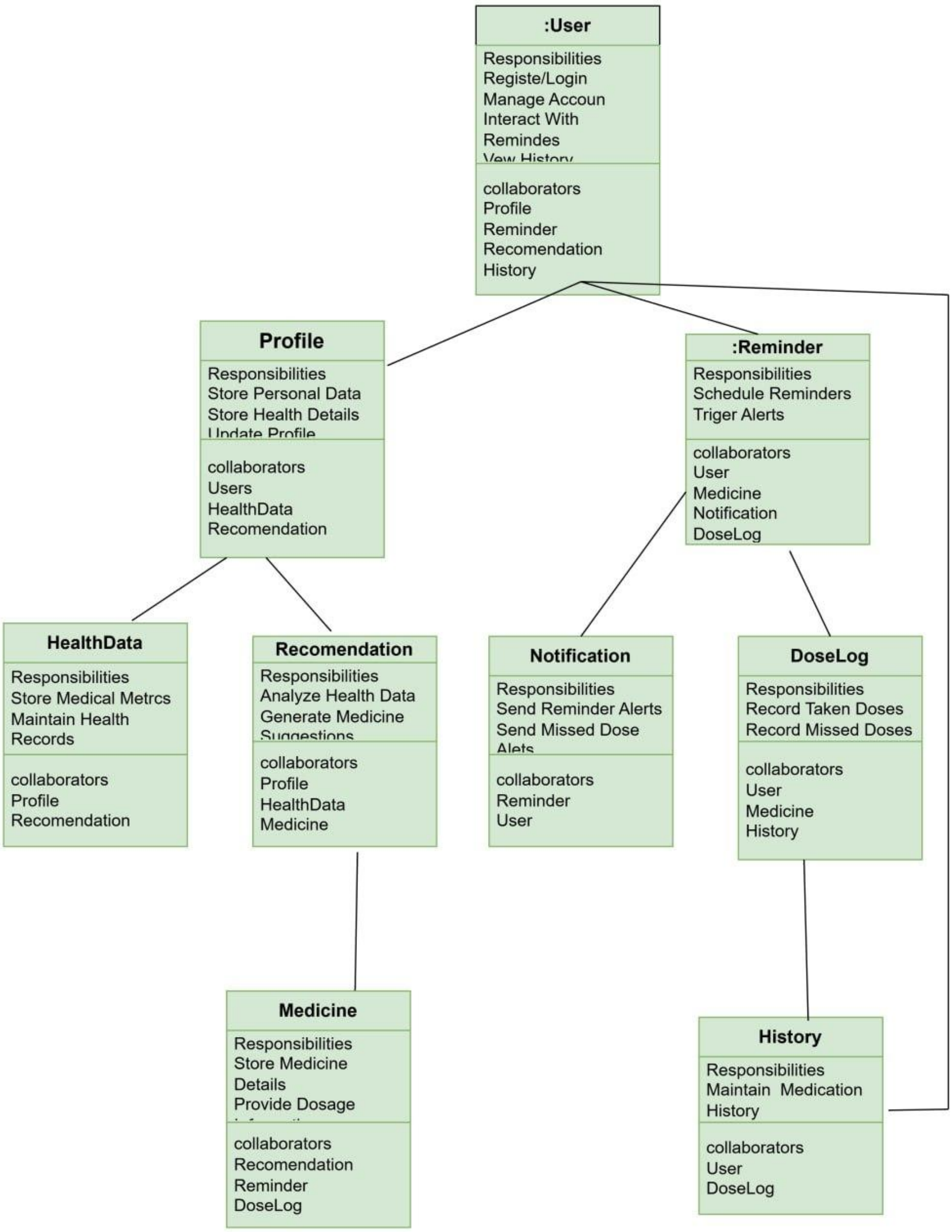

Figure 29: CRC analysis

2.6.1 Conceptual modelling: Class Diagram

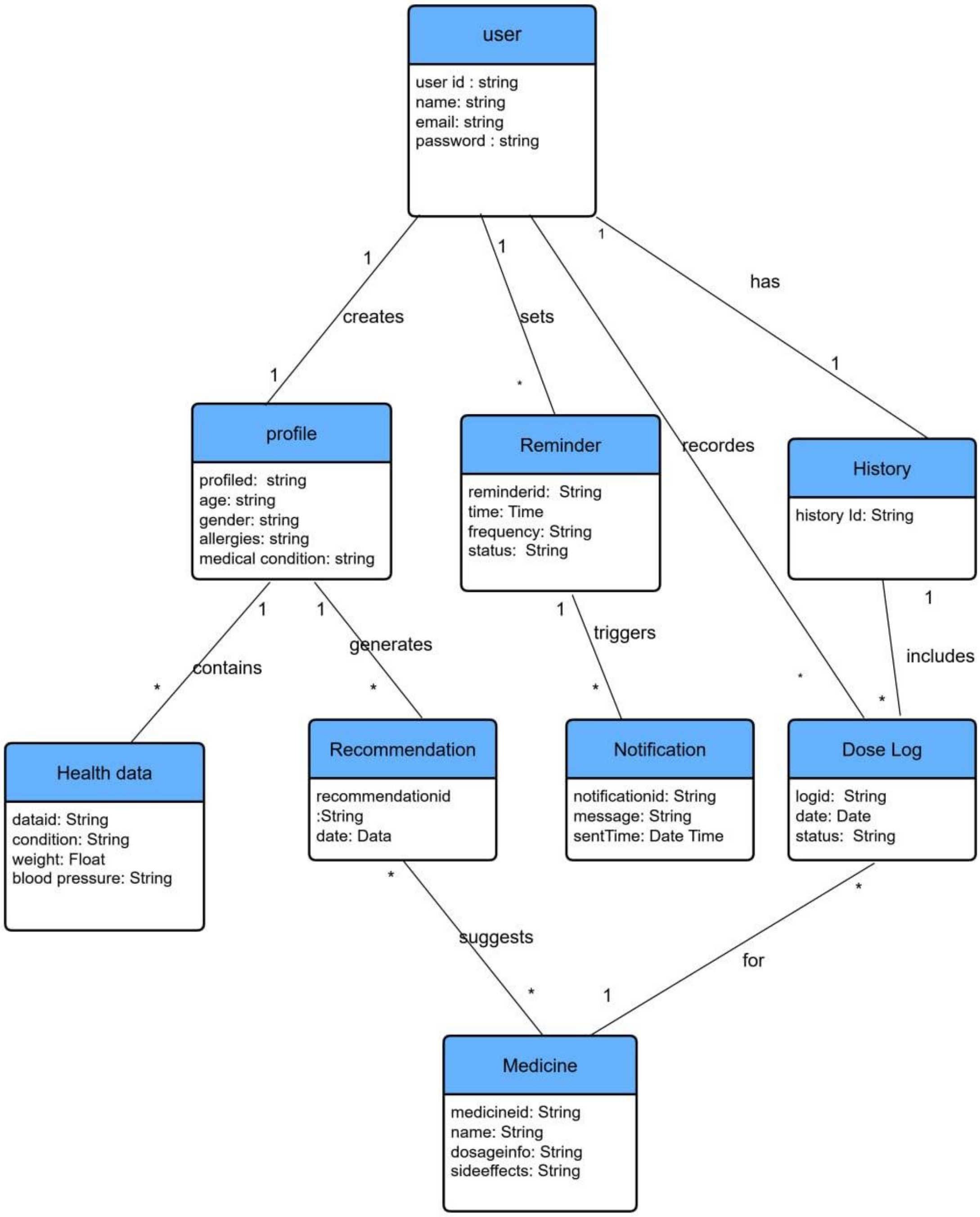

Figure 30 : conceptual model class diagram

2.6.2 Identifying Change Cases

Change Case	Description	Likelihood	Impact
Add a new user role	Introduce a role such as **Caregiver** or **Pharmacist** to help users manage medications and provide guidance.	Moderate	Modify the User class and update access control and permissions.
Extend recommendation engine	Enhance the recommendation system using machine learning to provide more personalized medicine suggestions.	High	Add new attributes and methods to the Recommendation class and integrate ML models.
Support medicine substitution	Allow the system to suggest alternative medicines when a prescribed drug is unavailable.	High	Update the Medicine and Recommendation classes to include substitute logic.
Introduce recurring reminders	Enable users to set complex reminder schedules such as weekly, monthly, or interval-based reminders.	High	Modify the Reminder class to include recurrence patterns and scheduling rules.
Add drug interaction alerts	Notify users about possible drug-to-drug interactions based on their	Moderate	Extend the Medicine and HealthData classes with interaction

	medication list.		attributes and validation logic.
Enhance notification channels	Support additional notification methods such as SMS, WhatsApp, or email alerts.	High	Update the Notification class to handle multiple delivery channels.
Offline reminder support	Allow reminders and dose logging to function without internet connectivity.	Moderate	Introduce local storage handling and sync logic in Reminder and DoseLog classes.
Improve adherence analytics	Provide detailed adherence reports and visual analytics for users.	Moderate	Extend the History and DoseLog classes with analytics-related attributes.
Multi-language support	Support multiple languages for medicine instructions and alerts.	High	Update UI resources and add language attributes to User and Notification classes.

Table 28: change case discription

2.6.3 User Interface Prototyping

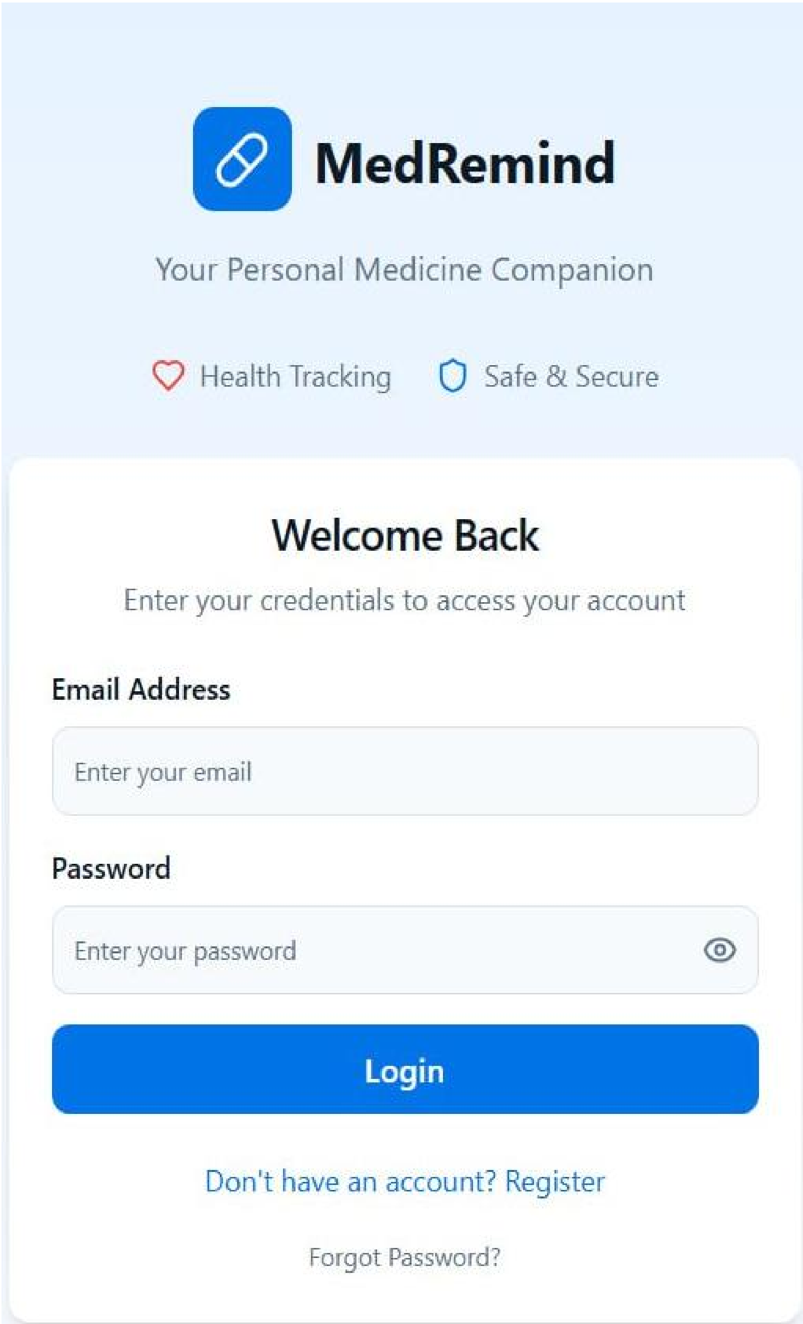

Figure 31: Login & Register UI

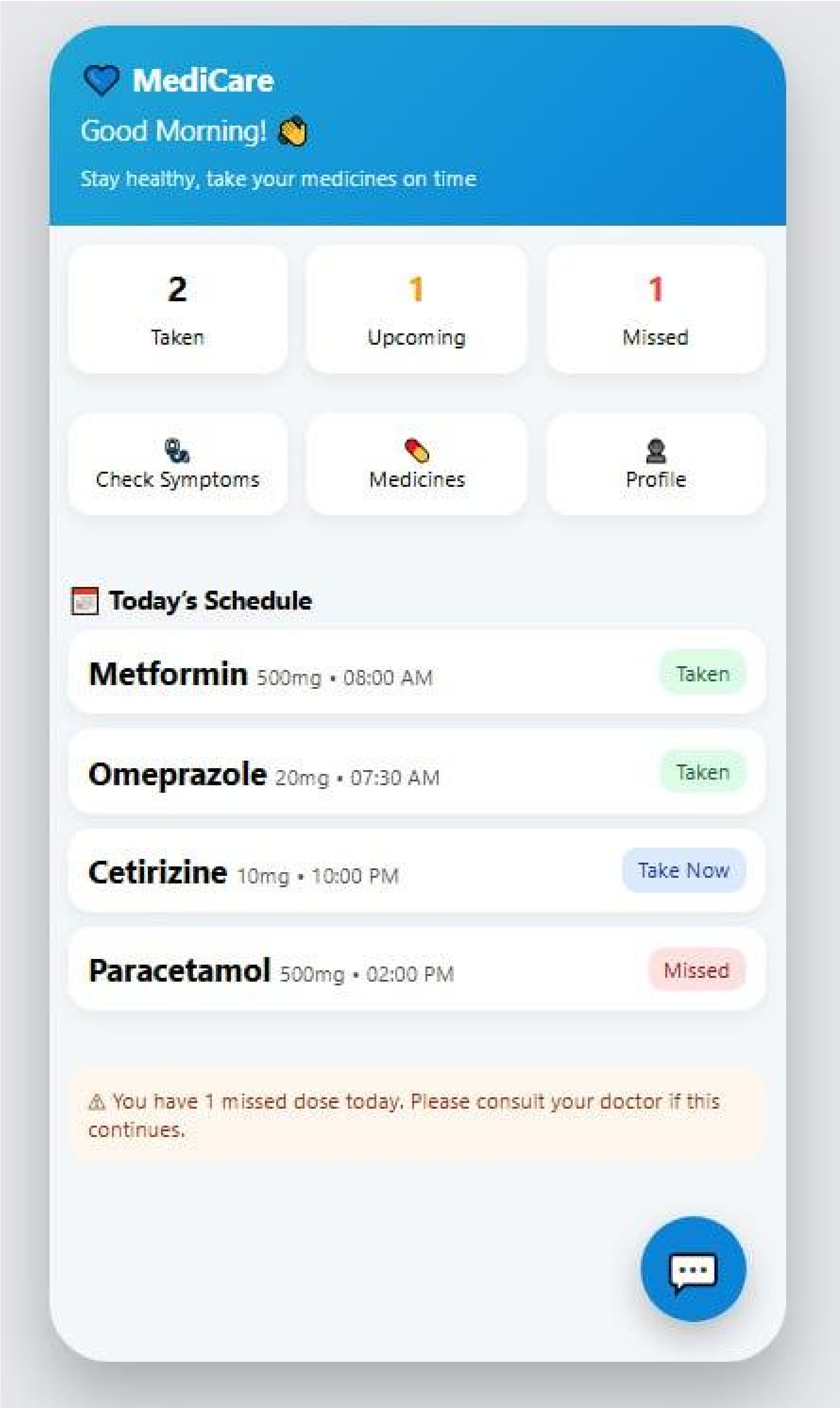

Figure 32: Dashboard UI

Chapter 3: System Design

3.1 Architectural Design

This chapter presents the system design of the *Medicine Reminder and Recommender App*. The design focuses on reliability, scalability, usability, and secure handling of medical information. The system is built using a **client–server architecture**, where the mobile application acts as the client and the backend system provides services such as medicine reminder management, symptom analysis, and medicine recommendation.

The architecture separates the system into independent layers, allowing each component to perform a specific function. This separation improves maintainability and makes the system flexible for future enhancements such as advanced recommendation logic or integration with pharmacy services.

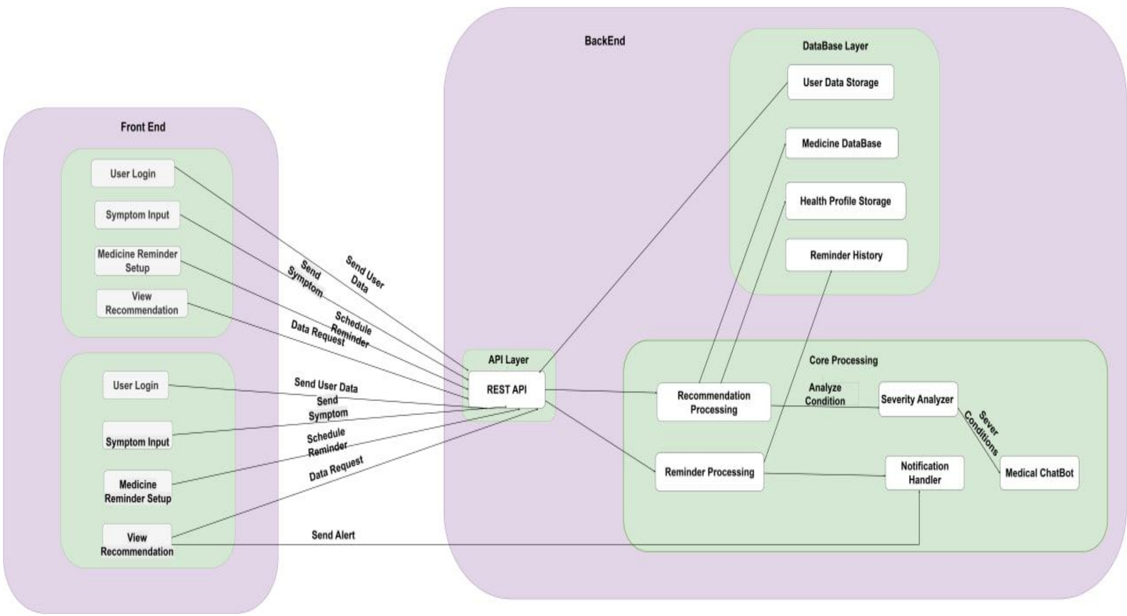

Figure 33 : architecture design

3.1.1 Component Modeling

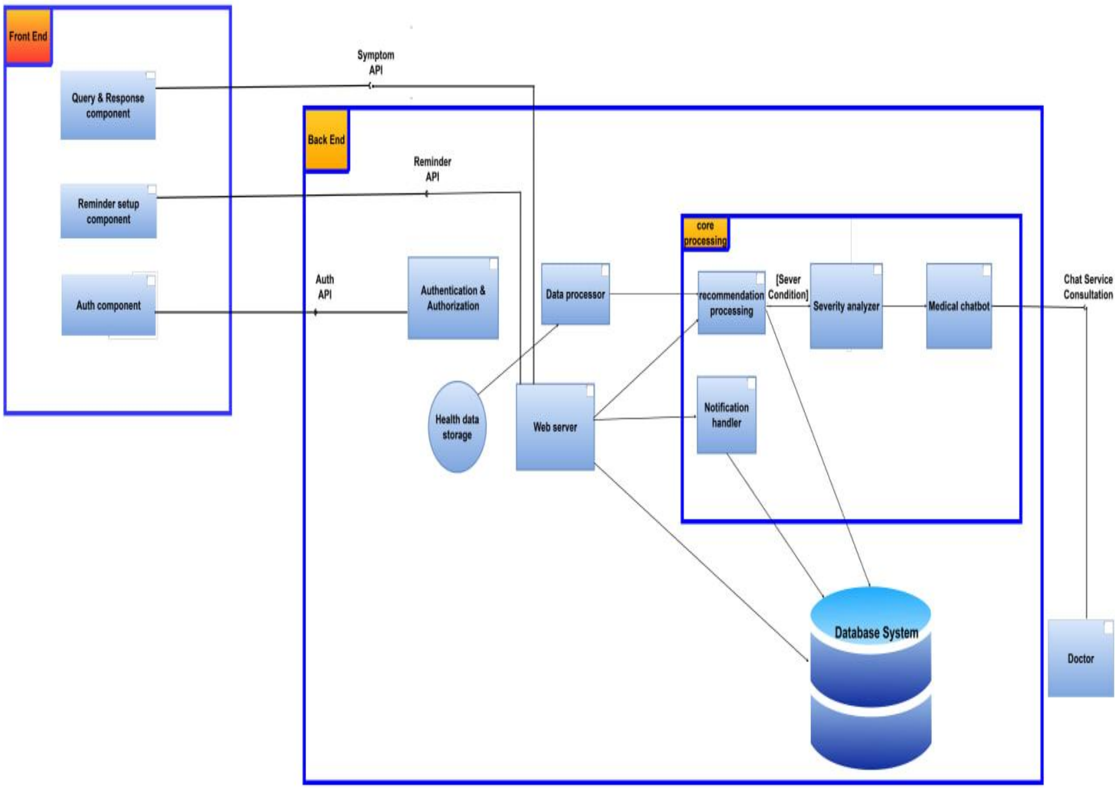

Figure 34 : component model

The system consists of the following main components:

- **Mobile Application (User Interface Layer)**
  This component provides the interface through which users interact with the system. It allows users to register, log in, manage health profiles, schedule medicine reminders, enter symptoms, and view recommendations and medicine history.
- **API Layer**
  The API layer manages communication between the mobile application and the backend services. It handles requests related to authentication, reminders, recommendations, and data retrieval.
- **Application Logic Layer**
  This layer contains the core business logic of the system. It processes user inputs,

manages reminder schedules, validates health data, and coordinates interactions between different modules.

- **Medicine Recommendation Engine**

  This component analyzes user-entered symptoms along with health profile data such as allergies and chronic conditions. Based on predefined rules and intelligent logic, it generates suitable medicine recommendations and safety warnings.

- **Notification Service**

  The notification service is responsible for sending medicine reminders and alerts to users at scheduled times. It also tracks whether a reminder was confirmed or missed.

- **Database Component**

  The database securely stores user information, health profiles, medicine data, reminder schedules, and medicine intake history.

3.1.2 Deployment Model

The application is deployed using a **client–server deployment model**. The mobile application runs on Android and iOS devices, while backend services are hosted on a cloud-based server.

Communication between the client and server is carried out through secure RESTful APIs. The notification service integrates with push notification platforms to ensure timely delivery of medicine reminders. This deployment approach ensures high availability, scalability, and efficient performance of the system.

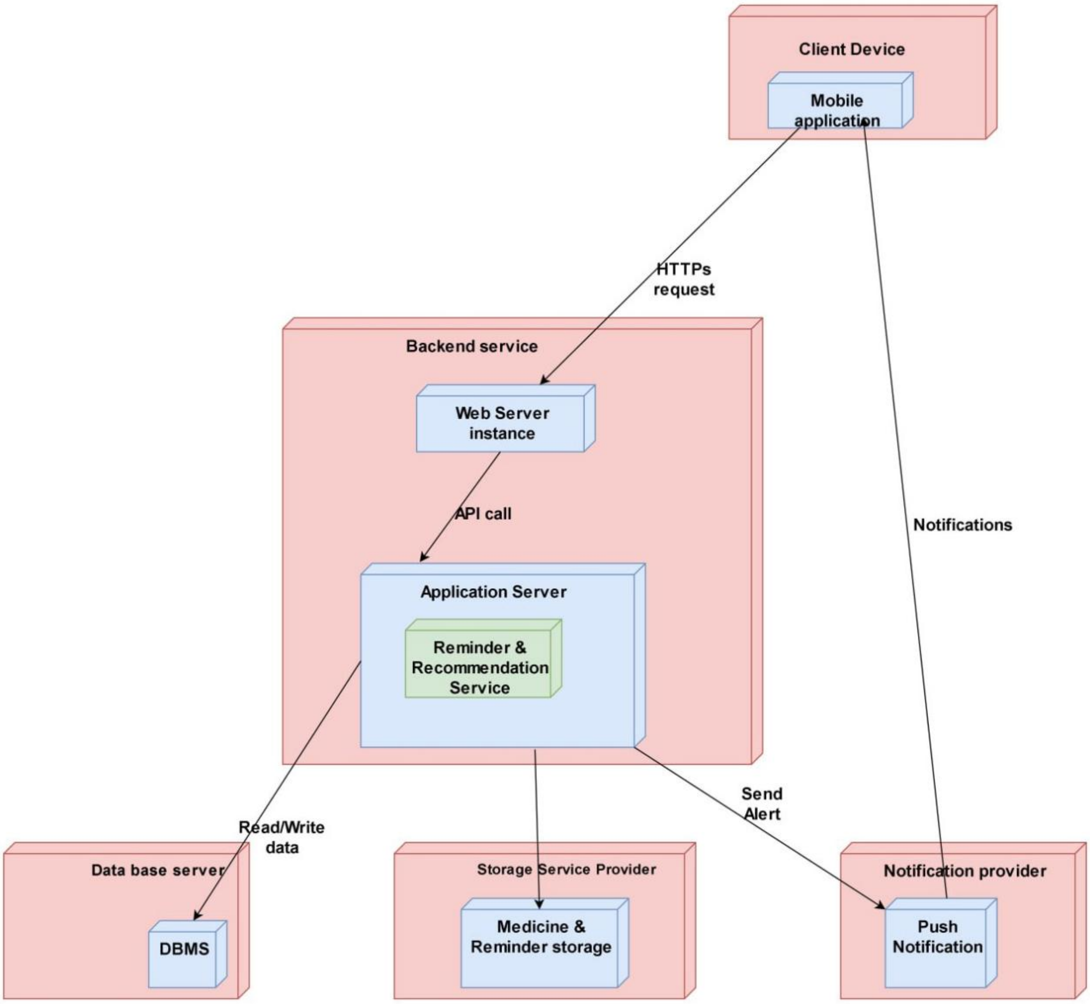

Figure 35 : Deployment model

3.2 Detailed Design

3.2.1 Design Class Model

The system is designed using **object-oriented principles**, where each class represents a real-world entity. The major classes in the system include:

- ✓ **User:** Stores user authentication details and basic account information.
- ✓ **HealthProfile:** Maintains health-related data such as age, allergies, and chronic conditions, which are used for personalized recommendations.

- ✓ **Medicine:** Represents medicine information including name, dosage, frequency, and usage instructions.
- ✓ **Reminder:** Handles the scheduling, tracking, and updating of medicine reminders.
- ✓ **Recommendation:** Stores recommended medicines generated based on symptom analysis and health data.
- ✓ **Notification:** Manages reminder alerts and system notifications.

These classes interact to support core system functionalities such as reminder scheduling, medicine recommendation, and intake tracking.

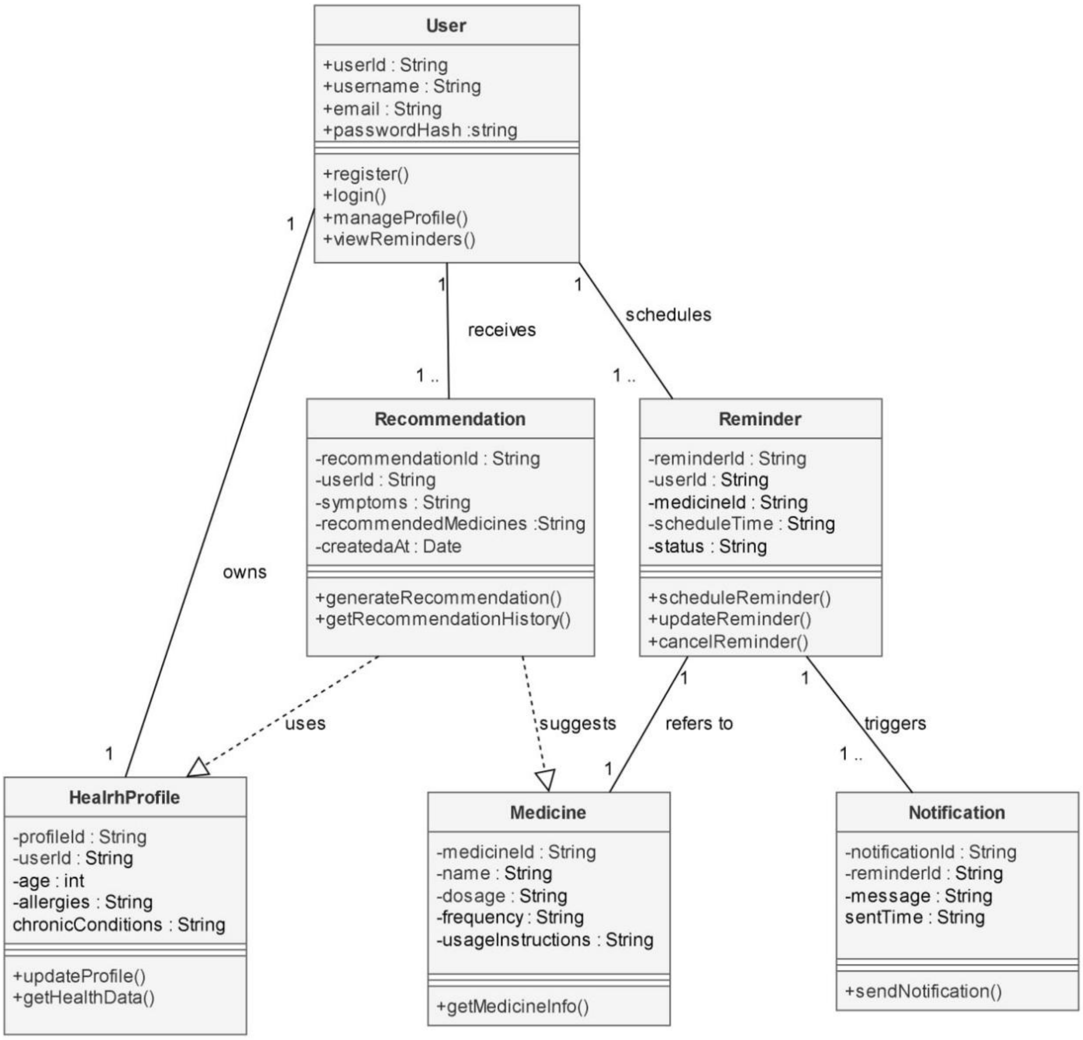

Figure 36 : Design class model

3.2.2 Persistent Model

The persistent model uses a **NoSQL database** to store system data in a flexible and scalable manner. The main collections include:

- ✓ **Users** – Stores account and authentication information
- ✓ **HealthProfiles** – Stores medical conditions and allergy details
- ✓ **Medicines** – Stores medicine descriptions and safety guidelines
- ✓ **Reminders** – Stores reminder schedules and dosage timing
- ✓ **History** – Stores records of confirmed and missed medicine intake

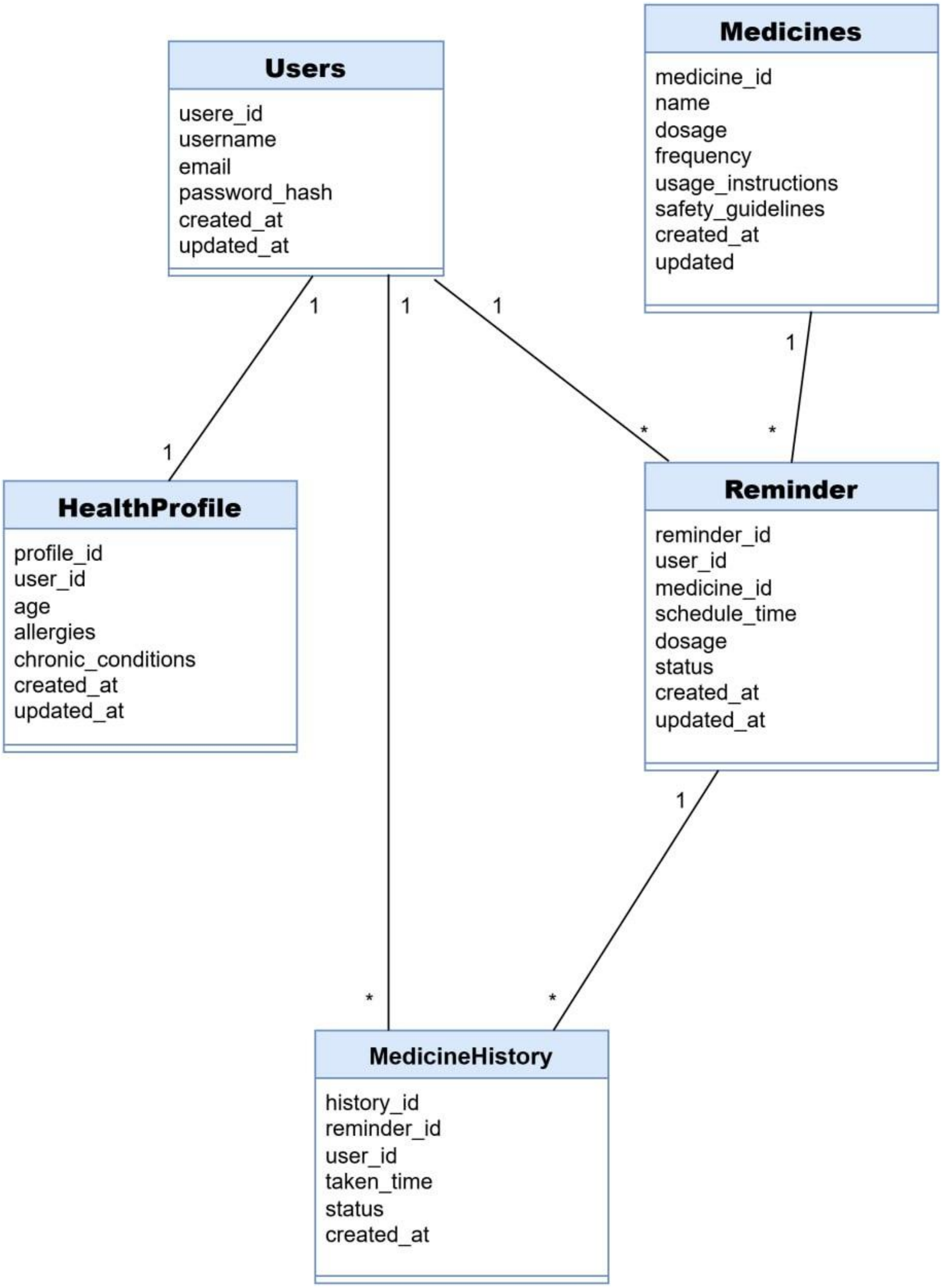

Figure 37 : persistence model

Here are some generated sample JSON structures for collections and documents:

A. Users collection

```
{
  "_id": "user_001",
  "name": "John Doe",
  "email": "john.doe@gmail.com",
  "passwordHash": "hashed_password_value",
  "role": "user",
  "createdAt": "2025-01-15T10:30:00Z",
  "lastLogin": "2025-01-18T08:15:00Z"
}
```

Figure 38 : users collection

B. Health profiles collection

```
{
 "_id": "hp_001",
 "userId: "user_001",
 "age": 60,
 "gender": "Male",
 "allergies": ["Penicillin"],
 "chronicConditions": ["Diabetes", "Hypertension"],
 "emergencyContact": {
  "name": "Jane Doe",
  "phone": "9876543210"
   },
   "lastUpdated": "2025-01-15T11:00:00Z"

 }
```

Figure 39 : Health profile collection

C. Medicines collection


```json
{
 "_id": "med_001",
 "medicineName": "Paracetamol",
 "dosageForm": "Tablet",
 "strength": "500mg",
 "usageInstructions": "Take after food",
 "sideEffects": ["Nausea", "Dizziness"],
 "warnings": ["Avoid alcohol"],
 "prescriptionRequired": false
}
```


Figure 40 : Medicines collection

D. Reminders collection


```json
{
 "_id": "rem_001",
 "userId": "user_001",
 "medicineId": "med_001",
 "medicineName": "Paracetamol",
 "dosage": "500mg",
 "frequency": "Twice a day",
 "times": ["08:00", "20:00"],
 "startDate": "2025-01-16",
 "endDate": "2025-01-20",
 "status": "active",
 "createdAt": "2025-01-15T12:00:00Z"
}
```


Figure 41 : Reminders collection

E. Medicine History collection


```json
{
 "_id": "hist_001",
 "userId": "user_001",
 "reminderId": "rem_001",
 "medicineName": "Paracetamol",
 "scheduledTime": "2025-01-16T08:00:00Z",
 "takenTime": "2025-01-16T08:05:00Z",
 "status": "taken"
}
```


Figure 42 : Medicine history collection

3.3 User Interface Design

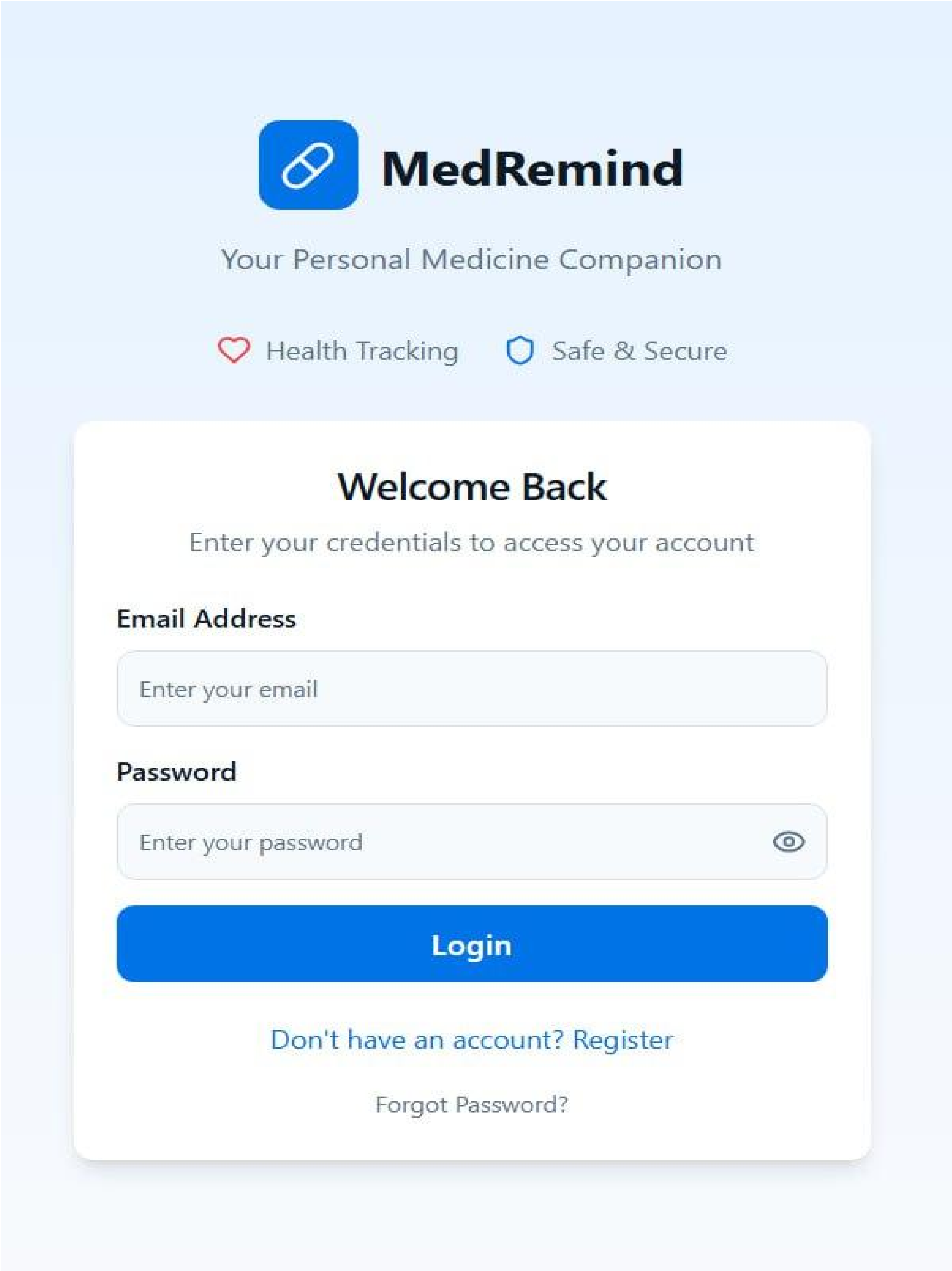

Figure 43: Login & Register UI

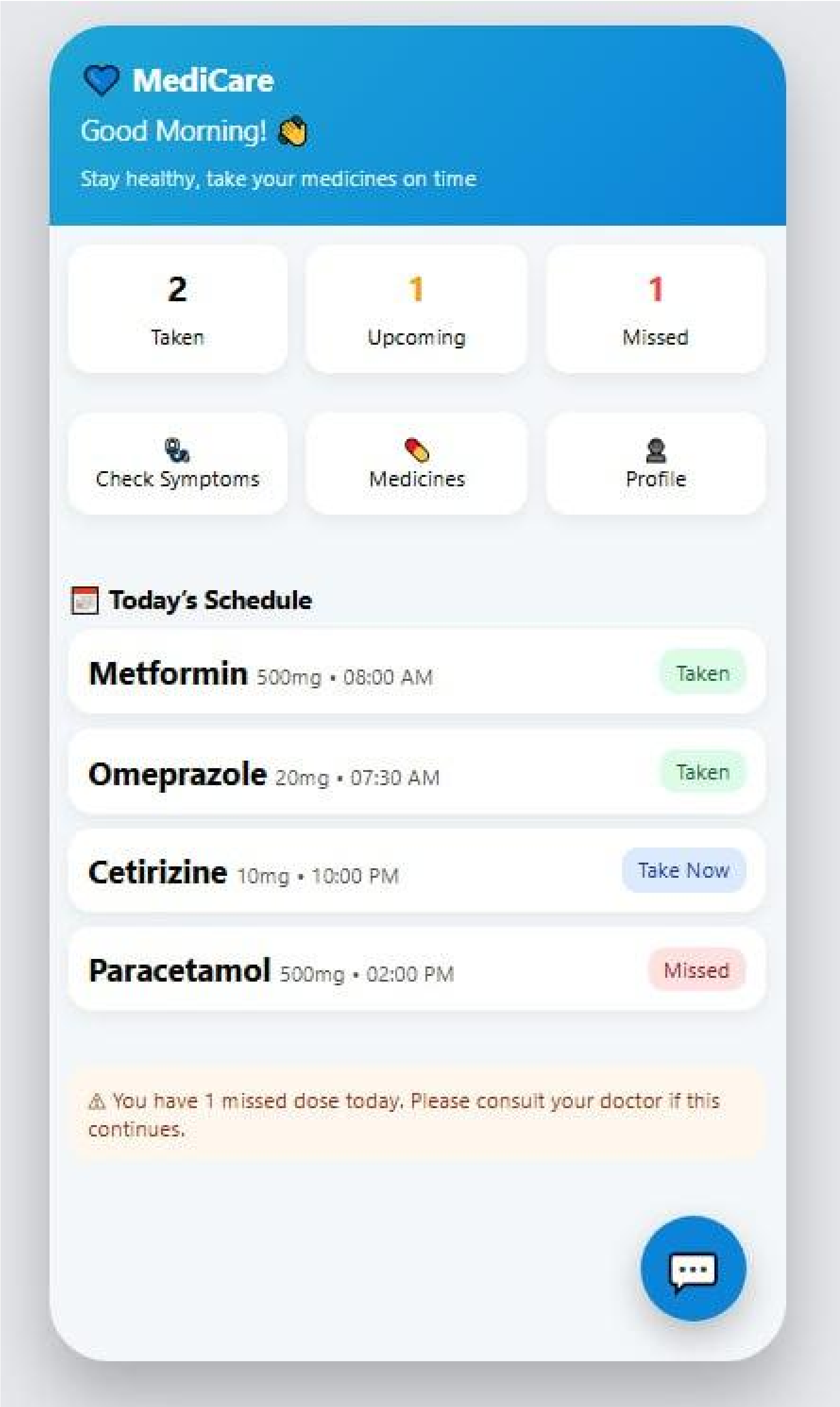

Figure 44: Dashboard UI

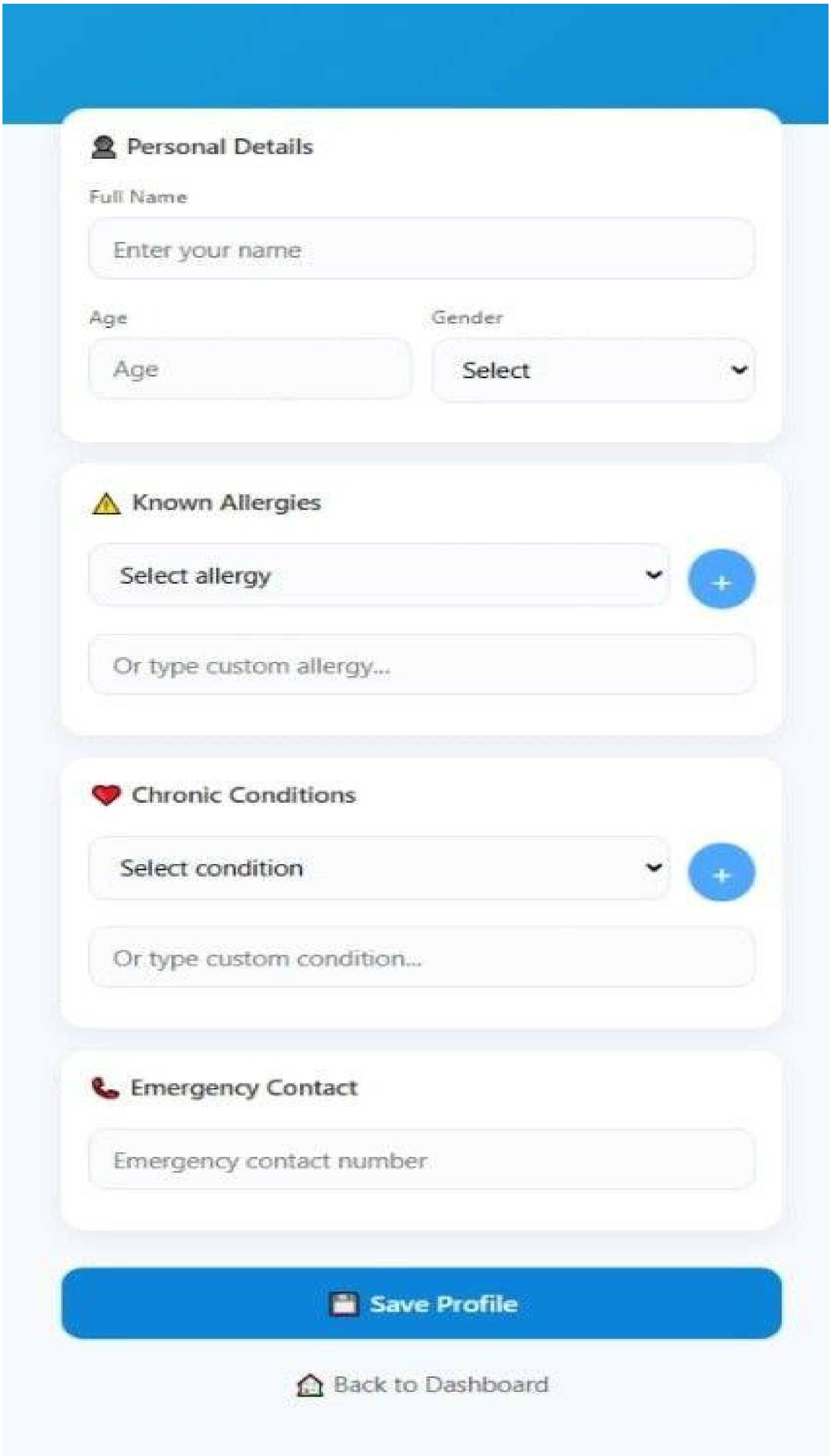

Figure 45: profile management UI

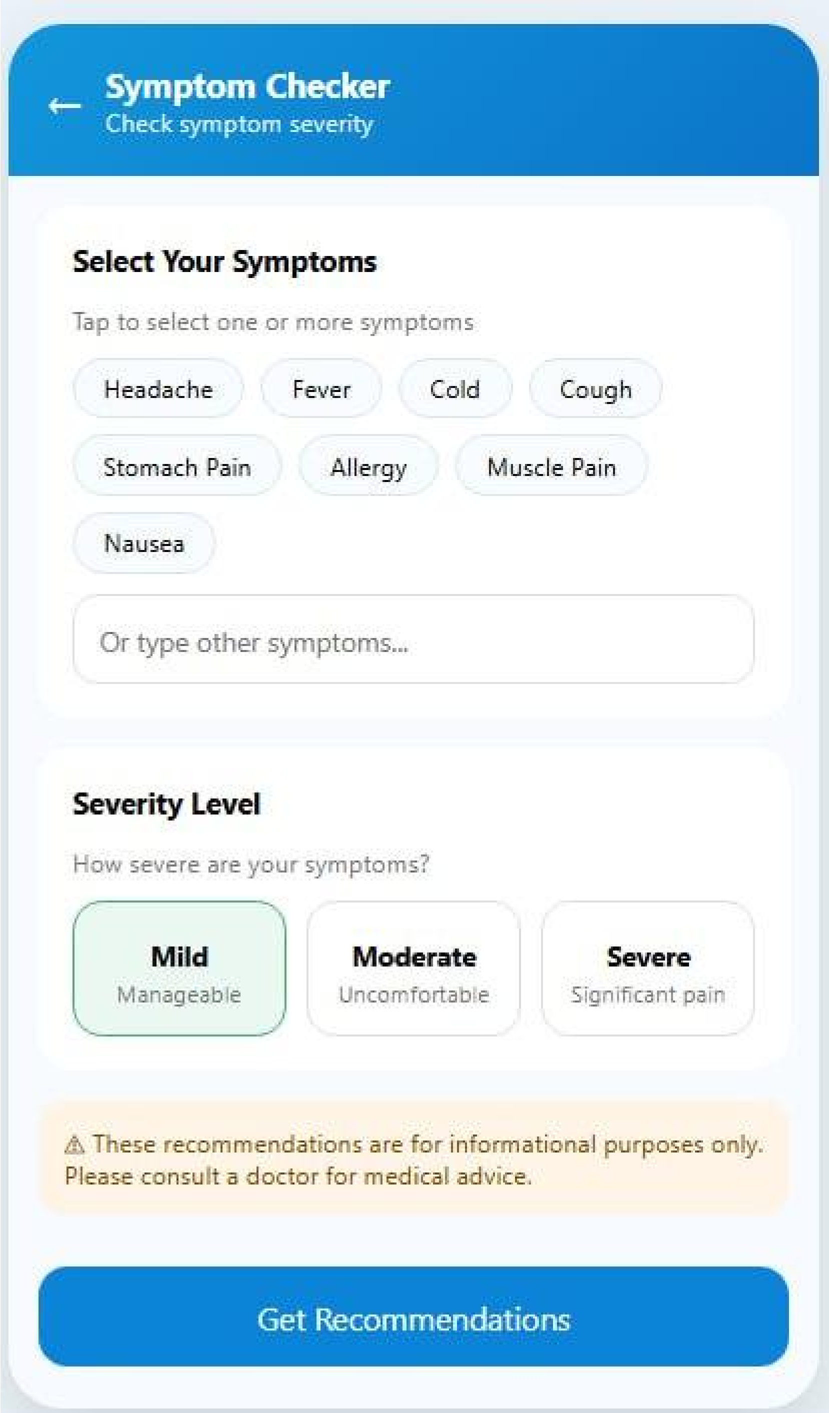

Figure 46: symptom checker UI

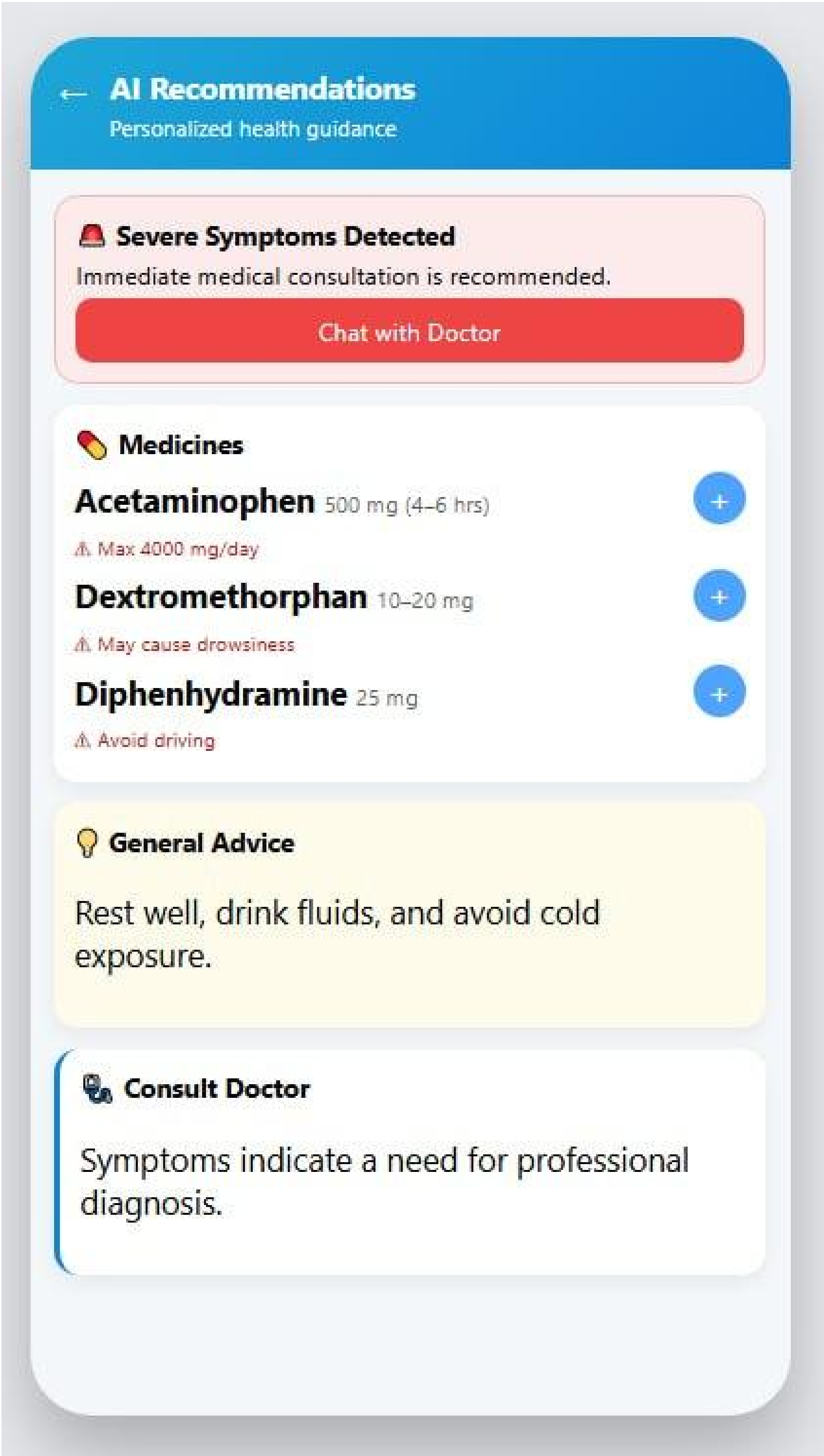

Figure 47: Recommendation UI

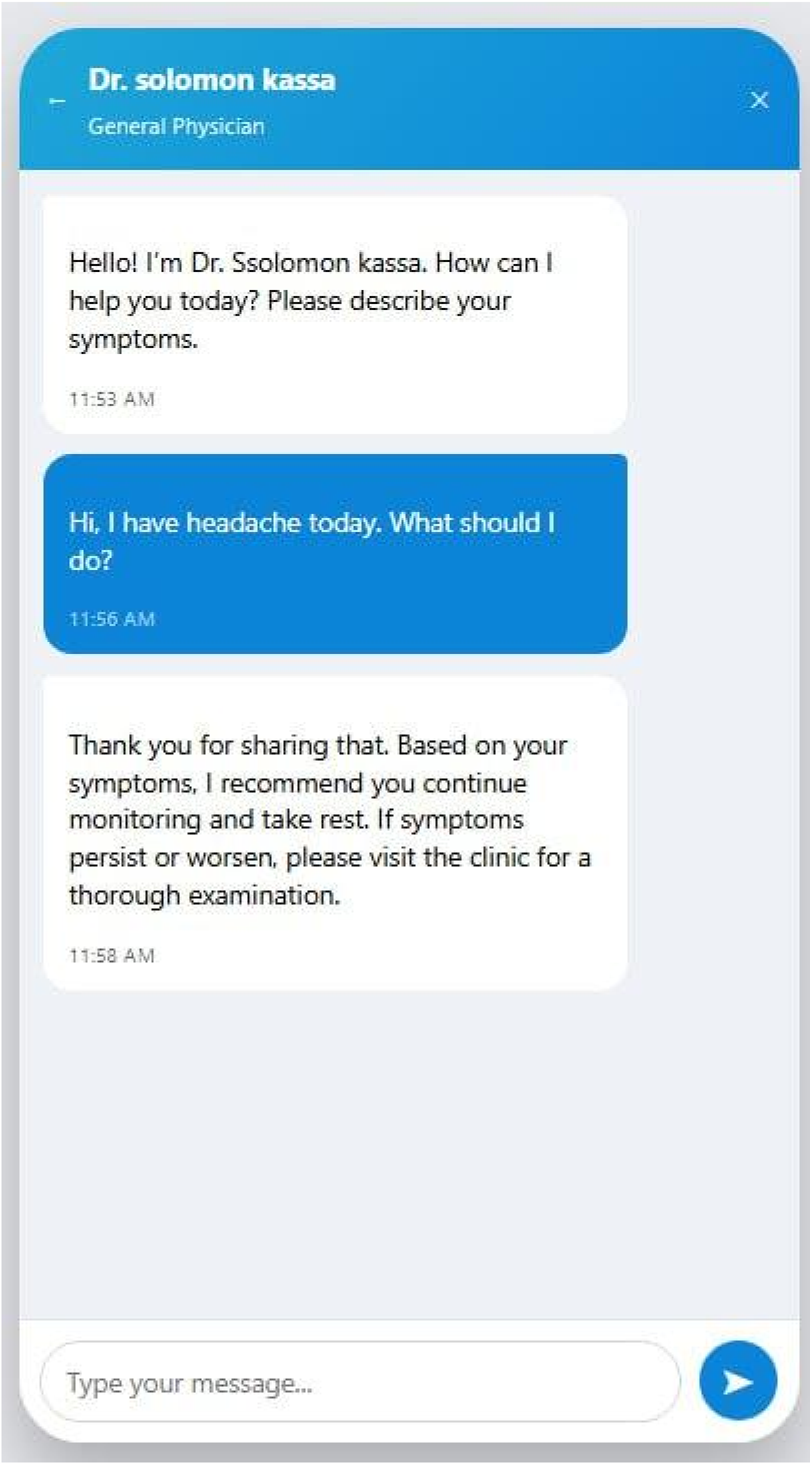

Figure 48:Chatbot UI

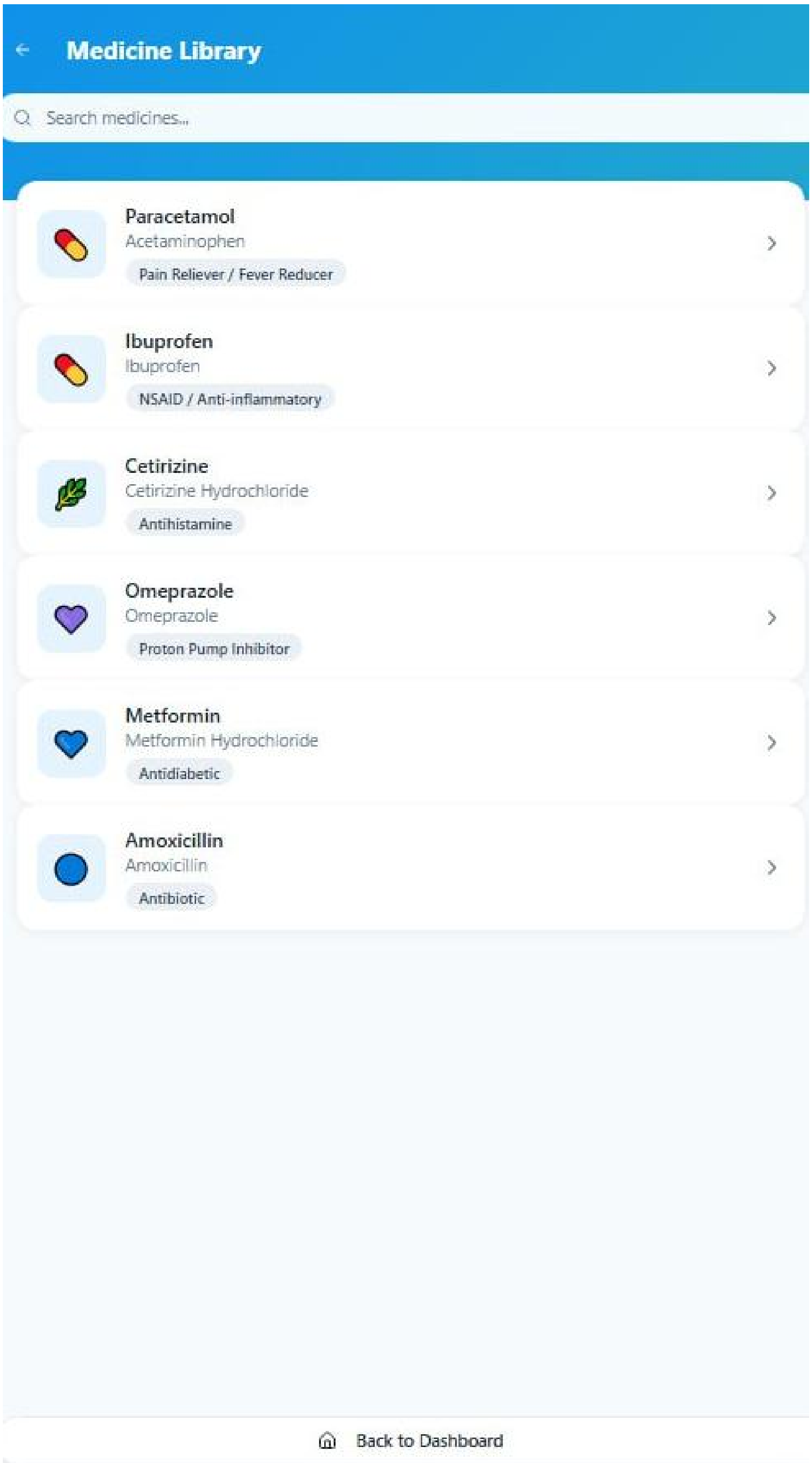

Figure 49:medicine library UI

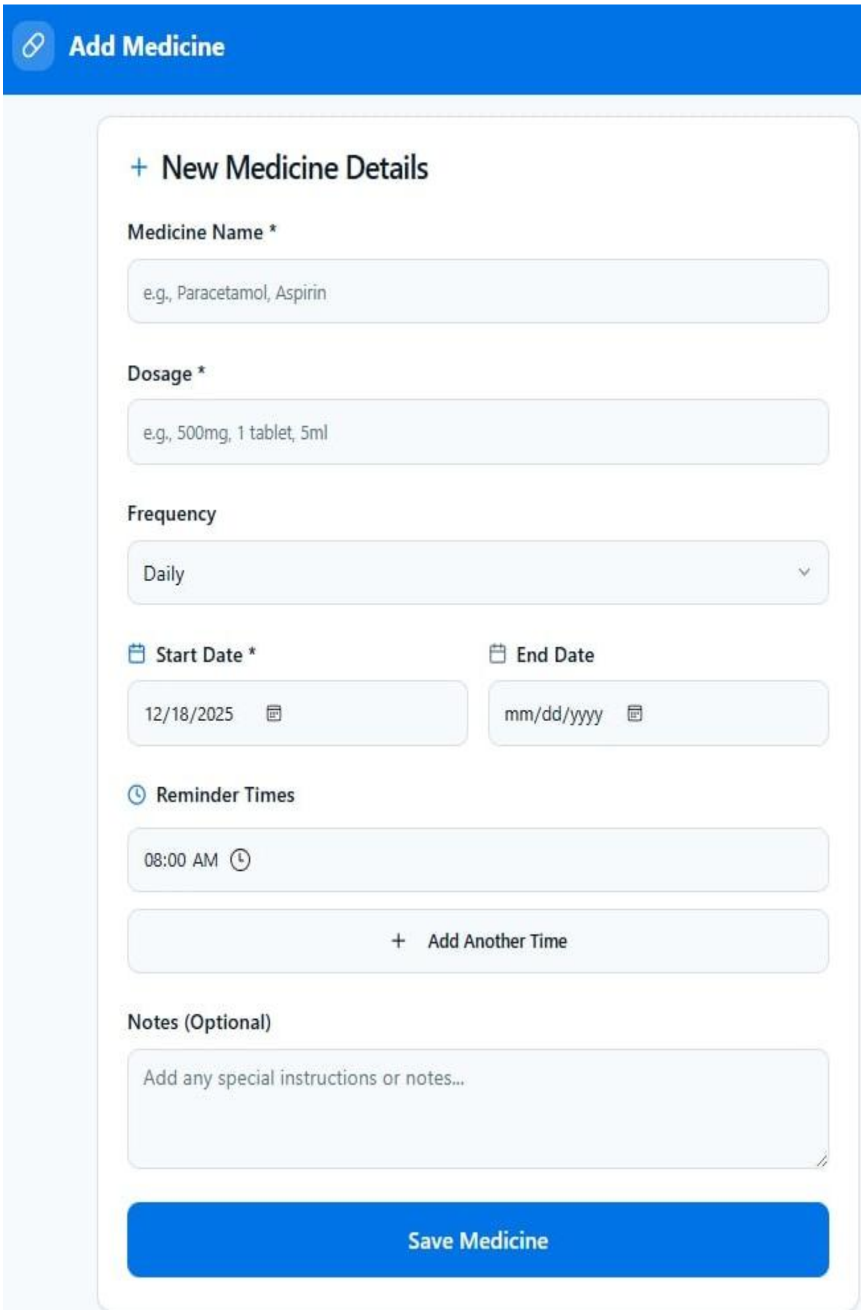

Figure 50: add medicine UI

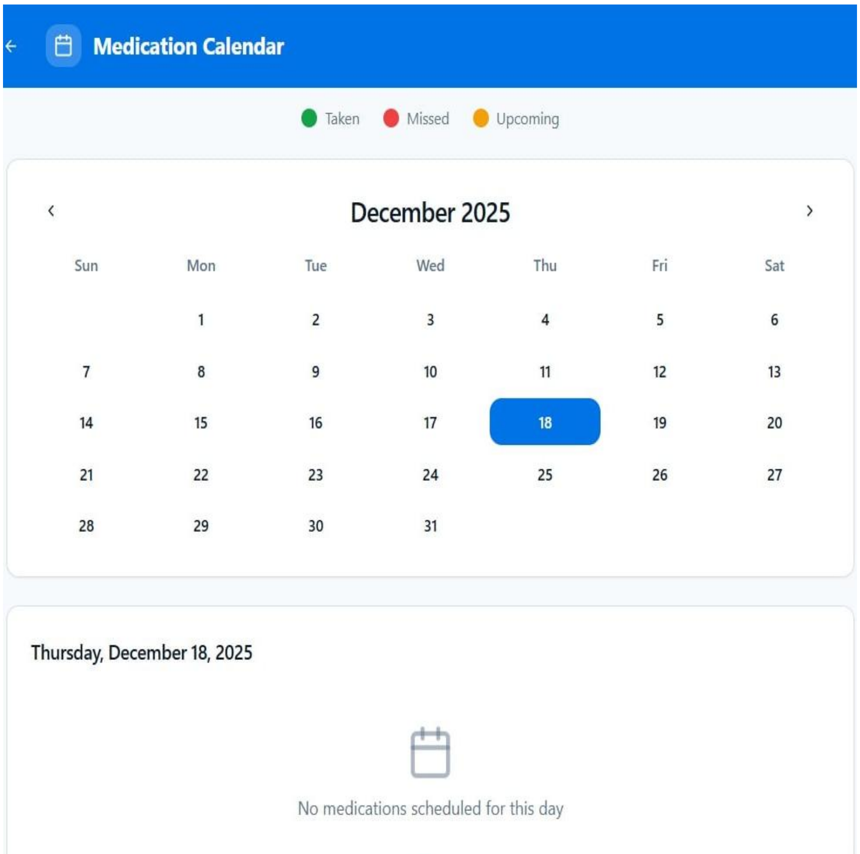

Figure 51: medication calendar UI

3.4 Access Control and Security

3.4.1 Access Control

User Authentication	Users must register and log in using valid credentials before accessing system features.
Authorization	Each user is authorized to access only their own personal and medical data.
Role Management	The system supports a basic role model where standard users can manage reminders and view recommendations.
Data Ownership Control	Users can only create, view, update, or delete their own reminders and health records.

Table 29 : access control

3.4.2 Security Measures

Data Confidentiality	Sensitive user and medical data are stored in encrypted form in the database.
Password Protection	User passwords are stored using secure hashing techniques instead of plain text.
Input Validation	All user inputs are validated to prevent invalid data and malicious attacks.
Access Protection	Authentication checks are enforced before any data access or modification.
Data Integrity	Controlled updates ensure that stored medical data remains accurate and consistent.
Privacy Protection	Personal health information is accessible only to the respective user.
System Reliability	Secure handling of reminders ensures accurate and timely notification delivery.

Table : 30 security measure

References

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Appendice

Section A: Medication Usage

Do you currently take medication regularly?

☐ Yes

☐ No

How many times per day do you take medicine?

☐ Once

☐ Twice

☐ Three times

☐ More than three times

Do you take more than one type of medicine at a time?

☐ Yes

☐ No

Who reminds you to take your medication?

☐ Myself

☐ Family member

☐ Caregiver

☐ Phone alarm

☐ No reminder

Section B: Medication Challenges

Have you ever missed a medication dose?

☐ Yes

☐ No

If yes, what is the main reason?

☐ Forgetfulness

☐ Busy schedule

☐ Lack of reminder

☐ Side effects

☐ Other: ___________

Have you ever taken the wrong medicine or dosage?

☐ Yes

☐ No

Do you know about possible drug interactions or allergies related to your medicines?

☐ Yes

☐ No